

Arachidonic Acid Metabolites and the Kidney

Raymond C. Harris | Ming-Zhi Zhang | Richard M. Breyer

CHAPTER OUTLINE

CELLULAR ORIGIN OF EICOSANOIDS, 357	PROSTANOIDS, 366
THE CYCLOOXYGENASE PATHWAY, 358	PROSTAGLANDINS, 376
RENAL COX-1 AND COX-2 EXPRESSION, 360	INVOLVEMENT OF CYCLOOXYGENASE METABOLITES IN RENAL PATHOPHYSIOLOGY, 377
RENAL COMPLICATIONS OF NONSTEROIDAL ANTIINFLAMMATORY DRUGS, 364	THE LIPOXYGENASE PATHWAY, 381
CARDIOVASCULAR EFFECTS OF COX-2 INHIBITORS, 366	THE CYTOCHROME P450 PATHWAY, 384

KEY POINTS

- The kidney is source of eicosanoids (prostaglandins, lipoxygenase metabolites, cytochrome P450 metabolites).
- Eicosanoids derived from arachidonic acid play important roles in kidney physiology and pathophysiology.
- Prostaglandins can regulate renal hemodynamics, the renin-angiotensin system and salt and water excretion.
- Eicosanoids are important mediators of systemic blood pressure.

CELLULAR ORIGIN OF EICOSANOIDS

Eicosanoids comprise a family of biologically active, oxygenated arachidonic acid (AA) metabolites. AA is a polyunsaturated fatty acid possessing 20 carbon atoms and four double bonds (C20:4) and is formed from linoleic acid (C18:2) by the addition of two carbons to the chain and further desaturation. In mammals, linoleic acid is derived strictly from dietary sources. Essential fatty acid deficiency occurs when dietary fatty acid precursors, including linoleic acid, are omitted, thereby depleting the hormone-responsive pool of AAs. Essential fatty acid deficiency thereby reduces the intracellular availability of AA in response to hormonal stimulation and abrogates many biologic actions of hormone-induced eicosanoid release.¹

Of an approximate 10 g of linoleic acid ingested/day, only about 1 mg/day is eliminated as end products of AA metabolism. Following its formation, AA is esterified into cell membrane phospholipids, principally at position 2 of the phosphatidylinositol fraction (i.e., sn-2-esterified AA), the major hormone-sensitive pool of AA that is susceptible to release by phospholipases.

Multiple stimuli lead to the release of membrane-bound AA via the activation of cellular phospholipases, principally

the phospholipase A2 family (PLA2).² This cleavage step is rate limiting in the production of biologically relevant arachidonate metabolites. In the case of PLA2 activation, membrane receptors activate guanine nucleotide-binding (G) proteins, leading to the release of AA directly from membrane phospholipids. Activation of PLC or PLD, on the other hand, releases AA via the sequential action of the phospholipase-mediated production of diacylglycerol (DAG), with the subsequent release of AA from DAG by DAG lipase.³ This pathway may also lead to the formation of the esterified AA metabolites arachidonylethanolamide (AEA) and 2-arachidonoylglycerol (2-AG), the endocannabinoids. These endocannabinoids can subsequently be converted to free AA by the action of monoacylglycerol lipases.⁴ When considering eicosanoid formation, the physiologic significance of AA release by these other phospholipases remains uncertain because, at least in the setting of inflammation, PLA2 action appears to be essential for the generation of biologically active AA metabolites.⁵

More than 15 proteins with PLA2 activity are known to exist, including secreted (sPLA2) and cytoplasmic PLA2 (cPLA2) isoforms.^{6,7} A mitogen-activated cytoplasmic PLA2 has been found to mediate AA release in a calcium-/calmodulin-dependent manner. Other hormones and growth factors,

including epidermal growth factor (EGF) and platelet-derived growth factors, activate PLA2 directly through tyrosine residue kinase activity, allowing the recruitment of co-activators to the enzyme without an absolute requirement for the intermediate action of Ca^{2+} -calmodulin or other cellular kinases.

Following de-esterification, AA is rapidly re-esterified into membrane lipids or avidly bound by intracellular proteins, hence making AA unavailable for further metabolism. Should it escape re-esterification and protein binding, free AA becomes available as a substrate for one of three major enzymatic transformations, the common result of which is the incorporation of oxygen atoms at various sites of the fatty acid backbone, with accompanying changes in its molecular structure (e.g., ring formation).^{8,9} This results in the formation of biologically active molecules, referred to as *eicosanoids*. The specific nature of the products generated is a function of the initial stimuli for AA release, as well as the metabolic enzyme available, which is determined, in part, by the cell type involved.^{9,10}

These products, in turn, either mediate or modulate the biologic actions of the agonist in question. AA release may also result from nonspecific stimuli, such as cellular trauma, including ischemia and hypoxia,¹¹ oxygen free radicals,¹² and osmotic stress.¹³ The identity of the specific AA metabolite generated in a particular cell system depends on the proximate stimulus and availability of the downstream AA metabolizing enzymes present in that cell.

Three major enzymatic pathways of free AA metabolism are present in the kidney—cyclooxygenases, lipoxygenases, and cytochrome P-450 (Fig. 13.1). The cyclooxygenase pathway mediates the formation of prostaglandins (PGs) and thromboxanes. The lipoxygenase pathway mediates the formation of mono-, di-, and trihydroxyeicosatetraenoic acids (HETEs), leukotrienes (LTs), and lipoxins (LXs). Whereas the cytochrome P-450–dependent oxygenation of AA mediates the formation of epoxyeicosatrienoic acids (EETs), their corresponding diols, HETEs, and monooxygenated AA derivatives. Fish oil diets, rich in ω -3 polyunsaturated fatty acids,¹⁴ interfere with metabolism via all three pathways by competing with AA oxygenation, resulting in the formation of biologically inactive end products.¹⁵ Interference with the production of

proinflammatory lipids has been hypothesized to underlie the beneficial effects of fish oil in immunoglobulin A (IgA) nephropathy, and other cardiovascular diseases.¹⁶ The following sections deal with the current understanding of the chemistry, biosynthesis, renal metabolism, mechanisms of release, receptor biology, signal transduction pathways, biologic activities, and functional significance of each of the metabolites generated by the three major routes of AA metabolism in the kidney.

THE CYCLOOXYGENASE PATHWAY

See Fig. 13.2.

MOLECULAR BIOLOGY

The cyclooxygenase (COX) enzyme system is the major pathway for AA metabolism in the kidney. COX (prostaglandin synthase G2/H2) is the enzyme responsible for the initial conversion of AA to prostaglandin G2 and subsequently to prostaglandin H2. The COX protein was first purified from ram seminal vesicles, and cDNA encoding COX was cloned in 1988. The protein is widely expressed, and the level of activity is not dynamically regulated. Other studies have supported the presence of a COX isoform that was dynamically regulated and responsible for increased prostanoid production in inflammation. This second inducible COX isoform was identified shortly after the cloning of the initial enzyme and designated as COX-2, whereas the initially isolated isoform is now designated as COX-1.^{8,17,18} COX-1 and COX-2 are encoded by distinct genes located on different chromosomes. The human COX-1 gene (PTGS1 [prostaglandin synthase 1]) is distributed over 40 kB on 11 exons on chromosome 9, whereas COX-2 is localized on chromosome 1 and spans approximately 9 kB. The genes are also subject to dramatically different regulatory signals.

REGULATION OF COX GENE EXPRESSION

At the cellular level, COX-2 expression is highly regulated by several processes that alter its transcription rate, message

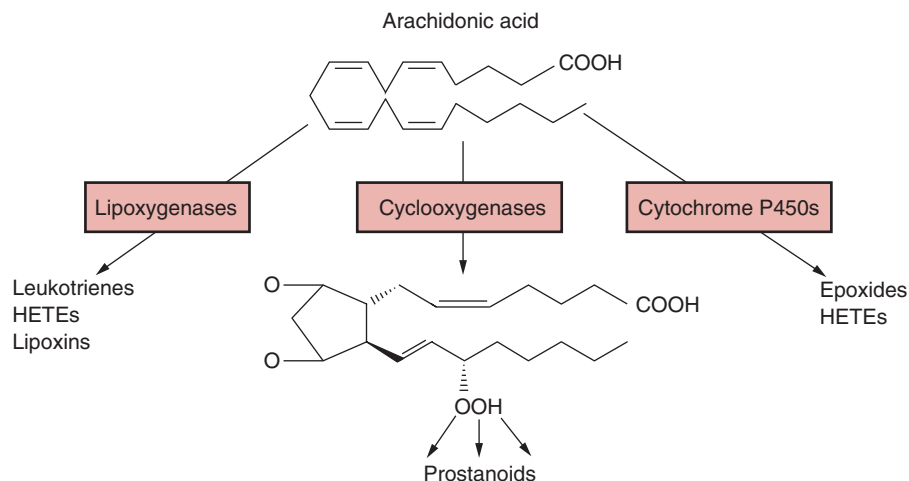


Fig. 13.1 Pathways of enzymatically mediated arachidonic acid metabolism. Arachidonic acid can be converted into biologically active compounds by cyclooxygenase (COX), lipoxygenase (LOX), or cytochrome P450 (CYP450)–mediated metabolism. *HETE*, Hydroxyeicosatetraenoic acid.

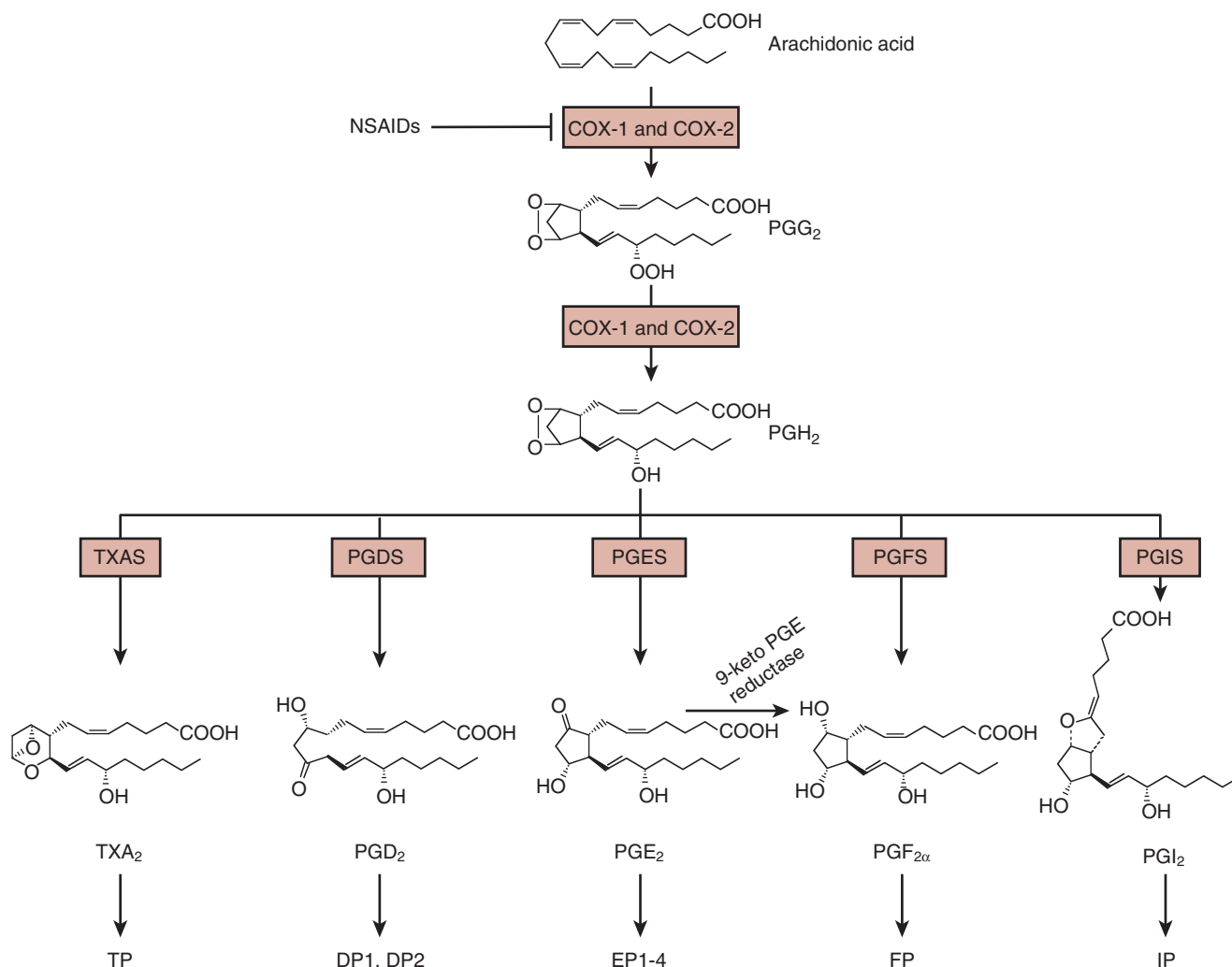


Fig. 13.2 Cyclooxygenase (COX) metabolism of arachidonic acid (AA). Both COX-1 and COX-2 convert AA to prostaglandin (PG) H₂ (PGH₂), which is then acted on by specific synthases to produce prostanoids that act at G protein–coupled receptors that increase or decrease cyclic adenosine monophosphate (cAMP) or increase intracellular calcium. *NSAID*, Nonsteroidal antiinflammatory drug; *TXAS*, thromboxane synthase; *PGDS*, prostaglandin D synthase; *PGES*, prostaglandin E synthase; *PGFS*, prostaglandin F synthase; *PGIS*, prostacyclin synthase.

export from the nucleus, messenger RNA (mRNA) stability, and efficiency of mRNA translation.^{19,20} These processes tightly control the expression of COX-2 in response to many of the same cellular stresses that activate arachidonate release (e.g., cell volume changes, shear stress, hypoxia),^{11,21} as well as a variety of cytokines and growth factors, including tumor necrosis factor (TNF), interleukin-1β, epidermal growth factor, and platelet-derived growth factor (PDGF). Activation of COX-2 gene transcription is mediated via the coordinated activation of several transcription factors that bind to and activate consensus sequences in the 5′ flanking region of the COX-2 gene for nuclear factor-kappa B (NF-κB), NF-IL-6/C-enhancer-binding protein (CEBP), and a cyclic adenosine monophosphate (cAMP) response element (CRE).²² The induction of COX-2 mRNA transcription by endotoxin (lipopolysaccharide) may also involve CRE²³ and NF-κB sites.²⁴

REGULATION OF COX EXPRESSION BY ANTIINFLAMMATORY STEROIDS

A molecular basis linking the antiinflammatory effects of COX-inhibiting nonsteroidal antiinflammatory drugs

(NSAIDs) and antiinflammatory glucocorticoids has long been sought. A novel mechanism for the suppression of arachidonate metabolism by corticosteroids involving the translational inhibition of COX formation had been suggested before the molecular recognition of COX-2. With the cloning of COX-2, it became well established that glucocorticoids suppress COX-2 expression and prostaglandin synthesis, an effect now viewed as central to the antiinflammatory effects of glucocorticoids. Posttranscriptional control of COX-2 expression represents another robust mechanism whereby adrenal steroids regulate COX-2 expression.²⁵ Accumulating evidence has suggested that COX-2 is modulated at multiple steps in addition to transcription rate, including stabilization of the mRNA and enhanced translation.^{19,26} Glucocorticoids, including dexamethasone, downregulate COX-2 mRNA, in part by destabilizing the mRNA.²⁶ The 3′-untranslated region of the COX-2 mRNA contains 22 copies of an AUUUA motif, which are known cis-RNA sequences important in destabilizing the COX-2 message in response to dexamethasone; other 3′ sequences appear to be important for COX-2 mRNA stabilization in response to IL-1β.²⁶ Effects of the 3′-untranslated region

(3'-UTR), as well as other factors regulating the efficiency of COX-2 translation, have also been suggested.¹⁹ The factors determining the expression of COX-1 are more obscure.

ENZYMATIC CHEMISTRY

Despite these differences, both prostaglandin (PG) synthases catalyze a similar reaction, resulting in the cyclization of C8 to C12 of the AA backbone, forming cyclic endoperoxide, accompanied by the concomitant insertion of two oxygen atoms at C15 to form PGG₂ (a 15-hydroperoxide). In the presence of a reduced glutathione-dependent peroxidase, PGG₂ is converted to the 15-hydroxy derivative, PGH₂. The endoperoxides (PGG₂ and PGH₂) have very short half-lives, about 5 minutes, and are biologically active in inducing aortic contraction and platelet aggregation.²⁷ However, under some circumstances, the formation of these endoperoxides may be strictly limited via the self-deactivating properties of the enzyme.

The expression of recombinant enzymes and determination of the crystal structure of COX-2 have provided further insight into the observed physiologic and pharmacologic similarities to, and differences from, COX-1. It is now clear that NSAIDs

work by inhibiting cyclooxygenases by sterically blocking access of AA to the heme-containing active enzymatic site.²⁸ Aspirin, also known as acetylsalicylic acid (ASA), predominantly targets COX-1. Particularly well conserved are COX-1 sequences surrounding the aspirin-sensitive serine residue at position 529 of COX-1, at which acetylation by aspirin irreversibly inhibits activity.²⁹ More recent evidence has shown that COX-1 and COX-2 are capable of forming heterodimers and sterically modulating each other's function.³⁰ The substrate-binding pocket of COX-2 is larger and therefore accepting of bulkier inhibitors and substrates. This difference has allowed the development and marketing of relatively and highly selective COX-2 inhibitors for clinical use as analgesics,³¹ antipyretics,³² and antiinflammatory agents.³¹ In addition to its central role in inflammation, aberrantly upregulated COX-2 expression has been implicated in the pathogenesis of a number of epithelial cell carcinomas³³ and in Alzheimer disease and other degenerative neurologic conditions.³⁴

RENAL COX-1 AND COX-2 EXPRESSION

See Fig. 13.3.

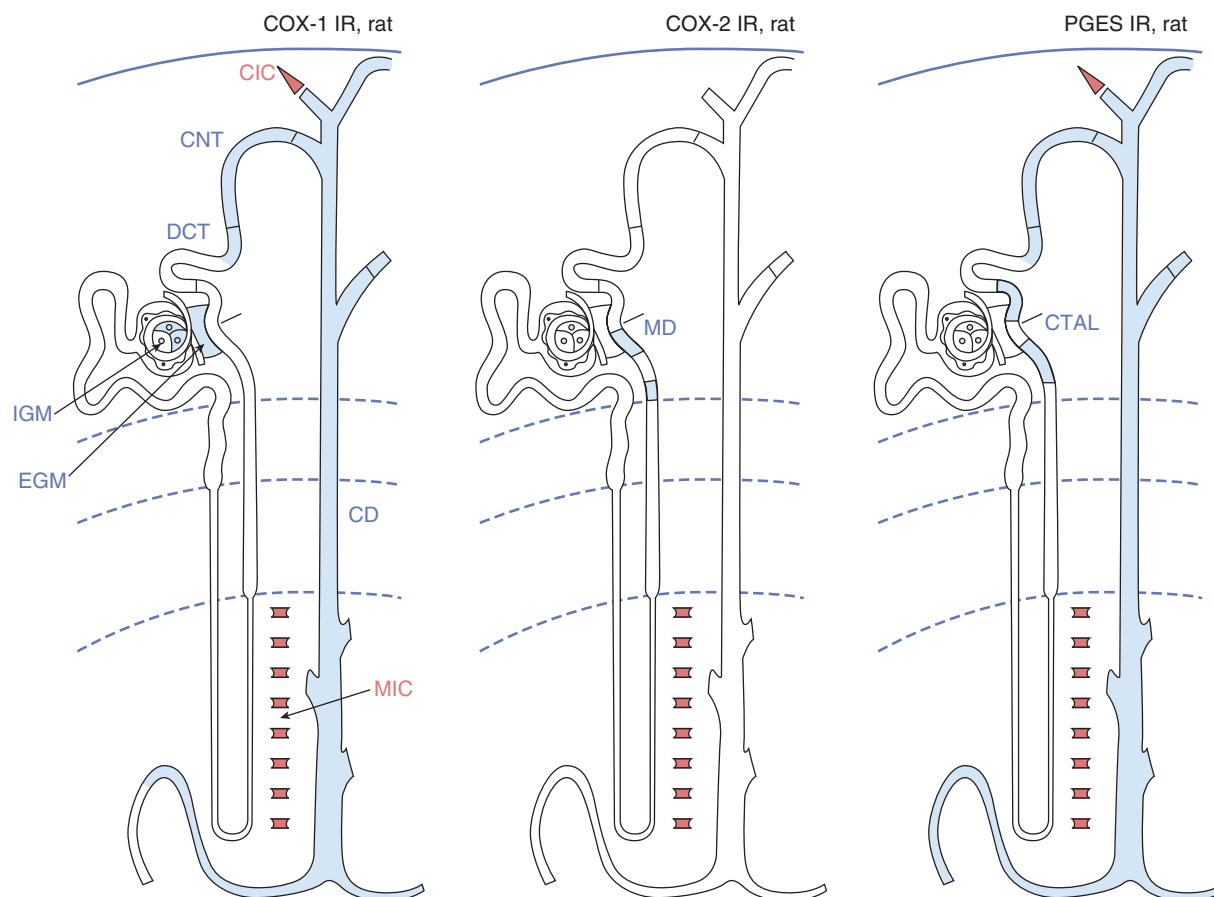


Fig. 13.3 Localization (indicated in blue shaded areas) of immunoreactive (IR) cyclooxygenase 1 and 2 (COX-1, COX-2) and microsomal prostaglandin E synthase (PGES) along the rat nephron. Eicosanoid expression in cortical (CIC) and medullary (MIC) interstitial cells is shaded in red. CD, Collecting duct; CNT, connecting tubule; CTAL, cortical thick ascending limb; DCT, distal convoluted tubule; EGM, extraglomerular mesangium; IGM, intraglomerular mesangium; MD, macula densa. (Modified from Campean V, Theilig F, Palleghe A, et al. Key enzymes for renal prostaglandin synthesis: site-specific expression in rodent kidney [rat, mouse]. *Am J Physiol Renal Physiol*. 2003;285:F19–F32.)

COX-2 EXPRESSION

COX-2 EXPRESSION IN THE KIDNEY

There is now definitive evidence for significant COX-2 expression in the mammalian kidney. This contrasts with the usual view that COX-2 is induced by cellular injury in other tissues, such as blood vessels. COX-2 mRNA and immunoreactive COX-2 are present at low but detectable levels in the normal adult mammalian kidney, where *in situ* mRNA hybridization and protein immunolocalization have demonstrated constitutive localized expression of COX-2 mRNA and immunoreactive protein in the cells of the macula densa and in a few cells in the cortical thick ascending limb (TAL) cells immediately adjacent to the macula densa.^{35,36} COX-2 expression is also abundant in the lipid-laden medullary interstitial cells in the tip of the papilla.^{35,37} Some investigators have reported that COX-2 may be expressed in inner medullary collecting duct cells or intercalated cells in the renal cortex in the collecting duct.³⁸ Nevertheless, COX-1 expression is also constitutive and clearly the most abundant isoform in the collecting duct, so the potential existence and physiologic significance of COX-2 co-expression in this segment remains uncertain.

COX-2 EXPRESSION IN THE RENAL CORTX

It is now well documented that COX-2 is expressed in macula densa–cortical thick ascending limb of Henle (cTALH) and in the mammalian kidney,^{1,36} including the human kidney, especially in kidneys of older adults,^{39,40} patients with diabetes mellitus, congestive heart failure,⁴¹ and Bartter-like syndrome.⁴²

The presence of COX-2 in the unique group of cells comprising the macula densa, points to a potential role for COX-2–derived prostanoids in regulating glomerular function.⁴³ Studies of the prostanoid-dependent control of the

glomerular filtration rate (GFR) by the macula densa suggest effects via both dilator and constrictor effects of prostanoids contributing to tubuloglomerular feedback (TGF).^{44,45} Some studies have suggested that COX-2–derived prostanoids are predominantly vasodilators.^{46,47} By inhibiting the production of dilator prostanoids contributing to the patency of adjacent afferent arteriole, COX-2 inhibition may contribute to the decline in the GFR observed in patients taking NSAIDs or selective COX-2 inhibitors⁴⁸ (see later).

The volume-depleted state is typified by low NaCl delivery to the macula densa, and COX-2 expression in the macula densa is increased in states associated with volume depletion (Fig. 13.4).³⁵ Of note, COX-2 expression in cultured macula densa cells and cTAL cells is also increased *in vitro* by reducing the extracellular Cl[−] concentration. Studies in which cortical thick limbs and associated glomeruli were removed and perfused from rabbits pretreated with a low-salt diet to upregulate macula densa COX-2 demonstrated COX-2–dependent release of PGE₂ from the macula densa in response to decreased chloride perfusate.⁴⁹ Furthermore the induction of COX-2 by low Cl[−] levels can be blocked by a specific p38 MAP kinase inhibitor.^{50,51} Finally, *in vivo*, renal cortical immunoreactive pp38 expression (the active form of p38) predominantly localizes to the macula densa and cTALH and increases in response to a low-salt diet.⁵⁰ These findings point to a molecular pathway whereby enhanced COX-2 expression occurring in circumstances associated with intracellular volume depletion could result from decreased luminal chloride delivery. The carbonic anhydrase inhibitor, acetazolamide, and dopamine may both indirectly regulate macula densa COX-2 expression by inhibiting proximal reabsorption and thereby increasing luminal macula densa chloride delivery.^{52,53} In mice deficient in the Na⁺-H⁺ exchanger subtype 2 (NHE2), the macula densa is shrunken, accompanied by increased COX-2 expression

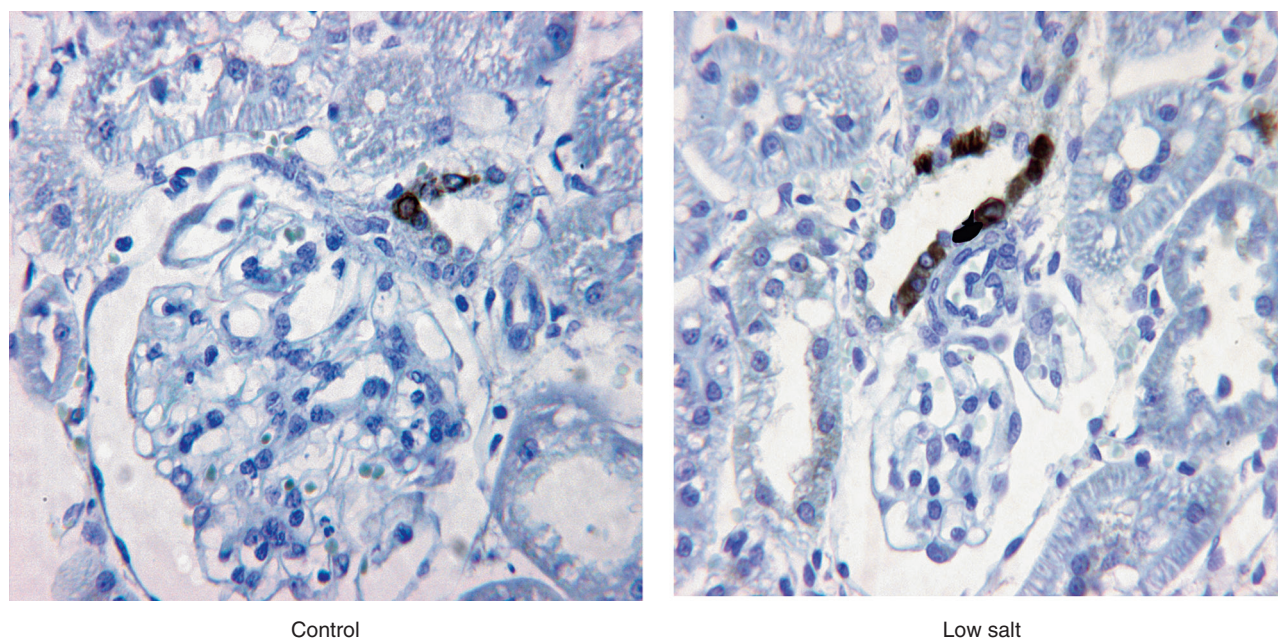


Fig. 13.4 Cyclooxygenase 2 (COX-2) expression is regulated in renal cortex in rats (original magnification: $\times 400$). Under basal conditions, sparse immunoreactive COX-2 is localized to the macula densa and surrounding cortical thick ascending limb. Following chronic administration of a sodium-deficient diet, macula densa–cortical thick ascending limb COX-2 expression increases markedly.

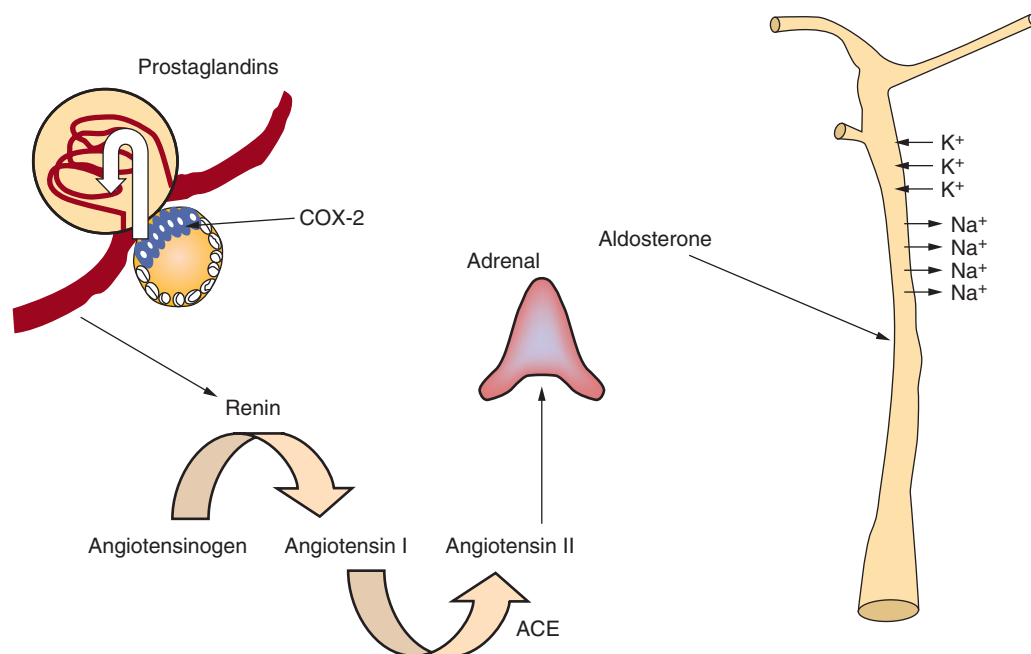


Fig. 13.5 Proposed intrarenal roles for vasodilatory prostaglandins to regulate renal function and blood pressure control. Prostaglandins released from the macula densa and/or the afferent arteriole can vasodilate the afferent arteriole and modulate renin release from juxtaglomerular cells. ACE, Angiotensin-converting enzyme; COX-2, cyclooxygenase 2.

and juxtaglomerular renin expression, suggesting that NHE2 appears to be the major isoform associated with macula densa cell volume regulation.⁵⁴

In the mammalian kidney, the macula densa is involved in regulating renin release by sensing alterations in luminal chloride via changes in the rate of $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransport (Fig. 13.5).⁵⁵ In vivo measurements in isolated perfused kidney and an isolated perfused juxtaglomerular preparation have all indicated that the administration of nonspecific COX inhibitors prevents the increases in renin release mediated by macula densa sensing of decreases in luminal NaCl .⁴⁴ Induction of a high-renin state by the imposition of a salt-deficient diet, angiotensin-converting enzyme (ACE) inhibition, diuretic administration, or experimental renovascular hypertension all significantly increase macula densa/cTALH COX-2 mRNA and immunoreactive protein levels.⁴³ COX-2-selective inhibitors blocked elevations in plasma renin activity, renal renin activity, and renal cortical renin mRNA in response to loop diuretics, ACE inhibitors, or a low-salt diet.^{43,56–58} and, in an isolated perfused juxtaglomerular preparation, increased renin release in response to lowering the perfusate NaCl concentration that was blocked by COX-2 inhibition.⁵⁹ In COX-2 knockout mice, increases in renin in response to low salt or ACE inhibitors were significantly blunted^{60,61} but were unaffected in COX-1 knockout mice.^{62,63} COX-2-derived PGE_2 , activating the type 4 (EP4) receptors on juxtaglomerular cells, has been shown to be important for the macula densa regulation of renin release.^{49,64} Macula densa COX-2-derived prostanoids appear to be predominantly involved in setting tonic levels of juxtaglomerular renin expression rather than necessarily mediating acute renin release,^{65,66} but PGE_2 from macula densa does stimulate CD44+ mesenchymal stromal-like cells to migrate to the juxtaglomerular apparatus (JGA) and increase renin pro-

duction, an effect mediated by EP4 receptors.⁶⁷ There is evidence that the effect of ACE inhibitors and angiotensin receptor blockers (ARBs) to increase macula densa COX-2 expression is mediated by feedback of angiotensin II on the macula densa, with AT1 and AT2 receptors inhibiting COX-2 expression.⁵⁶ In addition, prorenin and/or renin may stimulate macula densa COX-2 expression through activation of the prorenin receptor.⁶⁸

COX-2 inhibitors have also been shown to decrease renin production in models of renovascular hypertension,^{69,70} and studies in mice with targeted deletion of the prostacyclin receptor have suggested a predominant role for prostacyclin in mediating renin production and release in these models.⁷⁰ In a model of sepsis, COX-2 expression increased in macula densa and both cortical and medullary TAL. This increased COX-2 expression was mediated by Toll-like receptor 4 (TLR4) and, in $\text{TLR4}^{-/-}$ mice, JGA renin expression was absent.⁷¹

In addition to mediating juxtaglomerular renin expression, COX-2 metabolites may also modulate TGF. However, using different methodologies, investigators have reported that COX-2 metabolites predominantly modulate TGF by the production of vasodilatory prostanoids^{47,72} or mediate afferent arteriolar vasoconstriction by activating thromboxane receptors through the generation of thromboxane A2 and/or PGH_2 .⁷³ Further studies will be required to reconcile these divergent results.

There is evidence that macula densa COX-2 expression is sensitive not only to alterations in intravascular volume but also to alterations in renal metabolism. Specifically, the G protein-coupled receptor, GPR91, has been shown to be a receptor for succinate, an intermediate of the citric acid cycle (Krebs cycle).⁷⁴ GPR91 is expressed in macula densa, and both GPR91 and the intrarenal production of succinate are increased in diabetes. Studies have suggested that succinate

activation of GPR91 leads to increased macula densa COX-2 expression.^{75,76}

COX-2 EXPRESSION IN THE RENAL MEDULLA

The renal medulla is a major site of prostaglandin synthesis and abundant COX-1 and COX-2 expression (Fig. 13.6).⁷⁷ COX-1 and COX-2 exhibit differential compartmentalization within the medulla, with COX-1 predominating in the medullary collecting ducts and COX-2 predominating in medullary interstitial cells.⁴³ In the collecting duct, COX-2 has also been localized to intercalated cells but is absent in principal cells.⁷⁸ Intercalated cell COX-2 expression increases in response to angiotensin II (Ang II), and deletion exacerbates Ang II-mediated hypertension.⁷⁹ COX-2 may also be expressed in endothelial cells of the vasa recta supplying the inner medulla.

In medullary interstitial cells, dynamic regulation of COX-2 expression appears to be an important adaptive response to physiologic stresses, including water deprivation, increased dietary sodium, and exposure to endotoxins.^{38,77,80,81} In contrast, COX-1 expression is unaffected by water deprivation. Although hormonal factors could also contribute to COX-2 induction, shifting cultured renal medullary interstitial cells to hypertonic media (using NaCl or mannitol) is sufficient to induce COX-2 expression directly. Because prostaglandins play an important role in maintaining renal function during volume depletion or water deprivation, induction of COX-2 by hypertonicity provides an important adaptive response.

As is the case for the macula densa, medullary interstitial cell COX-2 expression is transcriptionally regulated in response to renal extracellular salt and tonicity. Water deprivation and a high-sodium diet both induce COX-2 expression in medullary interstitial cells by activating the nuclear factor-kappa B (NF- κ B) pathway.^{77,81} There is also evidence that

nitric oxide may modulate medullary COX-2 expression through mitogen-activated protein (MAP) kinase-dependent pathways.⁸²

The mechanisms underlying the upregulation of medullary COX-2 expression in response to volume expansion are probably multifactorial. There is clear evidence that increased medullary tonicity increases medullary COX-2 expression. Different studies have indicated a role for NF- κ B,⁷⁷ EGF receptor (EGFR) transactivation,⁸³ and mitochondria-generated reactive oxygen species (ROS).⁸⁴ Whether these represent parallel pathways or are all interrelated is not yet clear; however, it should be noted that the described EGFR transactivation is mediated by cleavage of the EGFR ligand, TGF- α , by ADAM17 (TACE), which is known to be activated by src, which can be activated by ROS. In addition to medullary COX-2, cortical COX-2 expression increases in salt-sensitive hypertension, especially in the glomerulus, and is inhibited by the superoxide dismutase mimetic, Tempol, or an ARB.⁸⁵ There is also recent evidence that COX-2 expression increases in renal macrophages in response to a high-salt diet, and selective inhibition of macrophage COX-2 expression exacerbates salt-sensitive hypertension.⁸⁶

COX-1 EXPRESSION

COX-1 EXPRESSION IN THE KIDNEY

Although well-defined factors regulating COX-2 and determining the role of COX-2 expression in the kidney are coming to light, the role of renal COX-1 remains more obscure. COX-1 is constitutively expressed in platelets,⁸⁷ in the renal microvasculature, and glomerular parietal epithelial cells (Fig. 13.7). In addition, COX-1 is abundantly expressed in the collecting duct, but there is little COX-1 expressed in the

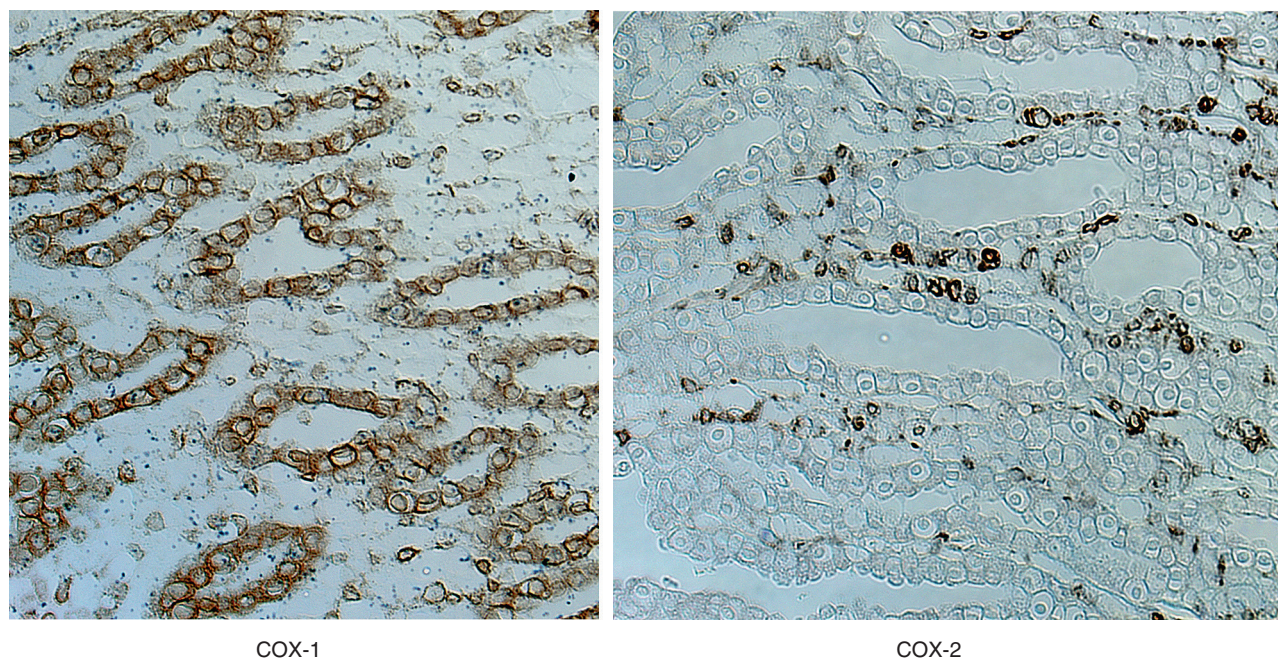


Fig. 13.6 Differential immunolocalization of cyclooxygenase 1 (COX-1) and COX-2 in the renal medulla of rodents (original magnification $\times 250$). COX-1 is predominantly localized to the collecting duct and is also found in a subset of medullary interstitial cells; COX-2 is predominantly localized to a subset of interstitial cells.

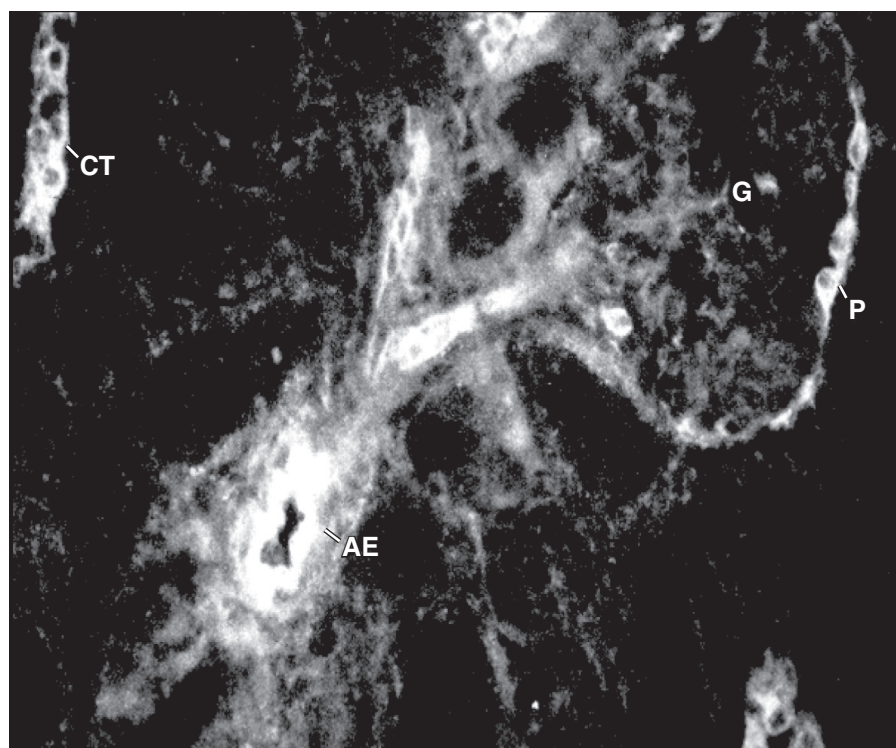


Fig. 13.7 Renal cortical cyclooxygenase 1 (COX-1) expression. Immunoreactive COX-1 is predominantly localized to the afferent arteriole (AE), glomerular mesangial cells (G), parietal glomerular epithelial cells (P), and the cortical collecting tubule (CT). (Modified from Yokoyama C, Yabuki T, Shimonishi M, et al. Prostacyclin-deficient mice develop ischemic renal disorders, including nephrosclerosis and renal infarction. *Circulation*. 2002;106(18):2397–403.)

proximal tubule or TAL.^{46,88} Although COX-1 expression levels do not appear to be dynamically regulated and, consistent with this observation, the COX-1 promoter does not possess a TATA box, vasopressin does increase COX-1 expression in collecting duct epithelial cells and in interstitial cells in the inner medulla.⁸⁸ The factors accounting for the tissue-specific expression of COX-1 are uncertain but may involve histone acetylation and the presence of two tandem Sp1 sites in the upstream promoter region of the gene.⁸⁹

RENAL COMPLICATIONS OF NONSTEROIDAL ANTIINFLAMMATORY DRUGS

SODIUM RETENTION, EDEMA, AND HYPERTENSION

The use of nonselective NSAIDs may be complicated by the development of significant Na^+ retention, edema, congestive heart failure, and hypertension.⁹⁰ These complications are also apparent in patients using COX-2 selective NSAIDs. Studies with celecoxib and rofecoxib have demonstrated that like nonselective NSAIDs, these COX-2 selective NSAIDs reduce urinary sodium (Na^+) excretion and are associated with modest Na^+ retention in otherwise healthy subjects.^{91,92} The evidence supporting a role for NSAID exacerbating hypertension, especially COX-2 inhibitors, continues to grow. COX-2 inhibition likely promotes salt retention via multiple

mechanisms (Fig. 13.8). A reduced GFR may limit the filtered Na^+ load and salt excretion.^{93,94} In addition, PGE₂ directly inhibits Na^+ absorption in the TAL and collecting duct.⁹⁵ The relative abundance of COX-2 in medullary interstitial cells places this enzyme adjacent to both these nephron segments, allowing for COX-2–derived PGE₂ to modulate salt absorption. COX-2 inhibitors decrease renal PGE₂ production^{91,96} and thereby may enhance renal sodium retention. Finally, a reduction in renal medullary blood flow by inhibition of vasodilator prostanoids may significantly reduce renal salt excretion and promote the development of edema and hypertension. COX-2–selective NSAIDs have been demonstrated to exacerbate salt-dependent hypertension.^{97,98} Similarly, patients with preexisting treated hypertension commonly experience hypertensive exacerbations with COX-2–selective NSAIDs.⁹² Taken together these data suggest that COX-2 selective NSAIDs have similar effects as nonselective NSAIDs with respect to salt excretion.

HYPERKALEMIA

Nonselective NSAIDs cause hyperkalemia due to suppression of the renin-aldosterone axis. Both a decreased GFR and inhibition of renal renin release may compromise renal K^+ excretion. Patients on a salt-restricted diet also have decreased urinary potassium excretion when treated with a COX-2–selective inhibitor,^{93,94} and COX-2–selective inhibitors may pose an equal or greater risk as nonselective NSAIDs for the development of hyperkalemia.⁹⁹

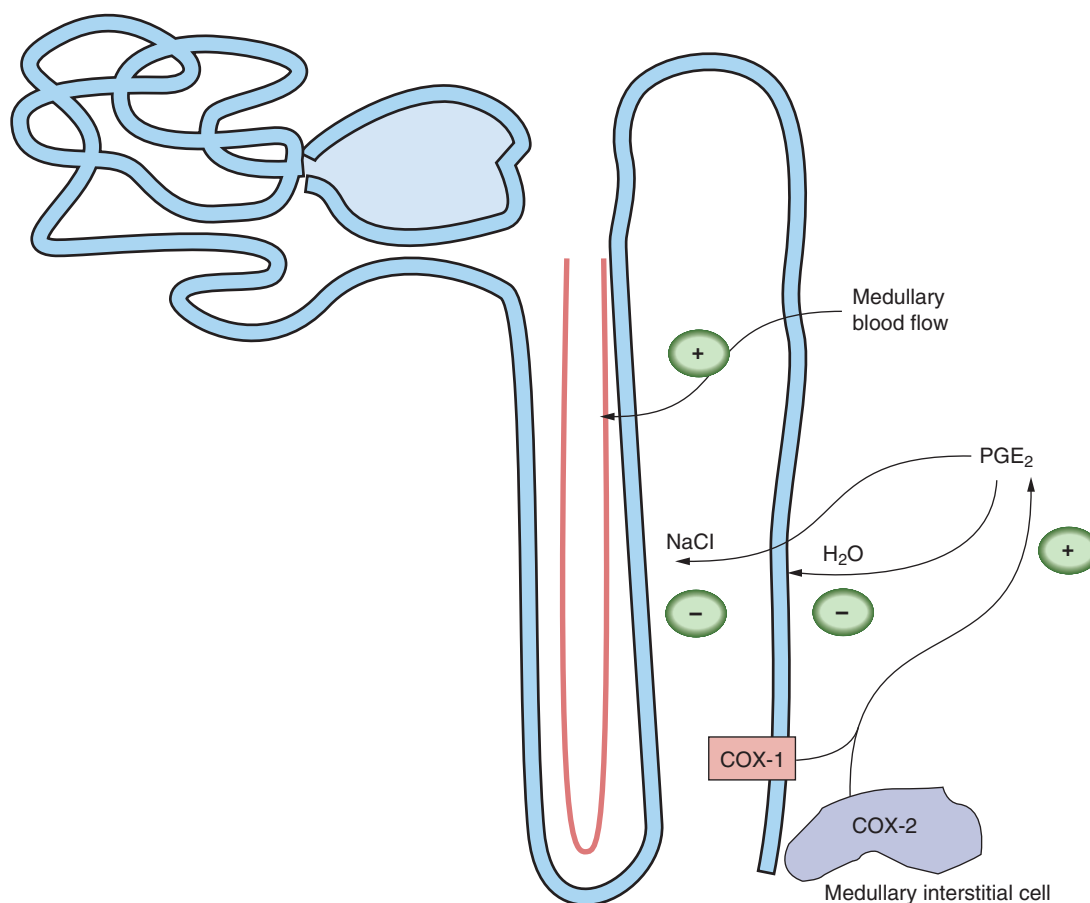


Fig. 13.8 Integrated role of prostaglandin E2 (*PGE*₂) on the regulation of salt and water excretion. *PGE*₂ can both increase medullary blood flow and directly inhibit NaCl reabsorption in the medullary thick ascending limb (mTAL) and water reabsorption in the collecting duct. COX-1, Cyclooxygenase 1; COX-2, cyclooxygenase 2.

PAPILLARY NECROSIS

Both acute and subacute forms of papillary necrosis have been observed with NSAID use.^{100–102} Acute NSAID-associated renal papillary injury is more likely to occur in the setting of dehydration, suggesting a critical dependence of renal function on COX metabolism in this setting.⁷⁷ Long-term use of NSAIDs has been associated with papillary necrosis and progressive renal structural and functional deterioration, much like the syndrome of analgesic nephropathy observed with acetaminophen, aspirin, and caffeine combinations.¹⁰¹ Experimental studies have suggested that renal medullary interstitial cells are an early target of injury in analgesic nephropathy.¹⁰³ COX-2 has been shown to be an important survival factor for cells exposed to a hypertonic medium.^{37,77,104,105} The coincident localized expression of COX-2 in these interstitial cells^{37,77} raises the possibility that like nonselective NSAIDs, long-term use of COX-2-selective NSAIDs may contribute to development of papillary necrosis.¹⁰⁶

ACUTE RENAL INSUFFICIENCY

Acute kidney injury (AKI) is a well-described complication of NSAID use.⁹⁰ This is generally considered to be a result of altered intrarenal microcirculation and glomerular

filtration secondary to the inability to produce beneficial endogenous prostanoids when the kidney is dependent on them for normal function. Like the traditional nonselective NSAIDs, COX-2-selective NSAIDs will also reduce glomerular filtration in susceptible patients.⁹⁰ Although rare overall, NSAID-associated renal insufficiency occurs in a significant proportion of patients with underlying volume depletion, renal insufficiency, congestive heart failure, diabetes, and old age.⁹⁰ These risk factors are additive and rarely are present in patients included in study cohorts used for a safety assessment of these drugs. It is therefore relevant that the COX-2-selective inhibitors celecoxib and rofecoxib cause a slight but significant fall in the GFR rate in salt-depleted but otherwise healthy subjects.^{93,94} Similar to nonselective NSAIDs, AKI can occur also secondary to COX-2-selective NSAIDs.^{48,107} Preclinical studies have supported the concept that the inhibition of COX-2-derived prostanoids generated in the macula densa contributes to a fall in the GFR by reducing the diameter of the afferent arteriole. In vivo video microscopy studies have documented a reduced afferent arteriolar diameter following the administration of a COX-2 inhibitor.⁴⁷ These animal data not only support the concept that COX-2 plays an important role in regulating the GFR but also the clinical observations that COX-2-selective inhibitors can cause renal insufficiency similar to that reported with nonselective NSAIDs.

INTERSTITIAL NEPHRITIS

The gradual development of renal insufficiency characterized by a subacute inflammatory interstitial infiltrate may occur after several months of continuous NSAID ingestion. Less commonly, the interstitial nephritis and renal failure may be fulminant. The infiltrate is typically accompanied by eosinophils; however, the clinical picture is typically much less dramatic than classic drug-induced allergic interstitial nephritis, lacking fever or rash.¹⁰⁸ This syndrome has also been reported with the COX-2-selective drug celecoxib.^{109,110} Dysregulation of the immune system is thought to play an important role in the syndrome, which typically abates rapidly following discontinuation of the NSAID or COX-2 inhibitor.

NEPHROTIC SYNDROME

Like interstitial nephritis, nephrotic syndrome typically occurs in patients chronically ingesting any one of a myriad of NSAIDs over the course of months.^{108,111} The renal pathology is usually consistent with minimal change disease, with foot process fusion of glomerular podocytes observed on electron microscopy (EM), but membranous nephropathy has also been reported.¹¹² Typically, the nephrotic syndrome occurs together with the interstitial nephritis.¹⁰⁸ Nephrotic syndrome without interstitial nephritis may occur, as well as immune-complex glomerulopathy, in a small subset of patients receiving NSAIDs. It remains uncertain whether this syndrome results from mechanism-based COX inhibition by these drugs, an idiosyncratic immune drug reaction, or a combination of both.

RENAL DYSGENESIS

Reports of renal dysgenesis and oligohydramnios in the offspring of women administered nonselective NSAIDs during the third trimester of pregnancy have implicated prostaglandins in the process of normal renal development.^{113,114} A similar syndrome of renal dysgenesis has been reported in mice, with targeted disruption of the COX-2 gene, as well as mice treated with the specific COX-2 inhibitor SC58236.¹¹⁵ Because neither COX-1^{-/-} mice or mice treated with the COX-1 selective inhibitor SC58560 exhibited altered renal development, a specific role for COX-2 in nephrogenesis has been suggested.^{116–118} A report of renal dysgenesis in the infant of a woman exposed to the COX-2-selective inhibitor nimesulide has suggested that COX-2 also plays a role in renal development in humans.¹¹³

The intrarenal expression of COX-2 in the developing kidney peaks in mice at postnatal day 4 and in the rat in the second postnatal week.^{115,119} It has not yet been determined if a similar pattern of COX-2 is seen in human kidneys. Although the most intense staining is observed in a small subset of cells in the nascent macula densa and cortical TAL, expression in the papilla has also been observed.^{115,119} *In vitro* studies have shown that exogenous PGE₂ promotes renal metanephric development¹²⁰ and is a critical growth factor for renal epithelia cells; studies in zebrafish have indicated that PGE₂, acting through EP₂ and EP₄, is a regulator of nephron formation in the zebrafish embryonic kidney.¹²¹ Studies have indicated that angiotensin II-AT₁ receptor signaling mediates COX-2-dependent postnatal development in mice.¹²²

CARDIOVASCULAR EFFECTS OF COX-2 INHIBITORS

EFFECTS OF COX-2 INHIBITION ON VASCULAR TONE

In addition to their propensity to reduce renal salt excretion and decrease medullary blood flow, NSAIDs and selective COX-2 inhibitors have been shown to exert direct effects on systemic resistance vessels. The acute pressor effect of angiotensin infusion in human subjects was significantly increased by pretreatment with the nonselective NSAID indomethacin at all Ang II doses. The administration of selective COX-2 inhibitors or COX-2 gene knockout has been shown to accentuate the pressor effects of Ang II in mice.⁴⁶ These studies have also demonstrated that Ang II-mediated blood pressure increases are markedly reduced by the administration of a selective COX-1 inhibitor or in COX-1 gene knockout mice.⁴⁶ These findings support the conclusion that COX-1-derived prostaglandins participate in, and are integral to, the pressor activity of Ang II, whereas COX-2-derived prostaglandins are vasodilators that oppose and mitigate the pressor activity of Ang II. Other animal studies have shown more directly that both NSAIDs and COX-2 inhibitors blunt arteriolar dilation and decrease flow through resistance vessels.¹²³

INCREASED CARDIOVASCULAR THROMBOTIC EVENTS

COX-2 is known to be induced in vascular endothelial cells in response to shear stress,¹²⁴ and selective COX-2 inhibition reduces circulating prostacyclin levels in normal human subjects.¹²⁵ Therefore, increasing evidence has indicated that COX-2-selective antagonism may carry increased thrombotic risks due to selective inhibition of the endothelial-derived antithrombotic prostacyclin without any inhibition of the prothrombotic platelet-derived thromboxane generated by COX-1.¹²⁶ Although animal studies have provided conflicting results about the role of COX-2 inhibition on development of atherosclerosis,^{127–131} there have been recent indications that COX-2 inhibition may destabilize atherosclerotic plaques.¹³² This has been suggested by studies indicating increased COX-2 expression and colocalization with microsomal PGE synthase-1 and metalloproteinases-2 and -9 in carotid plaques from individuals with symptomatic disease before endarterectomy.¹³³ Because of the concerns about increased cardiovascular risk, two selective COX-2 inhibitors, rofecoxib and valdecoxib, have been withdrawn from the market, and remaining coxibs and other NSAIDs have been relabeled to highlight the increased risk for cardiovascular events.

PROSTANOIDS

PROSTANOID SYNTHASES

Once PGH₂ is formed in the cell, it can undergo a number of possible transformations, yielding biologically active prostaglandins and thromboxane A₂. As seen in Fig. 13.9,

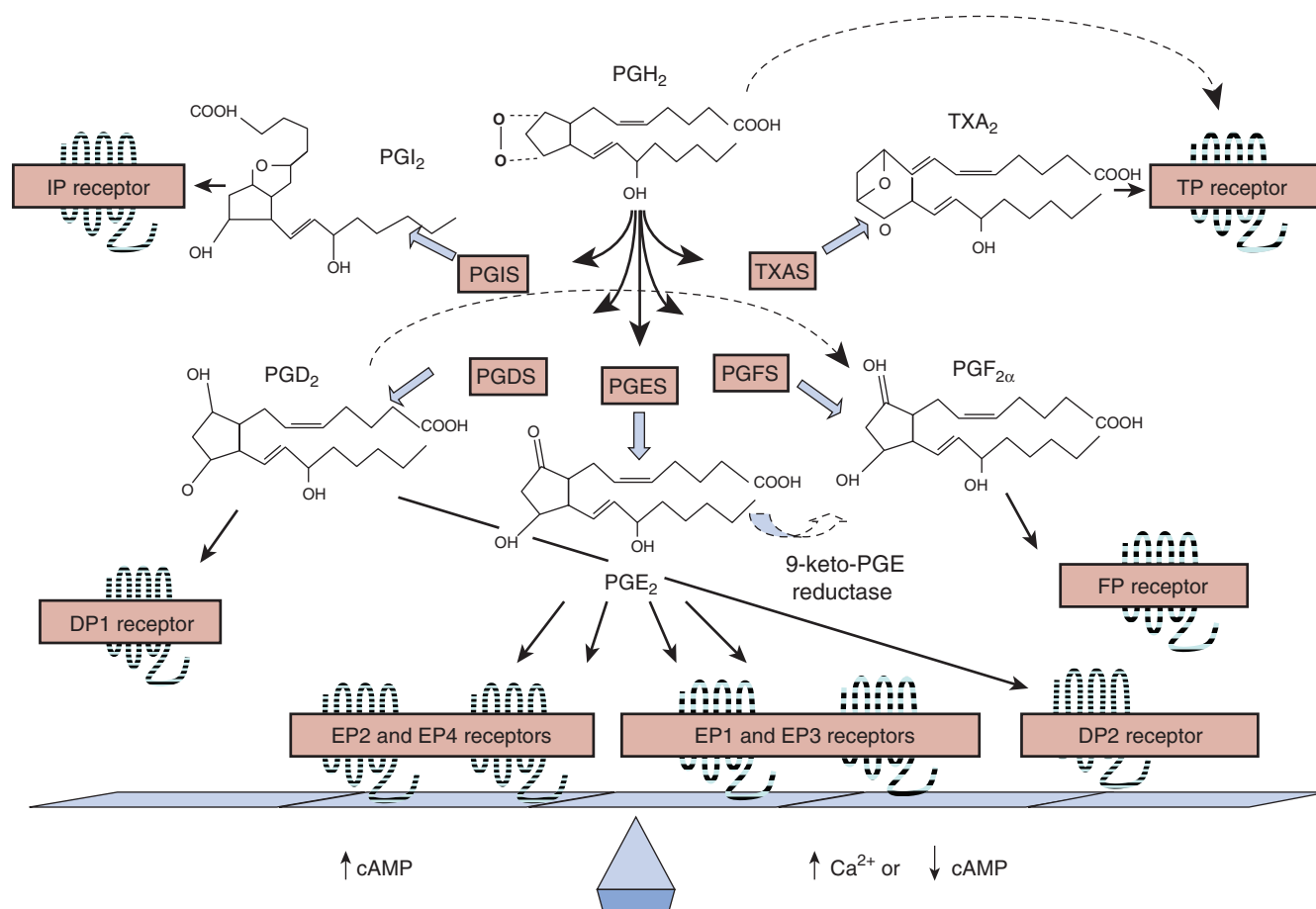


Fig. 13.9 Prostaglandin synthases. *cAMP*, Cyclic adenosine monophosphate; *PGDS*, prostaglandin D synthase; *PGES*, prostaglandin E synthase; *PGFS*, prostaglandin F synthase; *PGIS*, prostacyclin synthase; *TXAS*, thromboxane synthase.

in the presence of isomerase and reductase enzymes, PGH_2 is converted to PGE_2 and $\text{PGF}_{2\alpha}$, respectively. Thromboxane synthase converts PGH_2 into a bicyclic oxetane-oxane ring metabolite, thromboxane A_2 (TXA_2), a prominent reaction product in platelets and an established synthetic pathway in the glomerulus. Prostacyclin synthase, a 50-kDa protein located in plasma and nuclear membranes and found mostly in vascular endothelial cells, catalyzes the biosynthesis of prostacyclin (PGI_2). PGD_2 , the major prostaglandin product in mast cells, is also derived directly from PGH_2 , but its role in the kidney is uncertain. The enzymatic machinery and their localization in the kidney are discussed in detail later.

SOURCES AND NEPHRONAL DISTRIBUTION OF COX PRODUCTS

COX activity is present in arterial and arteriolar endothelial cells, including glomerular afferent and efferent arterioles.⁴³ The predominant metabolite from these vascular endothelial cells is PGI_2 .^{134,135} Whole glomeruli generate PGE_2 , PGI_2 , $\text{PGF}_{2\alpha}$, and TXA_2 .¹ The predominant products in rat and rabbit glomeruli are PGE_2 , followed by PGI_2 and $\text{PGF}_{2\alpha}$ and finally TXA_2 .

Analyses of individual cultured glomerular cell subpopulations have also provided insight into the localization of prostanoid synthesis. Cultured mesangial cells are capable of generating PGE_2 and, in some cases, $\text{PGF}_{2\alpha}$ and PGI_2

have also been detected.¹³⁶ Other studies have suggested that mesangial cells may produce the endoperoxide PGH_2 as a major COX product.¹³⁷ Glomerular epithelial cells also appear to participate in prostaglandin synthesis, but the profile of COX products generated in these cells remains controversial. Immunocytochemical studies of rabbit kidney have demonstrated intense staining for COX-1, predominantly in the parietal epithelial cells. Glomerular capillary endothelial cell PG generation profiles remain undefined but may well include prostacyclin.

The predominant synthetic site of PG synthesis along the nephron is the collecting duct (CD), particularly its medullary portion (MCT).¹³⁸ In the presence of exogenous arachidonic acid, PGE_2 is the predominant PG formed in the collecting duct, the variations among the other products being insignificant.¹ PGE_2 is also the major COX metabolite generated in medullary interstitial cells.¹³⁹ The role that specific prostanoid synthases may play in the generation of these products is outlined subsequently.

Thromboxane Synthase

TXA_2 is produced from PGH_2 by thromboxane synthase (TXAS), a microsomal protein of 533 amino acids with a predicted molecular weight of around 60 kDa. The amino acid sequence of the enzyme exhibits homology to the cytochrome P-450s and is now classified as CYP5A1.¹⁴⁰ The

human gene is localized on chromosome 7q and spans 180 kB. TXAS mRNA is highly expressed in hematopoietic cells, including platelets, macrophages, and leukocytes. TXAS mRNA is expressed in the thymus, kidney, lung, spleen, prostate, and placenta. Immunolocalization of TXAS demonstrates high expression in the dendritic cells of the interstitium, with lower expression in glomerular podocytes of human kidney.¹⁴¹ TXA2 synthase expression is regulated by dietary salt intake.¹⁴² Furthermore, the experimental use of ridogrel, a specific TXAS inhibitor, reduced blood pressure in spontaneously hypertensive rats.¹⁴³ The clinical use of TXA2 synthase inhibitors is complicated by the fact that its endoperoxide precursors (PGG2 and PGH2) are also capable of activating its downstream target, the TP receptor.²⁷

Prostacyclin Synthase

The biologic effects of prostacyclin are numerous. They include nociception, antithrombosis, and vasodilator actions, which have been targeted therapeutically to treat pulmonary hypertension.

PGI2 is produced by the enzymatic conversion of PGH2 via prostacyclin synthase (PGIS). The cloned cDNA contains a 1500–base pair open reading frame that encodes a 500–amino acid protein of approximately 56 kDa. The human prostacyclin synthase gene is present as a single-copy haploid genome and is localized on chromosome 20q. Northern blot analysis shows that prostacyclin synthase mRNA is widely expressed in human tissues and is particularly abundant in the ovary, heart, skeletal muscle, lung, and prostate. PGI synthase expression exhibits segmental expression in the kidney, especially in kidney inner medulla tubules and interstitial cells.

PGI2 synthase null mice¹⁴⁴ have reduced PGI2 levels in the plasma, kidneys, and lungs, documenting the role of this enzyme as an *in vivo* source of PGI2. Blood pressure and blood urea nitrogen and creatinine levels in the PGIS knockout mice were significantly increased, and renal pathologic findings included surface irregularity, fibrosis, cysts, arterial sclerosis, and hypertrophy of vessel walls. Thickening of the thoracic aortic media and adventitia were observed in aged PGI null mice.¹⁴⁴ Interestingly, this is a distinct phenotype from that reported for the IP receptor knockout mouse.¹⁴⁵ These differences point to the presence of additional IP-independent PGI2-activated signaling pathways. Regardless, these findings demonstrate the importance of PGI2 to the maintenance of blood vessels and to the kidney.

Prostaglandin D Synthase

Prostaglandin D2 is derived from PGH2 via the action of specific enzymes designated as PGD synthases. Two major enzymes are capable of transforming PGH2 to PGD₂, including a lipocalin-type PGD synthase and a hematopoietic-type PGD synthase.^{146,147} Mice lacking the lipocalin D synthase gene exhibit altered sleep and pain sensation.¹⁴⁸ PGD2 is the major prostanoid released from mast cells following challenge with immunoglobulin E (IgE). The kidney also appears capable of synthesizing PGD2. RNA for the lipocalin type PGD synthase has been reported to be widely expressed along the rat nephron, whereas the hematopoietic type of PGD synthase is restricted to the collecting duct.¹⁴⁹ Urinary excretion of lipocalin D synthase has been proposed as a biomarker predictive of renal injury,¹⁵⁰ and lipocalin D synthase knockout

mice appear to be more prone to diabetic nephropathy.¹⁵¹ However, the physiologic roles of these enzymes in the kidney remain less certain. Once synthesized, PGD2 is available to interact with the DP1 or DP2 receptors (see later) or undergo further metabolism to a PGF2 α -like compound.

Prostaglandin F Synthesis

PGF2 α is a major urinary COX product. It may be synthesized either directly from PGH2 via a PGF synthase¹⁵² or indirectly by metabolism of PGE2 via a 9-ketoreductase.¹⁵² Another more obscure pathway for PGF formation is by the action of a PGD2 ketoreductase, yielding a stereoisomer of PGF2 α , 11 β -PGF2 (11 β -PGF2 α).¹⁵² This reaction, and the conversion of PGD2 into an apparently biologically active metabolite (9 α ,11 β -PGF2 α) has been documented *in vivo*.¹⁵³ Interestingly, this isomer can also ligate and activate the FP receptor.¹⁵⁴ The physiologically relevant enzymes responsible for renal PGF2 α formation remain incompletely characterized.

Prostaglandin 9 Ketoreductase

Physiologically relevant transformations of COX products occur in the kidney via a nicotinamide adenine dinucleotide phosphate (NADPH)–dependent 9-ketoreductase, which converts PGE2 into PGF2 α . This enzymatic activity is typically cytosolic¹⁵² and may be detected in homogenates from the renal cortex, medulla, or papilla. The activity appears to be particularly robust in suspensions from the TALH. Renal PGE2 9-ketoreductase also exhibits 20 α -hydroxysteroid reductase activity that could affect steroid metabolism.¹⁵² This enzyme appears to be a member of the aldoketoreductase family 1C.¹⁵⁵

Interestingly, some studies have suggested that activity of a 9-ketoreductase may be modulated by salt intake and AT₂ receptor activation and may play an important role in hypertension.¹⁵⁶ Mice deficient in the AT₂ receptor exhibit salt-sensitive hypertension, increased PGE2 production, and reduced production of PGF2 α ,¹⁵⁷ consistent with reduced 9-ketoreductase activity. Other studies have suggested that dietary potassium intake may also enhance the activity of conversion from PGE2 to PGF2 α .¹⁵⁸ The intrarenal sites of expression of this enzymatic activity remain to be characterized.

Prostaglandin E Synthases

PGE2 is the other major product of COX-initiated AA metabolism in the kidney and is synthesized at high rates along the nephron, particularly in the collecting duct. Two membrane-associated PGE2 synthases have been identified, a 33-kDa and a 16-kDa membrane-associated enzyme.^{159,160} The initial report describing the cloning of a glutathione-dependent microsomal enzyme (the 16-kDa form) that specifically converts PGH2 to PGE2¹⁶⁰ noted that mRNA for this enzyme is highly expressed in reproductive tissues, as well as in kidney. Genetic disruption confirms that mPGES1^{−/−} mice exhibit a marked reduction in inflammatory responses compared with mPGES1^{+/+161} and indicate that mPGES1 is also critical for the induction of inflammatory fever.¹⁶²

Intrarenal expression of mPGES1 has been demonstrated and mapped to the collecting duct, with lower expression in medullary interstitial cells and the macula densa^{158,163} (see Fig. 13.3). Thus, in the kidney, this isoform colocalizes with both COX-1 and COX-2. In contrast, in inflammatory cells, this PGE synthase is co-induced with COX-2 and appears to

be functionally coupled to it.¹⁶⁴ Notably, the kidneys of mPGES1^{-/-} mice are normal and do not exhibit the renal dysgenesis observed in COX-2^{-/-} mice,^{117,165} nor do these mice exhibit perinatal death from a patent ductus arteriosus, which is observed with the prostaglandin EP4 receptor knockout mouse.¹⁶⁶

More recently, another membrane-associated PGE synthase, with a relative mass of around 33 kDa, was purified from the heart. The recombinant enzyme was activated by several thiol (SH)-reducing reagents, including dithiothreitol, glutathione (GSH), and betamercaptoethanol. Moreover, the mRNA distribution was high in the heart and brain and was also expressed in the kidney, but the mRNA was not expressed in the seminal vesicles. The intrarenal distribution of this enzyme is, at present, uncharacterized.¹⁵⁹

Other cytosolic proteins exhibit lower PGE synthase activity, including a 23-kDa glutathione S-transferase (GST) requiring cytoplasmic, PGES,¹⁶⁷ that is expressed in the kidney and lower genitourinary tract.¹⁶⁸ Some evidence has suggested that this isozyme may constitutively couple to COX-1 in inflammatory cells. In addition, several cytosolic glutathione-S-transferases have the capability to convert PGH₂ to PGE₂; however, their physiologic role in this process remains uncertain.¹⁶⁹

PROSTANOID RECEPTORS

See Figs. 13.10 and 13.11.

TP RECEPTORS

The TP receptor was originally purified by chromatography using a high-affinity ligand to capture the receptor.¹⁷⁰ This was

the first eicosanoid receptor cloned; it is a G protein-coupled transmembrane receptor capable of activating a calcium-coupled signaling mechanism. Other prostanoid receptors were cloned by finding cDNAs homologous to this TP receptor cDNA. Two alternatively spliced variants of the human thromboxane receptor have been described¹⁷¹ that differ in their carboxyl-terminal tail distal to arginine (Arg). Similar patterns of alternative splicing have been described for both the EP3 and FP receptors.¹⁷² Heterologous cAMP-mediated signaling of the thromboxane receptor may occur via its heterodimerization with the prostacyclin (IP) receptor.¹⁷³

Either the endoperoxide, PGH₂, or its metabolite, TXA₂, can activate the TP receptor.²⁷ Competition radioligand binding studies have demonstrated a rank order of potency on the human platelet TP receptor of the ligands I-BOP—S145 > SQ29548 > STA₂ > U-46619.^{174,175} Whereas I-BOP, STA₂, and U-46619 are agonists, SQ29548 and S145 are high-affinity TP receptor antagonists.¹⁷⁶ Studies have suggested that the TP receptor may mediate some of the biologic effects of the nonenzymatically-derived isoprostanes,¹⁷⁷ including modulation of tubuloglomerular feedback.¹⁷⁸ This latter finding may have significance in pathophysiologic conditions associated with increased oxidative stress.¹⁷⁹ Signal transduction studies have shown that the TP receptor activates phosphatidylinositol hydrolysis (PIP₂)-dependent Ca²⁺ influx.^{170,180} Quantitative polymerase chain reaction (PCR) analysis of mouse tissues has revealed that the highest level of TP mRNA expression is in the thymus, followed by the aorta, adrenal gland, vena cava, and spleen, with lower levels of expression in the pituitary gland, kidney, uterus, and brain.¹⁸¹

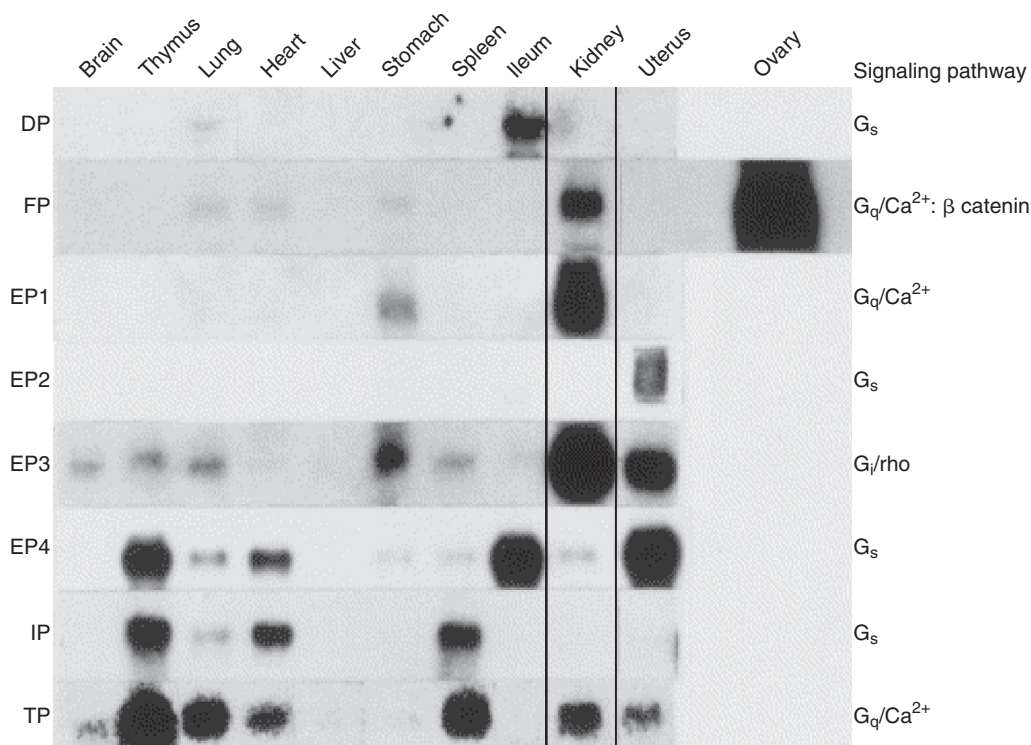


Fig. 13.10 Tissue distribution of prostanoid receptor mRNA. (Modified from Chaudhari A, Gupta S, Kirschenbaum M. Biochemical evidence for PGI₂ and PGE₂ receptors in the rabbit renal preglomerular microvasculature. *Biochim Biophys Acta*. 1990;1053(2-3):156–161.)

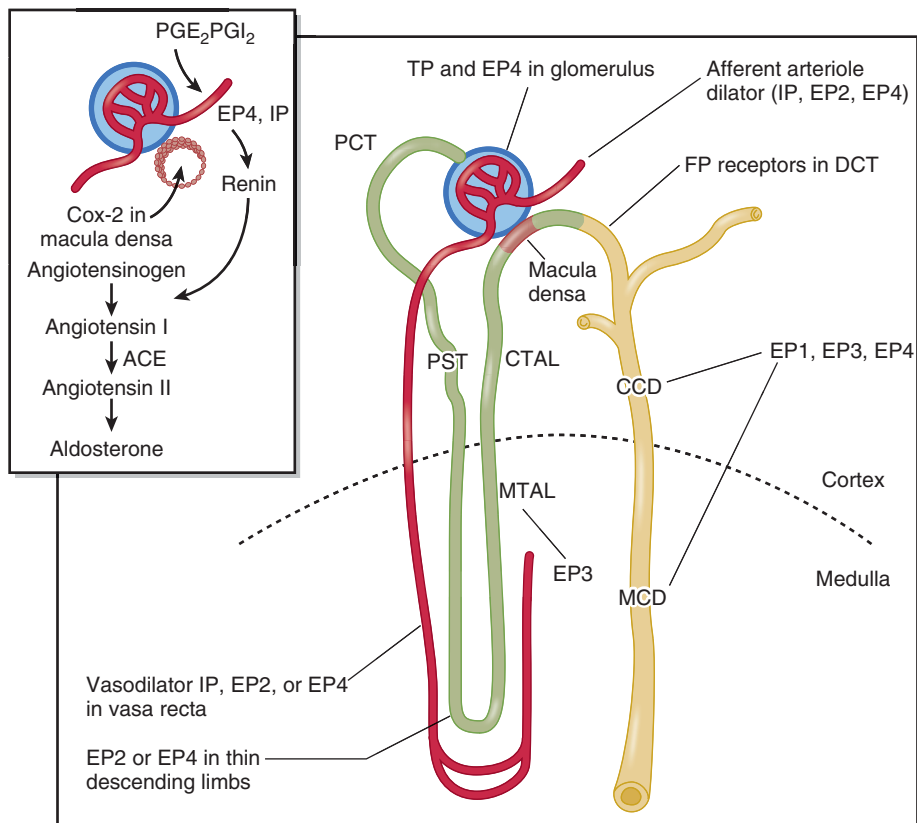


Fig. 13.11 Intrarenal localization of prostanoid receptors. Inset, Prostanoid signaling in the juxtaglomerular apparatus.

Table 13.1 Published Phenotypes of Prostanoid Receptor Knockout Mice

Receptor	Renal Expression	Renal Phenotype	Other Knockout Phenotypes
DP1	Minimal?	No	Reduced allergic asthma, reduced niacin flushing
DP2	Minimal?	++ Reduced fibrosis in UUO	Decreased cutaneous inflammatory responses
IP	++ Afferent arterioles	±	Reduced inflammation, increased thrombosis
TP	+ Glomerulus, tubules?	No	Prolonged bleeding time, platelet defect
FP	+++ Distal tubules	No	Failure of parturition
EP1	++++ MCD	No	Decreased Ang II hypertension
EP2	++ Interstitial stromal	++ Salt-sensitive hypertension	Impaired female fertility
EP3	++++ TAL, MCD	+	Impaired febrile response
EP4	+++ Glomerulus, + distal tubules	++ Reduced fibrosis in UUO	Perinatal death from persistent patent ductus arteriosus

Ang II, Angiotensin II; MCD, medullary collecting duct; TAL, thick ascending limb; UUO, unilateral ureter obstruction.

TX is a potent modulator of platelet shape change and aggregation, as well as smooth muscle contraction and proliferation. Moreover, a point mutation (Arg⁶⁰ to leucine [Leu]) in the first cytoplasmic loop of the TXA₂ receptor was identified in a dominantly inherited bleeding disorder in humans, characterized by a defective platelet response to TXA₂.¹⁸² Targeted gene disruption of the murine TP receptor also resulted in prolonged bleeding times and a reduction in collagen-stimulated platelet aggregation (Table 13.1). Conversely, overexpression of the TP receptor in vascular

tissue increases the severity of vascular pathology following injury.¹ Increased thromboxane synthesis has been linked to cardiovascular diseases, including acute myocardial ischemia, heart failure, and inflammatory renal diseases.¹

In the kidney, TP receptor mRNA has been reported in the glomeruli and vasculature. Radioligand autoradiography using ¹²⁵I-BOP has suggested a similar distribution of binding sites in mouse renal cortex, but additional renal medullary binding sites were observed.¹⁸³ These medullary TXA₂ binding sites are absent following disruption of the TP receptor gene,

suggesting that they also represent authentic TP receptors.¹⁸⁴ Glomerular TP receptors may participate in potent vasoconstrictor effects of TXA₂ analogues on the glomerular microcirculation associated with a reduced GFR.¹ Mesangial TP receptors coupled to phosphatidylinositol hydrolysis, protein kinase C activation, and glomerular mesangial cell contraction may also contribute to these effects.¹⁸⁵

An important role for TP receptors in regulating renal hemodynamics and systemic blood pressure has also been suggested. Administration of a TP receptor antagonist reduces blood pressure in spontaneously hypertensive rats (SHRs)¹⁴³ and in angiotensin-dependent hypertension.¹⁸⁶ The TP receptor also appears to modulate renal blood flow in Ang II-dependent hypertension¹⁸⁷ and in endotoxemia-induced renal failure.¹⁸⁸ Modulation of renal TP receptor mRNA expression and function by dietary salt intake has also been reported.¹⁸⁹ These studies have also suggested an important role for luminal TP receptors in the distal tubule to enhance glomerular vasoconstriction indirectly via effects on the macula densa and TGF.¹⁹⁰ However, other studies have revealed no significant difference in tubuloglomerular feedback between wild-type and TP receptor knockout mice.⁴⁵

A major phenotype of TP receptor disruption in mice and humans appears to be reduced platelet aggregation and prolonged bleeding time.¹⁸⁴ TX may also modulate the glomerular fibrinolytic system by increasing the production of an inhibitor of plasminogen activator-1 (PAI-1) in mesangial cells, which would promote fibrin accumulation.¹⁹¹ Although a specific renal phenotype in the TP receptor knockout mouse has not yet been reported, important pathogenic roles for TXA₂ and glomerular TP receptors in mediating renal dysfunction in glomerulonephritis, diabetes mellitus, and sepsis seem likely.

In an Ang II-dependent mouse model of hypertension, deletion of the TP receptor gene ameliorated hypertension and reduced cardiac hypertrophy, but had no effect on proteinuria.¹⁹² In NG-nitro-L-arginine methyl ester (L-NAME) hypertension, TP receptors again contributed to elevated blood pressure and cardiac hypertrophy. However, in this same model, TP receptors also provided unexpected protection against kidney injury—TP deletion led to an increase in worsening of histopathology and significant renal hypertrophy. This suggested that the TP receptor may play a renal protective role in some settings.¹⁹³

PROSTACYCLIN RECEPTORS

The cDNA for the IP receptor encodes a transmembrane protein of approximately 41 kDa. The IP receptor is selectively activated by the analogue cicaprost.¹⁹⁴ Iloprost and carbaprostacyclin potently activate the IP receptor but also activate the EP1 receptor. Selexipag is first in a class of long-acting IP agonists with good selectivity. Most evidence has suggested that the PGI₂ receptor signals via stimulation of cAMP generation; however, at 1000-fold higher concentrations, the cloned mouse PGI₂ receptor also signaled via PIP₂.¹⁹⁵ It remains unclear whether PIP₂ hydrolysis plays any significant role in the physiologic action of PGI₂.

The IP receptor mRNA is widely expressed and is especially abundant in mouse bone marrow, vasculature, spleen, and heart¹⁸¹ and in human kidney, liver, and lung.¹⁹⁶ In situ hybridization shows IP receptor mRNA predominantly in neurons of the dorsal root ganglia and vascular tissue,

including the aorta, pulmonary artery, and renal interlobular and glomerular afferent arterioles.¹⁹⁷ The expression of IP receptor mRNA in the dorsal root ganglia is consistent with a role for prostacyclin in pain sensation. Mice with IP receptor gene disruption exhibit a predisposition to arterial thrombosis, diminished pain perception, and inflammatory responses.¹⁴⁵

PGI₂ has been demonstrated to play an important vasodilator role in the kidney,¹⁹⁸ including in the glomerular microvasculature,¹⁹⁹ as well as regulating renin release.^{200,201} The capacity of PGI₂ and PGE₂ to stimulate cAMP generation in the glomerular microvasculature is distinct and additive,²⁰² demonstrating that the effects of these two prostanoids are mediated via separate receptors. IP receptor knockout mice also exhibit salt-sensitive hypertension.²⁰³ Prostacyclin is a potent stimulus of renal renin release, and studies using IP^{-/-} mice have confirmed an important role for the IP receptor in the development of renin-dependent hypertension of renal artery stenosis.⁷⁰ Selexipag has been shown to attenuate albuminuria in mouse models of diabetes in a nephrin-dependent manner.^{203a}

Renal epithelial effects of PGI₂ in the TAL have also been suggested,²⁰⁴ and IP receptors have been reported in the collecting duct,²⁰⁵ but the potential expression and role of prostacyclin in these segments are less well established. Of interest, in situ hybridization has also demonstrated significant expression of prostacyclin synthase in medullary collecting ducts,²⁰⁶ consistent with a role for this metabolite in this region of the kidney. In summary, although IP receptors appear to play an important role in regulating renin release and as a vasodilator in the kidney, their role in regulating renal epithelial function seems likely.

DP RECEPTORS

The DP1 receptor has been cloned and, like the IP and EP2/4 receptors, the DP receptor predominantly signals by increasing cAMP generation. The human DP receptor binds PGD₂ with a high-affinity binding of 300 pM and a lower affinity site of 13.4 nM.²⁰⁷ DP-selective PGD₂ analogues include the agonist BW 245C.²⁰⁸ DP receptor mRNA is highly expressed in leptomeninges, retina, and ileum but is not detected in the kidney.²⁰⁹ Northern blot analysis of the human DP receptor has demonstrated mRNA expression in the small intestine and retina,²¹⁰ whereas in the mouse, the DP receptor mRNA had highest expression in the olfactory epithelium, testes, and trachea.¹⁸¹ PGD₂ has also been shown to affect the sleep-wake cycle,²¹¹ pain sensation,¹⁴⁸ and body temperature.²¹² Peripherally, PGD₂ has been shown to mediate vasodilation as well as possibly inhibiting platelet aggregation. PGD₂ production is especially robust in mast cells. Inconsistent with this latter finding, the DP receptor knockout mice displayed reduced inflammation in the ovalbumin model of allergic asthma.²¹³ PGD₂ is a major mediator of niacin-induced flushing. Development of the antagonist laropiprant was undertaken to inhibit this niacin-induced vasodilation flushing response.²¹⁴ Although the kidney appears capable of synthesizing PGD₂, its role in the kidney remains poorly defined. Intrarenal infusion of PGD₂ resulted in a dependent increase in renal artery flow, urine output, creatinine clearance, and sodium and potassium excretion.²¹⁵

A second DP receptor was originally cloned as an orphan chemoattractant receptor from eosinophils and T cells (TH2 subset) and designated the CRTH2 receptor.²¹⁶ This receptor,

now designated as the DP2 receptor, bears no significant sequence homology to the family of prostanoid receptors discussed previously, and couples to increased cell calcium rather than increased cAMP. It binds agonists with an order of potency as follows: PGD₂ = PGJ₂ = 15 d PGJ₂ >> PGF₂α, PGE₂ > PGI₂, TAX₂. DP2 receptor action is blocked by the antagonist ramatroban, a drug used to treat allergic rhinitis and originally described as a TP receptor antagonist.²¹⁷ DP2 is widely expressed in a number of mouse tissues, including the kidney, and is most highly expressed in the uterus and testes.¹⁸¹ Deletion of the DP2 receptor gene was protective in a mouse unilateral ureteral obstruction (UO) model of fibrosis.²¹⁸ The recognition of this molecularly unrelated receptor allows for the possibility of the existence of a distinct and new family of prostanoid-activated membrane receptors.

FP RECEPTORS

The cDNA encoding the PGF₂α receptor (FP receptor) was cloned from a human kidney cDNA library and encodes a protein of 359 amino acid residues. The bovine and murine FP receptors, similarly cloned from the corpora lutea, encode proteins of 362 and 366 amino acid residues, respectively. Transfection of HEK293 cells with the human FP receptor cDNA conferred preferential ³H-P PGF₂α binding with a K_D of 4.3 ± 1.0 nM.^{176,219} Selective activation of the FP receptor may be achieved using fluprostenol or latanoprost.¹⁷⁶ ³H-PGF₂α binding was displaced by a panel of ligands with a rank order potency as follows: PGF₂α = fluprostenol > PGD₂ > PGE₂ > U46619 > iloprost.¹⁹⁴ When expressed in oocytes, PGF₂α or fluprostenol induced a Ca²⁺-dependent Cl⁻ current. Increased cell calcium has also been observed in fibroblasts expressing an endogenous FP receptor.²²⁰ FP receptors may also activate protein kinase C-dependent and Rho-mediated-PKC-independent signaling pathways.²²¹ An alternatively spliced isoform with a shorter carboxy-terminal tail has been identified that appears to signal via a similar manner as the originally described FP receptor.²²² Studies have also suggested that these two isoforms may exhibit differential desensitization and may also activate a glycogen synthase kinase/β-catenin-coupled signaling pathway.²²³

Tissue distribution of FP receptor mRNA shows highest expression in the ovary, followed by the heart, trachea, and kidney.¹⁸¹ Expression of the FP receptor in the corpora lutea is critical for normal birth, and homozygous disruption of the murine FP receptor gene results in failure of parturition in females, apparently due to failure of the normal preterm decline in progesterone levels.²²⁴ PGF₂α is a potent constrictor of smooth muscle in the uterus, bronchi, and blood vessels; however, an endothelial FP receptor may also play a dilator role.²²⁵ The FP receptor is also highly expressed in skin, where it may play an important role in carcinogenesis.²²⁶ A clinically important role for the FP receptor in the eye has been demonstrated to increase uveoscleral outflow and reduce ocular pressure. The FP-selective agonist latanoprost, an ester prodrug that is activated in the cornea, has been used clinically as an effective treatment for glaucoma.²²⁷ Bimatoprost is a structural analogue of prostaglandin F₂α (PGF₂α). Like other PGF₂α analogues, such as latanoprost, it increases the outflow of aqueous fluid from the eye and lowers intraocular pressure. It has a curious side effect of increasing eye growth. However, in contrast to latanoprost, it does not act on the FP receptor nor on any other known prostaglandin receptor.

The role of FP receptors in regulating renal function is only partially defined. FP receptor expression has been mapped to the cortical collecting duct in mouse and rabbit kidneys.²²⁸ FP receptor activation in the collecting duct inhibits vasopressin-stimulated water absorption via a pertussis toxin-sensitive (presumably Gi) dependent mechanism. Although PGF₂α increased cell Ca²⁺ levels in the cortical collecting duct, the FP-selective agonists latanoprost and fluprostenol did not increase calcium.²²⁹ Because PGF₂α can also bind to EP1 and EP3 receptors,^{194,230,231} these data suggest that the calcium increase activated by PGF₂α in the collecting duct may be mediated via an EP receptor. PGF₂α also increases Ca²⁺ in cultured glomerular mesangial cells and podocytes,^{232,233} suggesting that an FP receptor may modulate glomerular contraction. In contrast to these findings, the demonstration of glomerular FP receptors at the molecular level has not been forthcoming. Other vascular effects of PGF₂α have been described, including selective modulation of renal production of PGF₂α by sodium or potassium loading and AT2 receptor activation.¹⁵⁶

Some reports have uncovered a role of the FP receptor in regulating renin expression. Interestingly, FP agonists increased renin mRNA expression in the JGA in a dose-dependent manner but, unlike IP receptor agonists, it did not increase intracellular cAMP. Deletion of the FP receptor resulted in decreased renin levels and decreased systemic blood pressure. These data suggest that FP receptor blockade may be a novel target for the treatment of hypertension.²³⁴

MULTIPLE EP RECEPTORS

Four EP receptor subtypes have been identified.²³⁵ Although these four receptors uniformly bind PGE₂ with a higher affinity than other endogenous prostanoids, the amino acid homology of each is more closely related to other prostanoid receptors that signal through similar mechanisms.¹⁷⁵ Thus, the relaxant cAMP-coupled EP2 receptor is more closely related to other relaxant prostanoid receptors, such as the IP and DP receptors, whereas the constrictor/Ca²⁺-coupled EP1 receptor is more closely related to the other Ca²⁺-coupled prostanoid receptors, such as the TP and FP receptors.²³⁶ These receptors may also be selectively activated or antagonized by different analogues. EP receptor subtypes also exhibit differential expression along the nephron, suggesting distinct functional consequences of activating each EP receptor subtype in the kidney.²³⁷

EP1 Receptors

The human EP1 receptor cDNA encodes a 402-amino acid polypeptide that signals via IP₃ generation and increased cell Ca²⁺ with IP₃ generation. Studies of EP1 receptors may use one of several relatively selective antagonists, including ONO-871, SC19220, and SC53122. EP1 receptor mRNA has widespread expression, presumably from its vascular expression,¹⁸¹ and is expressed in the kidney >> gastric muscularis mucosae > adrenal.²³⁸ Renal EP1 mRNA expression determined by *in situ* hybridization is expressed primarily in the collecting duct and increases from the cortex to the papillae.²³⁸ Activation of the EP1 receptor increases intracellular calcium levels and inhibits Na⁺ and water reabsorption absorption in the collecting duct,²³⁸ suggesting that renal EP1 receptor activation might contribute to the natriuretic and diuretic effects of PGE₂.

Hemodynamic microvascular effects of EP1 receptors have also been reported. The EP1 receptor was originally described as a smooth muscle constrictor.²³⁹ The EP1 receptor may also be present in cultured glomerular mesangial cells,²⁴⁰ where it could play a role as a vasoconstrictor and a stimulus for mesangial cell proliferation. Although a constrictor PGE2 effect has been reported in the afferent arteriole of rat,²⁴¹ apparently produced by EP1 receptor activation,²⁴² there does not appear to be very high expression of the EP1 receptor mRNA in preglomerular vasculature or other arterial resistance vessels in mice or rabbits.²⁴³ Other reports have suggested that EP1 receptor knockout mice exhibit hypotension and hyperreninemia, supporting a role for this receptor in maintaining blood pressure.²⁴⁴

EP2 Receptors

Two cAMP-stimulating EP receptors, designated EP2 and EP4, have been identified. The EP2 receptor can be pharmacologically distinguished from the EP4 receptor by its sensitivity to butaprost.²⁴⁵ Before 1995, the cloned EP4 receptor was designated as the EP2 receptor, but then a butaprost-sensitive EP receptor was cloned,²⁴⁶ the original receptor was reclassified as the EP4 receptor and the newer butaprost sensitive protein as the EP2 receptor.²⁴⁷ A pharmacologically defined EP2 receptor has now also been cloned for the mouse, rat, rabbit, dog, and cow.²⁴⁸ The human EP2 receptor cDNA encodes a 358-amino acid polypeptide, which signals through increased cAMP. The EP2 receptor may also be distinguished from the EP4 receptor, the other major relaxant EP receptor, by its relative insensitivity to the EP4 agonist PGE1-OH and to the weak EP4 antagonist AH23848²⁴⁵ and the high-affinity EP4 antagonists ONO-AE3-208 and L-161982.²⁴⁹ The EP2 antagonist PF-04418948 has been described,²⁵⁰ as has the EP2/DP₁ antagonist TG4-155,²⁵¹ which should greatly facilitate the characterization of EP2 versus EP4 effects *in vivo*.

The precise distribution of the EP2 receptor mRNA has been partially characterized. This reveals a major mRNA species of around 3.1 kb, which is most abundant in the uterus, lung, and spleen, exhibiting only low levels of expression in the kidney.²⁴⁸ Studies using PCR across a range of tissue have demonstrated highest expression in the bone marrow > ovary > lung, consistent with these earlier findings.¹⁸¹ EP2 mRNA is expressed at much lower levels than EP4 mRNA in most tissues.²⁵² There is scant evidence to suggest segmental distribution of the EP2 receptor along the nephron.²⁴⁸ Interestingly, it is expressed in cultured renal interstitial cells, supporting the possibility that the EP2 receptor is predominantly expressed in this portion of the nephron.²⁴⁸ Studies in knockout mice have demonstrated a critical role for the EP2 receptor role in ovulation and fertilization.²⁵³ In addition, these studies have suggested a potential role for the EP2 receptor in salt-sensitive hypertension.²⁵³ This latter finding supports an important role for the EP2 receptor in protecting systemic blood pressure, perhaps via its vasodilator effect or its effects on renal salt excretion. Evidence for the latter role has been revealed in studies demonstrating that a high-salt diet increases PGE2 production, and infusion of EP-selective agonists identified the EP2 receptor as mediating PGE2-evoked natriuresis. Moreover, deletion of the EP2 receptor ablated the natriuretic effect of PGE2.²⁵⁴

EP3 Receptors

The EP3 receptor generally acts as a constrictor of smooth muscle.²⁵⁵ Ribonuclease protection and Northern blot analysis of mRNA levels have demonstrated relatively high levels of EP3 receptor expression in several tissues, including the kidney, uterus, adrenal, and stomach, with riboprobes hybridizing to major mRNA species at around 2.4 and around 7.0 kb.²⁵⁶ A metabolic pattern of expression was found by PCR with high levels of expression in the pancreas, as well as in brown fat tissue in addition to expression in the kidney.¹⁸¹ This receptor is unique in that there are multiple (more than eight) alternatively spliced variants, differing only in their C-terminal cytoplasmic tails.^{257–259} The EP3 splice variants bind PGE2 and the EP3 agonists MB28767 and sulprostone with similar affinity and, although they exhibit common inhibition of cAMP generation via a pertussis toxin-sensitive G_i-coupled mechanism, the tails may recruit different signaling pathways, including Ca²⁺-dependent signaling^{175,245} and the small G protein, rho.²⁶⁰ A Ptx-insensitive pathway for the inhibition of cAMP generation via G_z has also been described.²⁶¹ Differences in agonist-independent activity have been observed for several of the splice variants, suggesting that they may play a role in constitutive regulation of cellular events.²⁶² The physiologic roles of these different C-terminal splice variants and sites of expression within the kidney remain uncertain.

In situ hybridization has demonstrated that EP3 receptor mRNA is abundant in the TAL and collecting duct.²⁶³ This distribution has been confirmed by reverse transcriptase (RT)-PCR on microdissected rat and mouse collecting ducts and corresponds to the major binding sites for radioactive PGE2 in the kidney.²⁶⁴ An important role for a G_i-coupled PGE receptor in regulating water and salt transport along the nephron has been recognized for many years. PGE2 directly inhibits salt and water absorption in both microperfused TALs and collecting ducts (CDs). PGE2 directly inhibits Cl⁻ absorption in the mouse or rabbit medullary TAL from the luminal or basolateral surfaces.²⁶⁵ PGE2 also inhibits hormone-stimulated cAMP generation in TAL. Good and George have demonstrated that PGE2 modulates ion transport in the rat TAL by a pertussis toxin-sensitive mechanism.²⁶⁵ Interestingly, these effects also appear to involve protein kinase C activation,²⁶⁶ possibly reflecting activation of a novel EP3 receptor signaling pathway and possibly corresponding to alternative signaling pathways, as described earlier.²⁶⁰ Taken together, these data support a role for the EP3 receptor in regulating transport in the collecting duct and TAL.

Blockade of endogenous PGE2 synthesis by NSAIDs enhances urinary concentration. It is likely PGE2-mediated antagonism of vasopressin-stimulated salt absorption in the TAL, and water absorption in the collecting duct contributes to its diuretic effect. In the *in vitro* microperfused collecting duct, PGE2 inhibits both vasopressin-stimulated osmotic water absorption and vasopressin-stimulated cAMP generation.²²⁹ Furthermore, PGE2 inhibition of water absorption and cAMP generation are both blocked by pertussis toxin, suggesting effects mediated by the inhibitory G protein, G_i.²²⁹ When administered in the absence of vasopressin, PGE2 actually stimulates water absorption in the collecting duct from the luminal or basolateral side.²⁶⁷ These stimulatory effects of PGE2 on transport in the collecting duct appear to be related to activation of the EP4 receptor.²⁶⁷ Despite the presence of

this absorption-enhancing EP receptor, *in vivo* studies have suggested that in the presence of vasopressin, the predominant effects of endogenous PGE₂ on water transport are diuretic. Based on the preceding functional considerations, one would expect EP3^{-/-} mice to exhibit an inappropriately enhanced urinary concentration. Surprisingly, EP3^{-/-} mice exhibited a comparable urinary concentration following desmopressin (DDAVP), similar 24-hour water intake, and similar maximal and minimal urinary osmolality. The only clear difference was that in mice allowed free access to water, indomethacin increased urinary osmolality in normal mice but not in the knockout animals. These findings raise the possibility that some of the renal actions of PGE₂ normally mediated by the EP3 receptor have been co-opted by other receptors (e.g., the EP1 or EP4 receptor) in the EP3 knockout mouse. This remains to be formally tested.

The significance of EP3 receptor activation to animal physiology has been significantly advanced by the availability of mice with targeted disruption of this gene. Mice with targeted deletion of the EP3 receptor exhibit an impaired febrile response, suggesting that EP3 receptor antagonists could be effective antipyretic agents.²⁶⁸ Other studies have suggested that the EP3 receptor plays an important vasopressor role in the peripheral circulation of mice.²⁴³ Studies in knockout mice have also supported a potential role for the EP3 receptor as an important systemic vasopressor.^{243,269} In the intrarenal circulation, the PGE₂ has variable effects, acting as a vasoconstrictor in the larger proximal portion of the intralobular arteries and changing to a vasodilator effect in the smaller distal intralobular arteries and afferent arterioles.²⁷⁰

EP4 Receptor

Although, like the EP2 receptor, the EP4 receptor signals through increased cAMP,²⁷¹ it has been found to signal through a number of other pathways as well.²⁷² These other pathways include arrestin-mediated signaling, PI3 kinase, signaling β -catenin, and G_i coupling. The human EP4 receptor cDNA encodes a 488-amino acid polypeptide with a predicted molecular mass of ~53 kDa.²⁷³ Note that care must be taken in reviewing the literature prior to 1995, when this receptor was generally referred to as the EP2 receptor.²⁴⁷ In addition to the human receptor, EP4 receptors for the mouse, rat, rabbit, and dog have been cloned. EP4 receptors can be pharmacologically distinguished from the EP1 and EP3 receptors by insensitivity to sulprostone and from EP2 receptors by its insensitivity to butaprost and relatively selective activation by PGE₁-OH.¹⁷⁶ EP4-selective agonists (ONO-AE1-329, ONO-4819) and antagonists (ONOAC-3208, L-161982) have been generated²⁴⁵ and have been used to investigate the role of EP4 *in vivo*. Activation of the EP4 receptor was able to ameliorate the phenotype of a mouse model of nephrogenic diabetes insipidus.²⁷⁴

EP4 receptor mRNA is highly expressed relative to the EP2 receptor and widely distributed, with a major species of around 3.8 kb detected by Northern blot analysis in the thymus, ileum, lung, spleen, adrenal, and kidney.^{252,275,275a} Dominant vasodilator effects of EP4 receptor activation have been described in venous and arterial beds.^{208,255} A critical role for the EP4 receptor in regulating the perinatal closure of the pulmonary ductus arteriosus has also been suggested by studies of mice with targeted disruption of the EP4 receptor gene.^{166,276} On a 129-strain background, EP4^{-/-} mice had nearly

100% perinatal mortality due to persistent patent ductus arteriosus.²⁷⁶ Interestingly, when bred on a mixed genetic background, only 80% of EP4^{-/-} mice died, whereas around 21% underwent closure of the ductus and survived.¹⁶⁶ Preliminary studies in these survivors have supported an important role for the EP4 receptor as a systemic vasodepressor;²⁷⁷ however, their heterogeneous genetic background complicates the interpretation of these results because survival may select for modifier genes that not only allow ductus closure but also alter other hemodynamic responses.

Other roles for the EP4 receptor in controlling blood pressure have been suggested, including the ability to stimulate aldosterone release from zona glomerulosa cells.²⁷⁸ In the kidney, EP4 receptor mRNA expression is primarily in the glomerulus, where its precise function is uncharacterized,^{275,279} but might contribute to the regulation of the renal microcirculation as well as renin release.²⁸⁰ Studies in mice with genetic deletion of selective prostanoid receptors have indicated that EP4^{-/-} mice, as well as IP^{-/-} mice to a lesser extent, fail to increase renin production in response to loop diuretic administration, indicating that macula densa-derived PGE₂ increases renin primarily through EP4 activation.²⁸¹ This corresponds to other studies suggesting that EP4 receptors are expressed in cultured podocytes and juxtaglomerular apparatus cells.^{232,280} PGE₂ may mediate increased podocyte COX-2 expression through EP4-mediated increased cAMP, which activates P38 through an independent process.²⁸² Finally, the EP4 receptor in the renal pelvis may participate in regulation of salt excretion by altering afferent renal nerve output.²⁸³

REGULATION OF RENAL FUNCTION BY EP RECEPTORS

PGE₂ exerts a number of effects in the kidney, presumably mediated by EP receptors. PGE₂ not only dilates the glomerular microcirculation and vasa rectae, supplying the renal medulla,²⁸⁴ but also modulates salt and water transport in the distal tubule (Fig. 13.5).²⁸⁵ The maintenance of normal renal function during physiologic stress is particularly dependent on endogenous prostaglandin synthesis. In this setting, the vasoconstrictor effects of Ang II, catecholamines, and vasopressin are more effectively buffered by prostaglandins in the kidney than in other vascular beds, preserving normal renal blood flow, the GFR, and salt excretion. Administration of COX-inhibiting NSAIDs in the setting of volume depletion interferes with these dilator effects and may result in a catastrophic decline in the GFR, resulting in overt renal failure.²⁸⁶

Other evidence points to vasoconstrictor and prohypertensive effects of endogenous PGE₂. PGE₂ stimulates renin release from the juxtaglomerular apparatus,²⁸⁷ leading to a subsequent increase in the vasoconstrictor Ang II. In conscious dogs, chronic intrarenal PGE₂ infusion increases renal renin secretion, resulting in hypertension.²⁸⁸ Treatment of salt-depleted rats with indomethacin not only decreases plasma renin activity, but also reduces blood pressure, suggesting that PGs support blood pressure during salt depletion via their capacity to increase renin.²⁸⁹ Direct vasoconstrictor effects of PGE₂ on vasculature have also been observed.²⁴³ It is conceivable that these latter effects might predominate in circumstances where the kidney is exposed to excessively high perfusion pressures. Thus, depending on the setting, the primary effect of PGE₂ may be to increase or decrease

vascular tone, effects that appear to be mediated by distinct EP receptors.

RENAL CORTICAL HEMODYNAMICS

The expression of the EP4 receptor in the glomerulus suggests that it may play an important role in regulating renal hemodynamics. PGs regulate the renal cortical microcirculation and, as suggested previously, both glomerular constrictor and dilator effects of PGs have been observed.^{243,290} In the setting of volume depletion, endogenous PGE2 helps maintain the GFR by dilating the afferent arteriole.²⁹⁰ Some studies have suggested roles for EP and IP receptors coupled to increased cAMP generation in mediating vasodilator effects in the preglomerular circulation.^{44,280,291} PGE2 exerts a dilator effect on the afferent arteriole but not the efferent arteriole, consistent with the presence of an EP2 or EP4 receptor in the preglomerular microcirculation.

RENIN RELEASE

Other studies have suggested that the EP4 receptor may also stimulate renin release. Soon after the introduction of NSAIDs, it was recognized that endogenous PGs play an important role in stimulating renin release.⁴⁴ Treatment of salt-depleted rats with indomethacin not only decreases plasma renin activity, but also causes blood pressure to fall, suggesting that PGs support blood pressure during salt depletion via their capacity to increase renin. Prostanoids also play a central role in the pathogenesis of renovascular hypertension, and administration of NSAIDs lowers blood pressure in animals and humans with renal artery stenosis.²⁹² PGE2 induces renin release in isolated preglomerular juxtaglomerular apparatus cells.²⁸⁷ Like the effect of β -adrenergic agents, this effect appears to be through a cAMP-coupled response, supporting a role for an EP4 or EP2 receptor.²⁸⁷ EP4 receptor mRNA has been detected in microdissected JGAs,²⁹³ supporting the possibility that renal EP4 receptor activation contributes to enhanced renin release. Finally, regulation of plasma renin activity and intrarenal renin mRNA does not appear to be different in wild-type and EP2 knockout mice,²⁹⁴ arguing against a major role for the EP2 receptor in regulating renin release. Conversely, one report has suggested that EP3 receptor mRNA is localized to the macula densa, suggesting that this cAMP-inhibiting receptor may also contribute to the control of renin release.²⁷⁹

RENAL MICROCIRCULATION

The EP2 receptor also appears to play an important role in regulating afferent arteriolar tone.²⁹⁰ In the setting of systemic hypertension, the normal response of the kidney is to increase salt excretion, thereby mitigating the increase in blood pressure. This so-called “pressure natriuresis” plays a key role in the ability of the kidney to protect against hypertension.²⁹⁵ Increased blood pressure is accompanied by increased renal perfusion pressure and enhanced urinary PGE2 excretion.²⁹⁶ Inhibition of PG synthesis markedly blunts (although it does not eliminate) pressure natriuresis.²⁹⁷ The mechanism whereby PGE2 contributes to pressure natriuresis may involve changes in resistance of the renal medullary microcirculation.²⁹⁸ PGE2 directly dilates descending vasa recta, and increased medullary blood flow may contribute to the increased interstitial pressure observed as renal perfusion pressure increases, leading to enhanced salt excretion.²⁸⁴

The identity of the dilator PGE2 receptor controlling the contractile properties of the descending vasa recta remains uncertain, but EP2 or EP4 receptors seem likely candidates.²⁰⁸ Studies demonstrating salt-sensitive hypertension in mice with targeted disruption of the EP2 receptor²⁵³ have suggested that the EP2 receptor facilitates the ability of the kidney to increase sodium excretion, thereby protecting systemic blood pressure from a high-salt diet. Given its defined role in vascular smooth muscle,²⁵³ these effects of the EP2 receptor disruption seem more likely to relate to its effects on renal vascular tone. In particular, loss of a vasodilator effect in the renal medulla might modify pressure natriuresis and could contribute to hypertension in EP2 knockout mice. Nonetheless, a role for the EP2 or EP4 receptor in regulating renal medullary blood flow remains to be established. In conclusion, direct vasomotor effects of EP4 receptors, as well as effects on renin release, may play critical roles in regulating systemic blood pressure and renal hemodynamics.

EFFECTS ON SALT AND WATER TRANSPORT

COX-1 and COX-2 metabolites of arachidonate have important direct epithelial effects on salt and water transport along the nephron.²⁹⁹ Thus, functional effects can be observed that are thought to be independent of any hemodynamic changes produced by these compounds. Because biologically active AA metabolites are rapidly metabolized, they act predominantly in an autocrine or paracrine fashion, and thus, their locus of action will be close to their point of generation. One can expect, therefore, that direct epithelial effects of these compounds will result when they are produced by the tubule cells themselves or the neighboring interstitial cells, and the tubules possess an appropriate receptor for the ligand.

PROXIMAL TUBULE

Neither the proximal convoluted tubule nor the proximal straight tubule appear to produce amounts of biologically active COX metabolites of arachidonic acid. As will be discussed in a subsequent section, the dominant arachidonate metabolites produced by proximal convoluted and straight tubules are metabolites of the cytochrome P-450 pathway.³⁰⁰

Early whole-animal studies have suggested that PGE2 might have an action in the proximal tubule because of its effects on urinary phosphate excretion. PGE2 blocks the phosphaturic action of calcitonin infusion in thyroparathyroidectomized rats. Nevertheless, studies using *in vitro* perfused proximal tubules have failed to show an effect of PGE2 on sodium chloride or phosphate transport in the proximal convoluted tubule. More recent studies have suggested that PGE2 may play a key role in the phosphaturic action of FGF-23³⁰¹ because phosphaturia in hyp mice with X-linked hyperphosphaturia is associated with markedly increased urine PGE2 excretion, and phosphaturia was normalized by indomethacin.³⁰² Nevertheless, there are very little data on the actions of other COX metabolites in proximal tubules and scant molecular evidence for the expression of classic G protein-coupled prostaglandin receptors in this segment of the nephron.

LOOP OF HENLE

The nephron segments making up the loop of Henle also display limited metabolism of exogenous AA through the

COX pathway although, given the realization that COX-2 is expressed in this segment, it is of note that PGE₂ was uniformly greater in the cortical segment than in the medullary TAL. The TAL has been shown to exhibit PGE₂ receptors in high density.³⁰³ Studies have also demonstrated high expression levels of mRNA for the EP₃ receptor in medullary TAL of both rabbits and rats²³¹ (see earlier section on the EP₃ receptor). Subsequent to the demonstration that PGE₂ inhibits sodium chloride absorption in the medullary TAL of the rabbit TAL perfused in vitro, it was shown that PGE₂ blocks antidiuretic hormone (ADH) but not cAMP-stimulated sodium chloride absorption in the medullary TAL of the mouse. It is likely that the mechanism involves activation of G_i and inhibition of adenyl cyclase by PGE₂, possibly via the EP₃ receptors expressed in this segment.

COLLECTING DUCT SYSTEM

In vitro perfusion studies of rabbit cortical collecting tubule have demonstrated that PGE₂ directly inhibits sodium transport in the collecting duct when applied to the basolateral surface of this nephron segment. It is now apparent that PGE₂ uses multiple signal transduction pathways in the cortical collecting duct, including those that modulate intracellular cAMP levels and Ca²⁺. PGE₂ can either stimulate or suppress cAMP accumulation. The latter may also involve stimulation of phosphodiesterase. Although modulation of cAMP levels appears to play an important role in PGE₂ effects on water transport in the cortical collecting duct (see following section), it is less clear that PGE₂ affects sodium transport via the modulation of cAMP levels.²²⁹ PGE₂ has been shown to increase cell calcium possibly coupled with PKC activation in in vitro perfused cortical collecting ducts.³⁰⁴ This effect may be mediated by the EP₁ receptor subtype coupled to phosphatidylinositol hydrolysis.²³⁸

WATER TRANSPORT

Vasopressin-regulated water transport in the collecting duct is markedly influenced by COX products, especially PGs. When COX inhibitors are administered to humans, rats, or dogs, the antidiuretic action of arginine vasopressin is markedly augmented. Because vasopressin also stimulates endogenous PGE₂ production by the collecting duct, these results have suggested that PGE₂ participates in a negative feedback loop, whereby endogenous PGE₂ production dampens the action of arginine vasopressin (AVP).³⁰⁵ In agreement with this model, the early classical studies of Grantham and Orloff directly demonstrated that PGE₁ blunted the water permeability response of the cortical collecting duct to vasopressin.^{305a} In these early studies, the action of PGE₁ appeared to be at a pre-cAMP step. Interestingly, when administered by itself, PGE₁ modestly augmented basal water permeability. These earlier studies have been confirmed with respect to PGE₂. PGE₂ also stimulates basal hydraulic conductivity and suppresses the hydraulic conductivity response to AVP in the rabbit cortical collecting duct.^{306,307} Inhibition of both AVP-stimulated cAMP generation and water permeability appears to be mediated by the EP₁ and EP₃ receptors, whereas the increase in basal water permeability may be mediated by the EP₄ receptor.²⁶⁷ In contrast, EP₄ receptors may mediate vasopressin-independent water reabsorption because selective collecting duct deletion of EP₄ decreases aquaporin 2 expression and leads to a urine-concentrating defect.³⁰⁸

PROSTAGLANDINS

METABOLISM OF PROSTAGLANDINS

15-KETODEHYDROGENASE

The half-life of PGs is 3 to 5 minutes and that of TXA₂ is approximately 30 seconds. The elimination of PGE₂, PGF₂α, and PGI₂ proceeds through enzymatic and nonenzymatic pathways, whereas that of TXA₂ is nonenzymatic. The end products of all these degradative reactions generally possess minimal biologic activity, although this is not uniformly the case (see later). The principal enzyme involved in the transformation of PGE₂, PGI₂, and PGF₂α is 15-hydroxyprostaglandin dehydrogenase (PGDH), which converts the 15 alcohol group to a ketone.³⁰⁹

15-PGDH is an nicotinamide adenine dinucleotide (NAD⁺)/nicotinamide adenine dinucleotide phosphate (NADP⁺)–dependent enzyme that is 30 to 49 times more active in the kidney of the young rat (3 weeks of age) than in the adult. The K_m for PGE₂ is 8.4 μM and 22.6 μM for PGF₂α.³⁰⁹ It is mainly localized in the cortical and juxtamedullary zones,³¹⁰ with little activity detected in papillary slices. At baseline, it is found in the proximal tubule, TAL, and collecting duct. However, it was present in macula densa in COX-2 knockout mice and in the presence of a high-salt diet and in cultured macula densa cells, COX inhibition increases expression.³¹¹ Disruption of the 15-PGDH gene in mice results in persistent patent ductus arteriosus (PDA), thought to be a result of failure of circulating PGE₂ levels to fall in the immediate peripartum period.³¹² Thus administration of COX-inhibiting NSAIDs rescues the knockout mice by decreasing PGs and allowing the animals to survive.

Subsequent catalysis of 15-hydroxy products by a delta-13 reductase leads to the formation of 13,14-dihydro compounds. PGI₂ and TXA₂ undergo rapid degradation to 6-keto-PGF₁α and TXB₂, respectively.³⁰⁹ These stable metabolites are usually measured, and their rates of formation taken as representative of those of the parent molecules.

ω/ω-1-HYDROXYLATION OF PROSTAGLANDINS

Both PGA₂ and PGE₂ have been shown to undergo hydroxylation of the terminal or subterminal carbons by a cytochrome P450–dependent mechanism.³¹³ This reaction may be mediated by a CYP4A family member or CYP4F enzymes. Both CYP4A³¹⁴ and CYP4F members have been mapped along the nephron.³¹⁵ Some of these derivatives have been shown to exhibit biologic activity.

CYCLOPENTENONE PROSTAGLANDINS

The cyclopentenone PGs include PGA₂, a PGE₂ derivative, and PGJ₂, a derivative of PGD₂. Although it remains uncertain whether these compounds are actually produced in vivo, this possibility has received increasing attention because some cyclopentenone prostanoids have been shown to be activating ligands for nuclear transcription factors, including proliferator-activated receptor (PPAR)–δ and PPARγ.^{316–318} The realization that the antidiabetic thiazolidinedione drugs act through PPARγ to exert their antihyperglycemic and insulin-sensitizing effects³¹⁹ has generated intense interest in the possibility that the cyclopentenone PGs might serve as the

endogenous ligands for these receptors. Interestingly, DP2, unlike DP1 or indeed other members of the PGG protein-coupled receptor (GPCR) family, binds and is activated by PGD2 metabolites such as 15-deoxy- $\Delta^{12,14}$ -PGJ2, which acts at nanomolar concentrations.³²⁰ An alternative biologic activity of these compounds has been recognized in their capacity to covalently modify thiol groups, forming adducts with cysteine of several intracellular proteins, including thioredoxin 1, vimentin, actin, and tubulin.³²¹ Studies regarding the biologic activity of cyclopentenone prostanoids abound, and the reader is referred to several excellent sources in the literature.^{322–324} Although there is evidence supporting the presence of these compounds in vivo,³²⁵ it remains uncertain whether they can be formed enzymatically or are an unstable spontaneous dehydration product of the E and D ring PGs.³²⁶

NONENZYMATIC METABOLISM OF ARACHIDONIC ACID

It has long been recognized that oxidant injury can result in the peroxidation of lipids. In 1990, Morrow and colleagues reported that a series of PG-like compounds can be produced by free radical-catalyzed peroxidation of arachidonic acid that is independent of COX activity.³²⁷ These compounds, which are termed *isoprostanes*, have been increasingly used as a sensitive marker of oxidant injury in vitro and in vivo.³²⁸ In addition, at least two of these compounds, 8-iso-PGF2 α (15-F2-isoprostane) and 8-iso-PGE2 (15-E2-isoprostane) are potent vasoconstrictors when administered exogenously. 8-Iso-PGF2 α has been shown to constrict the renal microvasculature and decrease the GFR, an effect that is prevented by thromboxane receptor antagonism.³²⁹ However, the role of endogenous isoprostanes as mediators of biologic responses remains unclear.

PROSTAGLANDIN TRANSPORT AND URINARY EXCRETION

It is notable that most of the PG synthetic enzymes have been localized to the intracellular compartment, yet extracellular prostaglandins are potent autacoids and paracrine factors. Thus prostanoids must be transported extracellularly to achieve efficient metabolism and termination of their signaling. Similarly, enzymes that metabolize PGE2 to inactive compounds are also intracellular, requiring uptake of the PG for its metabolic inactivation. The molecular basis of these extrusion and uptake processes are slowly being defined.

As a fatty acid, PGs may be classified as an organic anion at a physiologic pH. Early microperfusion studies have documented that basolateral PGE2 could be taken up into proximal tubules cells and actively secreted into the lumen. Furthermore, this process could be inhibited by a variety of inhibitors of organic anion transport, including Para-aminohippurate (PAH), probenecid, and indomethacin. Studies of basolateral renal membrane vesicles have also supported the notion that this transport process occurs via an electroneutral anion exchanger. These studies are of note because renal PGs enter the urine in the loop of Henle, and late proximal tubule secretion could provide an important entry mechanism.¹

A molecule that mediates PGE2 uptake in exchange for lactate has been cloned and referred to as PGT, prostaglandin transporter.³³⁰ PGT is a member of the SLC21/SLCO: organic anion transport family, and its cDNA encodes a

transmembrane protein of 100 amino acids that exhibits broad tissue distribution (heart, placenta, brain, lung, liver, skeletal muscle, pancreas, kidney, spleen, prostate, ovary, small intestine, and colon).^{331–333} Immunocytochemical studies of PGT expression in rat kidneys have suggested expression primarily in glomerular endothelial and mesangial cells, arteriolar endothelial and muscularis cells, principal cells of the collecting duct, medullary interstitial cells, medullary vasa rectae endothelia, and papillary surface epithelium.³³⁴ PGT appears to mediate PGE2 uptake rather than release,³³⁵ allowing target cells to metabolize this molecule and terminate signaling.³³⁶ PGT expression is decreased with low salt and increased with high salt in the collecting duct, which may allow regulation of PG excretion by taking up more PGs excreted from the luminal surface, the site of PG transporter, thereby allowing more accumulation at the basolateral surface.³³⁷

Other members of the organic cation-anion-zwitterion transporter family SLC22 have also been shown to transport PGs³³⁸ and have been suggested to mediate PG excretion into the urine. Specifically, OAT1 and OAT3 are localized on the basolateral proximal tubule membrane, where they likely participate in the urinary excretion of PGE2.^{338,339} Conversely, members of the multidrug resistance protein (MRP) have been shown to transport PGs in an adenosine triphosphate (ATP)–dependent fashion.^{340,341} MRP2 (also designated as ABBC2) is expressed in kidney proximal tubule brush borders and may contribute to the transport (and urinary excretion) of glutathione-conjugated PGs.^{342,343} This transporter has more limited tissue expression, restricted to the kidney, liver, and small intestine, and could contribute not only to renal para-aminohippurate (PAH) excretion but also to PG excretion as well.³⁴⁴

INVOLVEMENT OF CYCLOOXYGENASE METABOLITES IN RENAL PATHOPHYSIOLOGY

EXPERIMENTAL AND HUMAN GLOMERULAR INJURY

GLOMERULAR INFLAMMATORY INJURY

COX metabolites have been implicated in functional and structural alterations in glomerular and tubulointerstitial inflammatory diseases.³⁴⁵ Essential fatty acid deficiency totally prevents the structural and functional consequences of the administration of nephrotoxic serum (NTS) to rats, an experimental model of antiglomerular basement membrane glomerulonephritis.³²⁹ Changes in arteriolar tone during the course of this inflammatory lesion are mediated principally by locally released COX and lipoxygenase (LO) metabolites of AA.³²⁹

TXA2 release appears to play an essential role in mediating the increased renovascular resistance observed during the early phase of this disease.¹ Subsequently, increasing rates of PGE2 generation may account for the progressive dilation of renal arterioles and increases in renal blood flow (RBF) at later stages of the disease. Consistent with this hypothesis, TXA2 antagonism ameliorated the falls in RBF and GFR 2 hours post-NTS administration, but not after 24 hours. During the latter heterologous phase of NTS, COX metabolites mediate the renal vasodilation and reduction in the glomerular ultrafiltration coefficient (K_f) that characterize this

phase.³²⁹ The net functional result of COX inhibition during this phase of experimental glomerulonephritis, therefore would depend on the relative importance of renal perfusion versus the preservation of K_f to the maintenance of the GFR. Evidence also indicates that COX metabolites are mediators of pathologic lesions and the accompanying proteinuria in this model.¹ COX-2 expression in the kidney increases in experimental anti-glomerular basement membrane (GBM) glomerulonephritis^{346,347} and after the systemic administration of lipopolysaccharide.³⁴⁸

A beneficial effect of fish oil diets (enriched in eicosapentanoic acid), with an accompanying reduction in the generation of COX products, has been demonstrated on the course of genetic murine lupus (MRL-lpr mice). In subsequent studies, enhanced renal TXA₂ and PGE₂ generation was demonstrated in this model, as well as in NZB mice, another genetic model of lupus.¹ In addition, studies in humans have demonstrated an inverse relation between TXA₂ biosynthesis and GFR and improvement of renal function following short-term therapy with a thromboxane receptor antagonist in patients with lupus nephritis.¹ More recently, studies have indicated that in humans, as well as NZB mice, COX-2 expression is upregulated in patients with active lupus nephritis, with colocalization to infiltrating monocytes, suggesting that monocytes infiltrating the glomeruli contribute to the exaggerated local synthesis of TXA₂.^{349,350} COX-2 inhibition selectively decreased thromboxane production, and chronic treatment of NZB mice with a COX-2 inhibitor and mycophenolate mofetil significantly prolonged survival.³⁵⁰ Taken together, these data, as well as others from animal and human studies, support a major role for the intrarenal generation of TXA₂ in mediating renal vasoconstriction during inflammatory and lupus-associated glomerular injury. In contrast, an EP₄-selective agonist was shown to reduce glomerular injury in a mouse model of anti-GBM disease.³⁵¹

The demonstration of a functionally significant role for COX metabolites in experimental and human inflammatory glomerular injury has raised the question of the cellular sources of these eicosanoids in the glomerulus. In addition to infiltrating inflammatory cells, resident glomerular macrophages, glomerular mesangial cells, and glomerular epithelial cells represent likely sources for eicosanoid generation. In the anti-Thy1.1 model of mesangioproliferative glomerulonephritis, COX-1 staining was transiently increased in diseased glomeruli at day 6 and was localized mainly to proliferating mesangial cells. COX-2 expression in the macula densa region also transiently increased at day 6.^{352,353} Glomerular COX-2 expression in this model has been controversial, with one group reporting increased podocyte COX-2 expression,³⁴⁷ and two other groups reporting minimal, if any, glomerular COX-2 expression.^{352,353} However, it is of interest that selective COX-2 inhibitors have been reported to inhibit glomerular repair in the anti-Thy1.1 model.³⁵³ In both anti-Thy1.1 and anti-GBM models of glomerulonephritis, the nonselective COX inhibitor, indomethacin, increased monocyte chemoattractant protein-1 (MCP-1), suggesting that prostaglandins may repress the recruitment of monocytes and macrophages in experimental glomerulonephritis.³⁵⁴

A variety of cytokines has been reported to stimulate PGE₂ synthesis and COX-2 expression in cultured mesangial cells. Furthermore, complement components, in particular C5b-9, which are known to be involved in the inflammatory models

described previously, have been implicated in the stimulation of PGE₂ synthesis in glomerular epithelial cells (GECs). Cultured GECs express predominantly COX-1, but exposure to C5b-9 significantly increases COX-2 expression¹.

GLOMERULAR NONINFLAMMATORY INJURY

Studies have suggested that prostanoids may also mediate altered renal function and glomerular damage following subtotal renal ablation, and glomerular PG production may be altered in such conditions. Glomeruli from remnant kidneys, as well as animals fed a high-protein diet, have increased prostanoid production.¹ These studies have suggested an increase in COX enzyme activity per se rather than, or in addition to, increased substrate availability, because increases in prostanoid production were noted when excess exogenous AA was added.

Following subtotal renal ablation, there are selective increases in renal cortical and glomerular COX-2 mRNA and immunoreactive protein expression, without significant alterations in COX-1 expression.³⁵⁵ This increased COX-2 expression was most prominent in the macula densa and surrounding cTALH. In addition, COX-2 immunoreactivity was also present in podocytes of remnant glomeruli, and increased PG production in isolated glomeruli from remnant kidneys was inhibited by a COX-2-selective inhibitor but was not decreased by a COX-1-selective inhibitor.³⁵⁵ Of interest, in the fawn-hooded rat, which develops spontaneous glomerulosclerosis, there is increased cTALH/macula densa COX-2 and neuronal nitric oxide synthase (nNOS) and juxtaglomerular cell renin expression preceding the development of sclerotic lesions.³⁵⁶ Studies have indicated that selective overexpression of COX-2 in podocytes in mice increases sensitivity to development of glomerulosclerosis, an effect that is mediated by TX receptor activation.³⁵⁷⁻³⁵⁹

When given 24 hours after subtotal renal ablation, a nonselective NSAID, indomethacin, normalized increases in renal blood flow and single-nephron GFR; similar decreases in hyperfiltration were noted when indomethacin was given acutely to rats 14 days after subtotal nephrectomy although in this latter study, the increased glomerular capillary pressure (P_{GC}) was not altered because both afferent and efferent arteriolar resistances increased.¹ Previous studies have also suggested that nonselective COX inhibitors may acutely decrease hyperfiltration in diabetes and inhibit proteinuria and/or structural injury¹; more recent studies have indicated that selective COX-2 inhibitors will decrease the hyperfiltration seen in experimental diabetes or increased dietary protein.^{360,361} Of note, NSAIDs have also been reported to be effective in reducing proteinuria in patients with refractory nephrotic syndrome.¹ Similarly, selective COX-2 inhibition decreased proteinuria in patients with both diabetic and nondiabetic renal disease, without alterations in blood pressure.³⁶²

The prostanoids involved have not yet been completely characterized, although it is presumed that vasodilatory prostanoids are involved in the mediation of the altered renal hemodynamics. Defective autoregulation of renal blood flow due to decreased myogenic tone of the afferent arteriole is seen after subtotal ablation or excessive dietary protein and is corrected by the inhibition of COX activity. In these hyperfiltering states, TGF is reset at a higher distal tubular flow rate.¹ Such a resetting dictates that afferent arteriolar vasodilation will be maintained in the face of increased distal

solute delivery. It was previously shown that the alterations in TGF sensitivity after reduction in renal mass are prevented with the nonselective COX inhibitor, indomethacin.¹ An important role has been suggested for neuronal nitric oxide synthase, which is localized to the macula densa, in the vasodilatory component of TGF.^{363–365} Of interest, studies by Ichihara and colleagues have determined that this nNOS-mediated vasodilation is inhibited by the selective COX-2 inhibitor, NS398, suggesting that COX-2-mediated prostanoids may be essential for arteriolar vasodilation.^{47,72}

Administration of COX-2-selective inhibitors decreased proteinuria and inhibited development of glomerular sclerosis in rats with reduced functioning renal mass.^{366,367} In addition, COX-2 inhibition decreased mRNA expression of TGF- β_1 and types III and IV collagen in the remnant kidney.³⁶⁶ Similar protection was observed with the administration of nitroflurbiprofen (NOF), an NO-releasing NSAID without gastrointestinal toxicity.³⁶⁸ Prior studies have also demonstrated that TXAS inhibitors retard the progression of glomerulosclerosis, with decreased proteinuria and glomerulosclerosis in rats with remnant kidneys and in diabetic nephropathy, in association with increased renal prostacyclin production and lower systolic blood pressure.³⁶⁹ Studies in models of types 1 and 2 diabetes have indicated that COX-2-selective inhibitors retard progression of diabetic nephropathy.^{370,371} Schmitz and colleagues have confirmed increases in TXB2 excretion in the remnant kidney and correlated decreased arachidonic and linoleic acid levels with increased thromboxane production because the TXAS inhibitor U63557A restores fatty acid levels and retards progressive glomerular destruction.³⁷²

Enhanced glomerular synthesis and/or urinary excretion of both PGE2 and TXA2 have been demonstrated in passive Heymann nephritis (PHN), and adriamycin-induced glomerulopathies in rats. Both COX-1 and COX-2 expression are increased in glomeruli with PHN.³⁷³ Both TXAS inhibitors and selective COX-2 inhibitors also decreased proteinuria in PHN.¹

In contrast to the putative deleterious effects of TX, the prostacyclin analogue, cicaprost, retarded renal damage in uninephrectomized dogs fed a high-sodium and high-protein diet, an effect that was not mediated by the amelioration of systemic hypertension.³⁷⁴ Similarly both EP2 and EP4 agonists decreased glomerular and tubulointerstitial fibrosis in a model of subtotal renal ablation.³⁶² Other studies have also indicated that in models of polycystic kidney disease, there is increased COX-2 expression and increased PGE2 and TX in cyst fluid. Either COX-2 inhibition or EP2 receptor inhibition decreased cyst growth and interstitial fibrosis.^{375,376}

Prostanoids have also been shown to alter extracellular matrix production by mesangial cells in culture. TXA2 stimulates matrix production by both TGF- β -dependent and TGF- β -independent pathways.³⁷⁷ PGE2 has been reported to decrease steady-state mRNA levels of alpha 1(I) and alpha 1(III) procollagens, but not alpha 1(IV) procollagen and fibronectin mRNA, and to reduce the secretion of all studied collagen types into the cell culture supernatants. Of interest, this effect did not appear to be mediated by cAMP.³⁷⁸ PGE2 has also been reported to increase production of matrix metalloproteinase-2 and mediate Ang II-induced increases in MMP-2.³⁷⁹ Whether vasodilatory prostaglandins mediate decreased fibrillar collagen production and increased matrix degrading activity in glomeruli in vivo has not yet been studied;

however, there is compelling evidence in nonrenal cells that prostanoids may mediate or modulate matrix production.³⁸⁰ Cultured lung fibroblasts isolated from patients with idiopathic pulmonary fibrosis exhibit decreased ability to express COX-2 and synthesize PGE2.³⁸¹

ACUTE KIDNEY INJURY

When cardiac output is compromised, as in extracellular fluid volume depletion or congestive heart failure, systemic blood pressure is preserved by the action of high circulating levels of systemic vasoconstrictors (e.g., norepinephrine, Ang II, AVP). Amelioration of their effects within the renal vasculature serves to blunt the development of otherwise concomitant marked depression of renal blood flow. The intrarenal generation of vasodilator products of AA, including PGE2 and PGI2, is a central part of this protective adaptation. Increased renal vascular resistance induced by exogenously administered Ang II or renal nerve stimulation (increased adrenergic tone) is exaggerated during the concomitant inhibition of prostaglandin synthesis. Experiments in animals with volume depletion have demonstrated the existence of intrarenal AVP-prostaglandin interactions, similar to those described previously for Ang II.¹ Studies in patients with congestive heart failure have confirmed that enhanced prostaglandin synthesis is crucial in protecting kidneys from various vasoconstrictor influences in this condition.

Renal dysfunction accompanying the acute administration of endotoxin in rats is characterized by progressive reductions in RBF and GFR in the absence of hypotension. Renal histology in these animals is normal, but cortical generation of COX metabolites is markedly elevated. A number of reports have provided evidence for a role for TXA2-induced renal vasoconstriction in this model of renal dysfunction.³⁸² In addition, roles for PGs and TXA2 in modulating or mediating renal injury have been suggested in ischemia and reperfusion,³⁸³ and models of toxin-mediated acute tubular injury, including those induced by uranyl nitrate,³⁸⁴ amphotericin B,³⁸⁵ aminoglycosides,³⁸⁶ and glycerol.³⁸⁷ In experimental acute renal failure, administration of vasodilator PGs has been shown to ameliorate injury.³⁸⁸ Similarly, the administration of nonselective or COX-2-selective NSAIDs exacerbates experimental ischemia-reperfusion injury.³⁸⁹

COX-2 expression decreases in the kidney in response to acute ischemic injury.³⁹⁰ The role of COX products in ischemia-reperfusion injury is controversial. Roles for PGs and TXA2 in modulating or mediating renal injury have been suggested in ischemia-reperfusion³⁸³ and in models of toxin-mediated acute tubular injury, including those induced by uranyl nitrate,³⁸⁴ amphotericin B,³⁸⁵ aminoglycosides,³⁸⁶ and glycerol.³⁸⁷ Furthermore, fibrosis resulting from prolonged ischemic injury has been shown to be ameliorated by non-specific COX inhibition.³⁹¹ In contrast, renal injury in response to ischemia-reperfusion is worsened by COX-2-selective inhibitors or in COX-2^{-/-} mice,³⁸⁹ and administration of vasodilator PGs has been shown to ameliorate injury,³⁸⁸ possibly through a PPAR α -dependent mechanism.³⁹²

URINARY TRACT OBSTRUCTION

Following the induction of chronic (>24 hours) ureteral obstruction, renal PG and TXA2 synthesis are markedly

enhanced, particularly in response to stimuli such as endotoxins or bradykinins. Enhanced prostanoid synthesis, especially TX, likely arises from infiltrating mononuclear cells, proliferating fibroblast-like cells, interstitial macrophages, and interstitial medullary cells.³⁴⁵ Selective COX-2 inhibitors may prevent renal damage in response to unilateral ureteral obstruction.^{393,394} However, PGE₂ acting through the EP₄ receptor can limit tubulointerstitial fibrosis resulting from UUO.³⁹⁵ Prostaglandins derived from medullary COX-2 are mediators of the early phase of diuresis seen after the relief of ureteral obstruction because COX-2 inhibition prevents the acute (24-hour) phase of postobstructive diuresis. However, more persistent, chronic postobstructive diuresis is not PG-dependent but results from the downregulation of NKCC2 and decreases aquaporin-2 phosphorylation and translocation to the collecting duct membrane.³⁹⁶

ALLOGRAFT REJECTION AND CYCLOSPORINE NEPHROTOXICITY

ALLOGRAFT REJECTION

Acute administration of a TXA₂ synthesis inhibitor is associated with significant improvement in rat renal allograft function.³⁹⁷ A number of other experimental and clinical studies have also demonstrated increased TXA₂ synthesis during allograft rejection,^{398,399} leading some to suggest that increased urinary TXA₂ excretion may be an early indicator in renal and cardiac allograft rejection.

CALCINEURIN INHIBITOR NEPHROTOXICITY

Numerous investigators have demonstrated effects for cyclosporine A (CYA) on renal PG and TXA₂ synthesis and provided evidence for a major role for renal and leukocyte TXA₂ synthesis in mediating acute and chronic CYA nephrotoxicity in rats.⁴⁰⁰ Fish oil-rich diets, TXA₂ antagonists, or administration of CYA in fish oil as vehicle have all been shown to reduce renal TXA₂ synthesis and may therefore afford protection against nephrotoxicity. Moreover, CYA has been reported to decrease renal COX-2 expression.⁴⁰¹

HEPATIC CIRRHOSIS AND HEPATORENAL SYNDROME

Patients with cirrhosis of the liver show an increased renal synthesis of vasodilating PGs, as indicated by the high urinary excretion of PGs and/or their metabolites. Urinary excretion of 2-3-dinor-6-keto PGF₁ α , an index of systemic PGI₂ synthesis, is increased in patients with cirrhosis and hyperdynamic circulation, thus raising the possibility that systemic synthesis of PGI₂ may contribute to the arterial vasodilation of these patients. Inhibition of COX activity in these patients may cause a profound reduction in renal blood flow and GFR, a reduction in sodium excretion, and an impairment of free water clearance.⁴⁰² The sodium-retaining properties of NSAIDs are particularly exaggerated in patients with cirrhosis of the liver, attesting to the dependence of renal salt excretion on vasodilatory PGs. In the kidneys of rats with cirrhosis, COX-2 expression increases, while COX-1 expression is unchanged; however, in these animals, selective inhibition of COX-1 leads to impaired renal hemodynamics and natriuresis, whereas COX-2 inhibition has no effect.^{403,404}

Diminished renal PG synthesis has been implicated in the pathogenesis of the severe sodium retention seen in hepatorenal syndrome, as well as in the resistance to diuretic therapy.^{405,406} There is reduced renal synthesis of vasodilating PGE₂ in the presence of activation of endogenous vasoconstrictors and a maintained or increased renal production of TXA₂.^{402,407} Therefore, an imbalance between vasoconstricting systems and the renal vasodilator PGE₂ has been proposed as a contributing factor to the renal failure observed in this condition. However, administration of exogenous prostanoids to patients with cirrhosis is not effective in ameliorating renal function or preventing the deleterious effect of NSAIDs.⁴⁰²

DIABETES MELLITUS

In the streptozotocin-induced model of diabetes in rats, COX-2 expression is increased in the cTALH-macula densa region,^{360,370} as well as in podocytes,⁴⁰⁸ possibly mediated by epigenetic processes.⁴⁰⁹ COX-2 immunoreactivity has also been detected in the macula densa region in human diabetic nephropathy.⁴¹ Studies have suggested that COX-2 dependent vasodilator prostanoids play an important role in the hyperfiltration seen early in diabetes mellitus,^{410,360,411-413} as well as in response to a high-protein diet.⁴¹⁴ The increased COX-2 expression appears to be mediated at least in part by increased ROS production in diabetes, because the superoxide dismutase analogue, Tempol, blocks the increased expression.⁴¹⁵

It has been found that the chronic administration of a selective COX-2 inhibitor significantly decreases proteinuria and reduces extracellular matrix deposition, as indicated by decreases in immunoreactive fibronectin expression and mesangial matrix expansion.^{370,416} In addition, COX-2 inhibition reduced expression of TGF- β , PAI-1, and vascular endothelial growth factor (VEGF) in the kidneys of the diabetic hypertensive animals. Increasing intrarenal dopamine production also ameliorates diabetic nephropathy progression, at least in part by inhibiting renal cortical COX-2 expression.⁴¹⁷ The vasoconstrictor TXA₂ may play a role in the development of albuminuria and basement membrane changes with diabetic nephropathy (DN). In addition, the administration of a selective PGE₂ EP₁ receptor antagonist prevented the development of experimental DN,⁴¹⁸ whereas EP₄ receptor activation may exacerbate DN.⁴¹⁹ In contrast to the beneficial effect of global inhibition of COX-2, selective inhibition of renal macrophage COX-2 actually exacerbates diabetic renal injury.⁴²⁰

PREGNANCY

Most, but not all, investigators have not reported increases in vasodilator PG synthesis or suggested an essential role for prostanoids in the mediation of the increased GFR and renal plasma flow (RPF) of normal pregnancy;⁴²¹ however, diminished synthesis of PGI₂ has been demonstrated in humans and in animal models of pregnancy-induced hypertension,⁴²² which is associated with decreased expression of COX-2 and PGI₂ synthase in placental villi.⁴²³ In animal models, the inhibition of TXA₂ synthetase has been associated with resolution of the hypertension, suggesting a possible pathophysiologic role.⁴²⁴ A moderate beneficial effect of reducing TXA₂ generation, while preserving PGI₂ synthesis, by low-dose aspirin therapy (60–100 mg/day) has been demonstrated

in patients at high risk for pregnancy-induced hypertension and preeclampsia.^{425,426}

LITHIUM NEPHROTOXICITY

Lithium chloride is a mainstay of treatment in psychiatry for bipolar illness. However, it is routinely complicated by polyuria and even frank nephrogenic diabetes insipidus. In vitro and in vivo studies have demonstrated lithium-induced renal medullary interstitial cell COX-2 protein expression via inhibition of glycogen synthase kinase-3 β (GSK-3 β). COX-2 inhibition prevented lithium-induced polyuria. COX-2 inhibition also resulted in the upregulation of aquaporin 2 (AQP2) and Na-K-2Cl co-transporter (NKCC2).^{427,428}

ROLE OF REACTIVE OXYGEN SPECIES AS MEDIATORS OF COX-2 ACTIONS

In addition to NADPH oxidase, nitric oxide synthase, and xanthine oxidase, COX-2 can also be a source of oxygen radicals.⁴²⁹ COX-2 enzymatic activity is commonly accompanied by associated oxidative mechanisms (co-oxidation) and free radical production.⁴³⁰ The catalytic activity of COX consists of a series of radical reactions that use molecular oxygen and generate intermediate ROS.⁴³¹ Elevated levels of COX-2 protein are associated with increased ROS production and apoptosis in cultured renal cortical cells⁴³² and human mesangial cells.⁴³³ It has been suggested that COX-2-mediated lipid peroxidation, rather than PGs, can induce DNA damage via adduct formation.⁴³⁴ A COX-2 specific inhibitor, NS-398, was able to reduce the oxidative activity with prevention of oxidant stress.⁴³⁵

In addition to ROS generated by COX per se, prostanoids may also activate intracellular pathways that generate ROS.

Locally generated ROS may damage cell membranes, leading to lipid peroxidation and release of AA. Prostanoids released during inflammatory reactions cause rapid degenerative changes in some cultured cells, and their potential cytotoxic effect has been suggested to occur by accelerating intracellular oxidative stress. TX⁴³⁶ and PGE2 acting through the EP1 receptor⁴³⁷ have been reported to induce NADPH oxidase and ROS production. Of interest, PGE2 acting through the EP4 receptor inhibits macrophage oxidase activity.^{438,439} As mentioned previously, there is also evidence for crosstalk between COX-2 and ROS, such that ROS may induce COX-2 expression.⁴³⁴ Interestingly, during aging, there is ROS-mediated NF- κ B expression, which increases COX-2 expression in the kidney.⁴⁴⁰ Furthermore, this appears to induce a vicious cycle, because COX-2 then serves as a source of ROS. This interaction of COX-derived PG and ROS production has been posited to play a role in the development of hypertension.⁴⁴¹ The amount of renal ROS resulting from COX activity increases with age, so that up to 25% of total kidney ROS production in aged rat kidneys is inhibited by NSAID administration.

THE LIPOXYGENASE PATHWAY

The lipoxygenase enzymes metabolize AA to form LTs, HETEs, and LXs (Fig. 13.12). These lipoxygenase metabolites are primarily produced by leukocytes, mast cells, and macrophages in response to inflammation and injury. There are three lipoxygenase enzymes—5-, 12-, and 15-lipoxygenase—so named for the carbon of AA where they insert an oxygen. The lipoxygenases are products of separate genes and have distinct distributions and patterns of regulation. Glomeruli, mesangial cells, cortical tubules, and vessels also produce

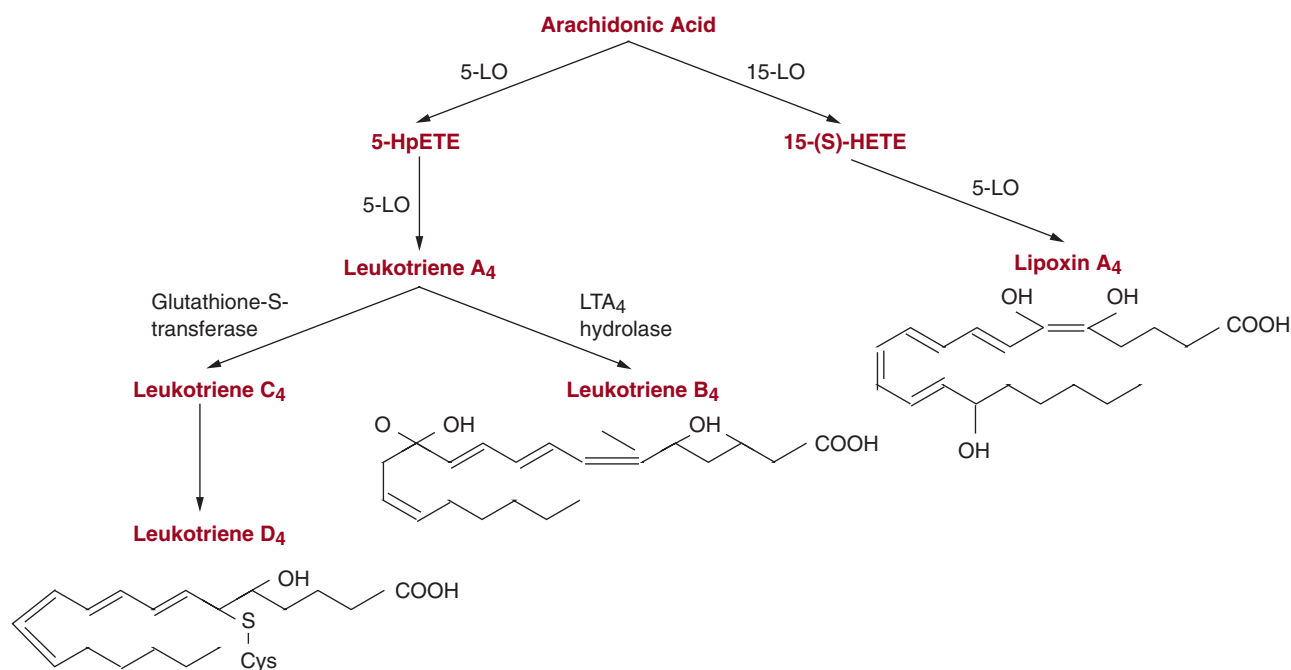


Fig. 13.12 Pathways of lipoxygenase (LOX) metabolism of arachidonic acid. 5-LO, 5-Lipoxygenase; 15-(S)-HETE, 15(S)-hydroxyeicosatetraenoic acid; 5-HpETE, 5-hydroperoxyeicosatetraenoic acid.

the 12-lipoxygenase (12-LOX) product, 12(*S*)-HETE, and the 15-LOX product, 15-HETE. Studies have localized 15-LOX mRNA primarily to the distal nephron and 12-LOX mRNA to the glomerulus. 5-LOX mRNA and 5-LOX-activating protein (FLAP) mRNA were expressed in the glomerulus and the vasa recta.⁴⁴² In polymorphonuclear leukocytes (PMNs), macrophages, and mast cells, 5-LOX mediates the formation of leukotrienes.⁴⁴³ 5-LOX, which is regulated by FLAP, catalyzes the conversion of AA to 5-HpETE and then to leukotriene A₄ (LTA₄).⁴⁴⁴ LTA₄ is then further metabolized to the peptidyl leukotrienes (LTC₄ and LTD₄) by glutathione-S-transferase or to LTB₄ by LTA₄ hydrolase. Although glutathione-S-transferase expression is limited to inflammatory cells, LTA₄ hydrolase is also expressed in glomerular mesangial cells and endothelial cells.⁴⁴⁵ PCR analysis has actually demonstrated ubiquitous LTA₄ hydrolase mRNA expression throughout the rat nephron.⁴⁴² LTC₄ synthase mRNA could not be found in any nephron segment.⁴⁴²

Two cysteinyl leukotriene receptors (CysLTR) have been cloned and identified as members of the G protein-coupled superfamily of receptors. They have been localized to vascular smooth muscle and endothelium of the pulmonary vasculature.^{446–448} In the kidney, the CysLTR type 1 is expressed in the glomerulus, whereas CysLTR type 2 mRNA has not been detected in any nephron segment to date.⁴⁴²

The peptidyl leukotrienes are potent mediators of inflammation and vasoconstrictors of vascular, pulmonary, and gastrointestinal smooth muscle. In addition, they increase vascular permeability and promote mucous secretion.⁴⁴⁹ Because of the central role that peptidyl leukotrienes play in the inflammatory trigger of asthma exacerbation, effective receptor antagonists have been developed and are now an important component of asthma treatment.⁴⁵⁰

In the kidney, LTD₄ administration has been shown to decrease RBF and GFR, and peptidyl leukotrienes are thought to be mediators of decreased RBF and GFR associated with acute glomerular inflammation. Micropuncture studies have revealed that the decreases in GFR are the result of afferent and arteriolar vasoconstriction, with more pronounced efferent vasoconstriction and a decrease in K_f .¹ In addition both LTC₄ and LTD₄ increase the proliferation of cultured mesangial cells.

The LTB₄ receptor is also a seven-transmembrane G protein-coupled receptor. On PMNs, receptor activation promotes chemotaxis, aggregation, and attachment to endothelium. In the kidney LTB₄ mRNA is localized to the glomerulus.⁴⁴² A second, low-affinity LTB₄ receptor is also expressed,⁴⁵¹ which may mediate calcium influx into PMNs, thereby leading to activation. LTB₄ receptor blockers lessen acute renal ischemic-reperfusion injury⁴⁵² and nephrotoxic nephritis in rats,⁴⁵³ and PMN infiltration and structural and functional evidence of organ injury by ischemia and reperfusion are magnified in transgenic mice overexpressing the LTB₄ receptor.⁴⁵⁴ In addition to the activation of cell surface receptors, LTB₄ has also been shown to be a ligand for the nuclear receptor PPAR α .⁴⁵⁵

15-LOX leads to the formation of 15-S-HETE. In addition, dual oxygenation in activated PMNs and macrophages by 5- and 15-LOX leads to formation of the lipoxins. LX synthesis also can occur via transcellular metabolism of the leukocyte-generated intermediate, LTA₄, by 12-LOX in platelets or adjoining cells, including glomerular endothelial cells.^{456,457}

15-S-HETE is a potent vasoconstrictor in the renal microcirculation;⁴⁵⁸ however, 15-LOX-derived metabolites antagonize proinflammatory actions of leukotrienes, both by inhibiting PMN chemotaxis, aggregation, and adherence and by counteracting the vasoconstrictive effects of the peptidyl leukotrienes.^{459,460} Administration of 15-S-HETE reduced LTB₄ production by glomeruli isolated from rats with acute nephrotoxic, serum-induced glomerulonephritis, and it has been proposed that 15-LOX may regulate 5-LOX activity in chronic glomerular inflammation because it is known that in experimental glomerulonephritis, lipoxin A₄ (LXA₄) administration increases renal blood flow and GFR, mainly by inducing afferent arteriolar vasodilation, an effect mediated in part by release of vasodilator PGs.¹ LXA₄ also antagonized the effects of LTD₄ to decrease the GFR, although not RBF, even though administration of LXA₄ and LXB₄ directly into the renal artery induced vasoconstriction. Glomerular micropuncture studies have revealed that LXA₄ leads to moderate decreases in K_f .⁴⁵⁹ Lipoxins signal through a specific G protein-coupled receptor denoted ALXR. This receptor is related at the nucleotide sequence level to chemokine and chemotactic peptide receptors, such as N-formyl peptide receptor.⁴⁶¹ It is also noteworthy that in isolated perfused canine renal arteries and veins, LTC₄ and LTD₄ were found to be vasodilators, which were partially dependent on an intact endothelium; this was mediated by nitric oxide production.⁴⁶²

A potential interaction between COX- and LOX-mediated pathways has been reported. Although aspirin inhibits PG formation by COX-1 and COX-2, aspirin-induced acetylation converts COX-2 to a selective generator of 15-(*S*)-HETE. This product can then be released, taken up in a transcellular route by PMNs and converted to 15-epilipoxins, which have similar biologic actions as the lipoxins.⁴⁶³

Similar to 15-HETE, 12(*S*)-HETE also potently vasoconstricts glomerular and renal vasculature.⁴⁵⁶ 12(*S*)-HETE increases protein kinase C and depolarizes cultured vascular smooth muscle cells. Afferent arteriolar vasoconstriction and increases in smooth muscle calcium in response to 12(*S*)-HETE, were partially inhibited by voltage-gated, L-type calcium channel inhibitors.⁴⁶⁴ 12(*S*)-HETE has also been proposed to be an angiogenic factor, because in cultured endothelial cells, 12-LOX inhibition reduces cell proliferation, and 12-LOX overexpression stimulates cell migration and endothelial tube formation.⁴⁶⁵ 12- and 15-LOX inhibitors and elective elimination of the leukocyte 12-LOX enzyme also ameliorate the development of diabetic nephropathy in mice.⁴⁶⁶ There is also interaction between 12- and 15-LOX pathways and TGF- β -mediated pathways in the diabetic kidney.⁴⁶⁷ 12(*S*)-HETE has also been proposed to be a mediator of renal vasoconstriction by Ang II, with inhibition of the 12-LOX pathway attenuating Ang II-mediated afferent arteriolar vasoconstriction and decreased renal blood flow.⁴⁶⁸ LOX inhibition also blunted renal arcuate artery vasoconstriction by norepinephrine and KCl.⁴⁶⁹ However, 12-LOX products have also been implicated as inhibitors of renal renin release.^{470,471}

Although the major significance of LOX products in the kidney derives from their release from infiltrating leukocytes or resident cells of macrophage or monocyte origin, there is evidence to suggest that intrinsic renal cells are capable of generating LTs and LXs, either directly or through transcellular metabolism of intermediates.⁴⁷² Human and rat glomeruli

can generate 12- and 15-HETE, although the cells of origin are unclear. LTB₄ can be detected in supernatants of normal rat glomeruli, and its synthesis could be markedly diminished by mechanisms that deplete glomeruli of resident macrophages, such as irradiation or fatty acid deficiency. In addition, 5-, 12-, and 15-HETEs were detected from pig glomeruli, and their structural identity was confirmed by mass spectrometry¹. 12-LOX products are increased in mesangial cells exposed to hyperglycemia and in diabetic nephropathy.⁴⁷³ There also appears to be crosstalk between 12- and 15-LOX and COX-2. Both are increased with diabetes or high glucose levels and, in cultured cells, 12(S)-HETE increases COX-2, whereas PGE₂ increases 12- and 15-LOX. Knockdown of 12- and 15-LOX expression with short hairpin RNA (shRNA) decreases COX-2 expression, whereas 12- and 15-LOX overexpression increases COX-2 expression.⁴³⁰

Glomeruli subjected to immune injury release LTB₄,⁴⁷⁴ and LTB₄ generation was suppressed by resident macrophage depletion. Synthesis of peptido-LTs by inflamed glomeruli has also been demonstrated,⁴⁷⁵ but leukocytes could not be excluded because its primary source LXA₄ is generated by immune-injured glomeruli.⁴⁷⁶ Rat mesangial cells generate LXA₄ when provided with LTA₄ as substrate, thereby providing a potential intraglomerular source of LXs during inflammatory reactions. In nonglomerular tissue, 12-HETE production has been reported from rat cortical tubules and epithelial cells and 12- and 15-HETE from rabbit medulla¹.

BIOLOGIC ACTIVITIES OF LIPOXYGENASE PRODUCTS IN THE KIDNEY

In early experiments, the systemic administration of LTC₄ in the rat and administration of LTC₄ and LTD₄ in the isolated perfused kidney revealed potent renal vasoconstrictor actions of these eicosanoids. Subsequently, micropuncture measurements revealed that LTD₄ exerts preferential constrictor effects on postglomerular arteriolar resistance and depresses K_f and GFR. The latter is likely due to receptor-mediated contraction of glomerular mesangial cells, which has been demonstrated for LTC₄ and LTD₄ in vitro (see earlier). These actions of LTD₄ in the kidney are consistent with its known smooth muscle contractile properties. LTD₄, a potent chemotactic and leukocyte-activating agent, is devoid of constrictor action in the normal rat kidney. Lipoxin A₄ dilates afferent arterioles when infused into the renal artery, without affecting efferent arteriolar tone. This results in elevations in intraglomerular pressure and plasma flow rate, thereby augmenting the GFR.¹

INVOLVEMENT OF LIPOXYGENASE PRODUCTS IN RENAL PATHOPHYSIOLOGY

Increased generation rates of LTC₄ and LTD₄ have been documented in glomeruli from rats with immune complex nephritis and mice with spontaneously developing lupus nephritis.^{443,476} Moreover, results from numerous physiologic studies using specific LTD₄ receptor antagonists have provided strong evidence for the release of these eicosanoids during glomerular inflammation. In four animal models of glomerular immune injury (anti-GBM nephritis, anti-Thy1.1 antibody-mediated mesangiolysis, passive Heymann nephritis, and murine lupus nephritis), acute antagonism of LTD₄ by

receptor binding competition or inhibition of LTD₄ synthesis led to highly significant increases in GFR in nephritic animals.⁴⁷⁷ The principal mechanism underlying the improvement in GFR was reversal of the depressed values of K_f, which is characteristically compromised in immune-injured glomeruli. In other studies in PHN, Katoh and colleagues provided evidence that endogenous LTD₄ not only mediates reductions in K_f and GFR, but that LTD₄-evoked increases in intraglomerular pressure underlie, to a large extent, the accompanying proteinuria.⁴⁷⁷ Cysteinyl leukotrienes have been implicated in cyclosporine nephrotoxicity.⁴⁷⁸ Of interest, 5-LOX deficiency accelerates renal allograft rejection.⁴⁷⁹

LTB₄ synthesis, measured in the supernates of isolated glomeruli, is markedly enhanced early in the course of several forms of glomerular immune injury.⁴⁸⁰ Cellular sources of LTB₄ in injured glomeruli include PMNs and macrophages. All studies concur as to the transient nature of LTB₄ release. LTB₄ production decreases 24 hours after onset of the inflammation, which coincides with macrophage infiltration, a major source of 15-LOX activity.⁴⁸¹ 15-HPETE incubation decreased lipopolysaccharide-induced TNF expression in a human monocytic cell line,⁴⁸² and HVJ-liposome-mediated glomerular transfection of 15-LOX in rats decreased markers of injury (e.g., blood urea nitrogen [BUN], proteinuria) and accelerated functional (GFR, RBF) recovery in experimental glomerulonephritis (GN).⁴⁸³ In addition, MK501, a FLAP antagonist, restored size selectivity and decreased glomerular permeability in acute GN.⁴⁸⁴

The suppression of LTB₄ synthesis beyond the first 24 hours of injury is rather surprising, because both PMNs and macrophages are capable of effecting the total synthesis of LTB₄; they contain the two necessary enzymes that convert AA to LTB₄—namely, 5-LOX and LTA₄ hydrolase. It has therefore been suggested, based on in vitro evidence, that the major route for LTB₄ synthesis in inflamed glomeruli is through transcellular metabolism of leukocyte-generated LTA₄ to LTB₄ by LTA₄ hydrolase present in glomerular mesangial, endothelial, and epithelial cells. Because the transformation of LTA₄ to LTB₄ is rate-limiting, regulation of LTB₄ synthetic rate might relate to the regulation of LTA₄ hydrolase gene expression or catalytic activity in these parenchymal cells, rather than to the number of infiltrating leukocytes. In any case, leukocytes represent an indispensable source for LTA₄, the initial 5-LOX product and the precursor for LTB₄, because endogenous glomerular cells do not express the 5-LOX gene.⁴⁸⁵ Thus, it was demonstrated that the PMN cell-specific activator, N-formyl-Met-Leu-Phe, stimulated LTB₄ production from isolated perfused kidneys harvested from NTS-treated rats to a significantly greater degree than from control animals treated with nonimmune rabbit serum.⁴⁸⁶ The renal production of LTB₄ correlated directly with renal myeloperoxidase activity, suggesting interdependence of LTB₄ generation and PMN infiltration.

The acute and long-term significance of LTB₄ generation in conditioning the extent of glomerular structural and functional deterioration has been highlighted in studies in which LTB₄ was exogenously administered or in which its endogenous synthesis was inhibited. Intrarenal administration of LTB₄ to rats with mild NTS-induced injury was associated with an increase in PMN infiltration, reduction in renal plasma flow rate, and marked exacerbation of the fall in the GFR, with the latter correlating strongly with the number of

infiltrating PMNs and glomeruli, whereas inhibition of 5-LOX led to preservation of the GFR and abrogation of proteinuria.⁴⁸⁶ Similarly, both 5-LOX knockout mice and wild-type mice treated with the 5-LOX inhibitor zileuton had reduced renal injury in response to ischemia and reperfusion.⁴⁸⁷ Thus although devoid of vasoconstrictor actions in the normal kidney, increased intrarenal generation of LTB₄ during early glomerular injury amplifies leukocyte-dependent reductions in glomerular perfusion and filtration rates and inflammatory injury, likely due to enhancement of PMN recruitment and activation.

12(S)-HETE has been reported to increase AT1 receptor (AT1R) mRNA and protein expression in cultured rat mesangial cells by stabilizing AT1R mRNA and enhancing the profibrotic effects of Ang II.⁴⁸⁸ Ang II AT1R blockade has been shown to inhibit the development of diabetic nephropathy via the upregulation of glomerular nephrin and P-cadherin expression through inhibition of 12- and 15-LOX activation in rats.⁴⁸⁹ Genetic or pharmacologic inhibition of 12- and 15-LOX led to decreases in 12(S)-HETE production, proteinuria, renal oxidative stress, and collagen deposition in type I diabetes.⁴⁹⁰ Recently, 12(S)-HETE was found to increase profibrotic gene expression and enhance the permissive histone lysine modification at their promoters by upregulation of protein levels of SET7, a histone H3 lysine 4 methyltransferase, its nuclear translocation, and enhancement at profibrotic gene promoters in mesangial cells.⁴⁹¹

Both LOX and leukotriene signaling pathways have been reported to be involved in the development of cisplatin-mediated acute kidney injury in Wistar albino rats.⁴⁹² Urinary 12(S)- and 15(S)-HETE levels have been shown to correlate positively with elevated serum creatinine levels after kidney

transplantation.⁴⁹³ In the JCR:LA-corpulent rat, a model of the metabolic syndrome, fish oil (ω -3 polyunsaturated fatty acid) supplements markedly reduced albuminuria and glomerulosclerosis in association with decreases in 5(S)-, 12(S)-, and 15(S)-HETE.⁴⁹⁴ The nucleotide-binding oligomerization domain-like receptor containing pyrin domain 3 (NLRP3) inflammasome promotes renal inflammation and contributes to chronic kidney disease through the generation of proinflammatory cytokines IL- β and IL-18.⁴⁹⁵ Docosahexaenoic acid (DHA, in the form of fish oil), a well-known ω -3 polyunsaturated fatty acid, inhibits inflammation and exerts a beneficial action in numerous inflammatory human diseases. Recent studies have indicated that the LOX-mediated DHA metabolite, resolving D1, attenuates podocyte injury during hyperhomocysteinemia by inhibiting NLRP3 inflammasome activation.⁴⁹⁶

THE CYTOCHROME P450 PATHWAY

Following their elucidation and characterization as endogenous metabolites of AA, numerous studies have investigated the possibility that cytochrome P450 (CYP450) AA metabolites subserve physiologic and/or pathophysiologic roles in the kidney. In whole-animal physiology, these compounds have been implicated in the mediation of release of peptide hormones, regulation of vascular tone, and regulation of volume homeostasis. On the cellular level, CYP AA metabolites have been proposed to regulate ion channels and transporters and to act as mitogens. See Fig. 13.13.

CYP450 monooxygenases are mixed-function oxidases that use molecular oxygen and NADPH as cofactors^{497,498} and will

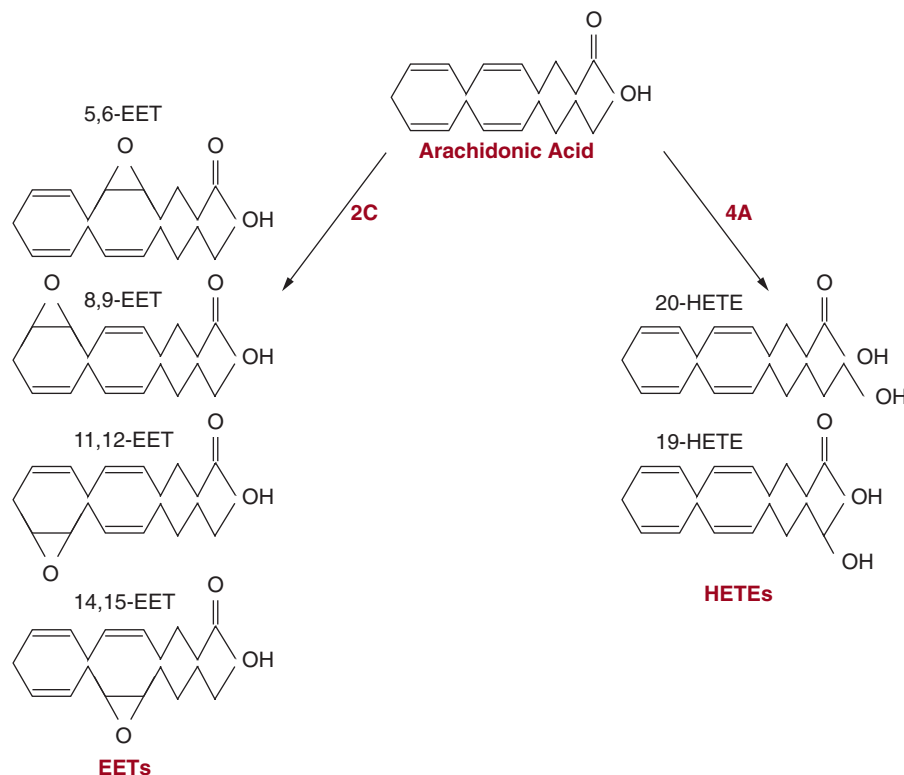


Fig. 13.13 Pathways of CYP450 metabolism of arachidonic acid. *EET*, Epoxyeicosatrienoic acid; *HETE*, hydroxyeicosatetraenoic acid.

add an oxygen molecule to AA in a region- and stereo-specific geometry. CYP450 monooxygenase pathways metabolize AA to generate HETEs and epoxyeicosatrienoic acids (EETs); EETs can be hydrolyzed to dihydroxyeicosatrienoic acids (DHETs).^{497,499,500} The kidney displays one of the highest CYP450 activities of any organ and produces CYP450 AA metabolites in significant amounts.^{497,500,501} HETEs are formed primarily via CYP450 hydroxylase enzymes and EETs, and DHETs are formed primarily via CYP450 epoxygenase enzymes.⁵⁰¹ The CYP450 4A gene family is the major pathway for synthesis of hydroxylase metabolites, especially 20-HETE and 19-HETE,^{300,501} whereas the production of epoxygenase metabolites is primarily via the 2C gene family.^{497,502} A member of the 2J family that is an active epoxygenase is also expressed in the kidney.⁵⁰³

CYP450 enzymes have been localized to vasculature and tubules.³⁰⁰ The 4A family of hydroxylases is expressed in preglomerular renal arterioles, glomeruli, proximal tubules, the TALH, and macula densa.⁵⁰⁴

The 2C and 2J families of epoxygenases are expressed at the highest levels in the proximal tubule and collecting duct.^{503,505} When isolated nephron segments expressing CYP450 protein have been incubated with AA, the production of CYP450 AA metabolites can be detected. 20-HETE and EETs are both produced in the afferent arterioles,⁵⁰⁶ glomerulus,⁵⁰⁷ and proximal tubule.⁵⁰⁸ 20-HETE is the predominant CYP450 AA metabolite produced by the TALH and in the pericytes surrounding vasa recta capillaries,⁵⁰⁷ whereas EETs are the predominant CYP450 AA metabolites produced by the collecting duct.⁵⁰⁹

Renal production of both epoxygenase and hydroxylase metabolites has been shown to be regulated by hormones and growth factors, including Ang II, endothelin, bradykinin, parathyroid hormone, and epidermal growth factor.^{300,498,502}

Alterations in dietary salt intake also modulate CYP450 expression and activity.⁵¹⁰ Alterations in the production of CYP450 metabolites have also been reported with uninephrectomy, diabetes mellitus, and hypertension.^{300,499} Glycerol-containing epoxygenase metabolites are produced endogenously and serve as high-affinity ligands for cannabinoid receptors, implicating these compounds as endocannabinoids.⁵¹¹

VASCULATURE

20-TRIHIDROXYEICOSATETRAENOIC ACID

In rat and dog renal arteries and afferent arterioles, 20-HETE is a potent vasoconstrictor,⁵⁰⁶ whereas it is a vasodilator in rabbit renal arterioles. The vasoconstriction is associated with membrane depolarization and a sustained rise in intracellular calcium. 20-HETE is produced in the smooth muscle cells, and its afferent arteriolar vasoconstrictive effects are mediated by the closure of K_{Ca} channels through a tyrosine kinase- and ERK-dependent mechanism (Fig. 13.14). Recent studies have indicated that 20-HETE-induced hypertension is a mediator of kidney disease in diabetes.⁵¹²

An interaction between CYP450 AA metabolites and nitric oxide has also been demonstrated. NO can inhibit the formation of 20-HETE in renal vascular smooth muscle (VSM) cells. A significant portion of NO's vasodilator effects in the preglomerular vasculature appear to be mediated by the inhibition of tonic 20-HETE vasoconstriction, and inhibition of 20-HETE formation attenuates the pressor response and fall in RBF seen with NO synthase inhibition.^{513,514}

EPOXIDES

Unlike CYP450 hydroxylase metabolites, epoxygenase metabolites of AA increase RBF and GFR.^{300,498,502} 11,12-EET and 14,15-EET vasodilate the preglomerular arterioles

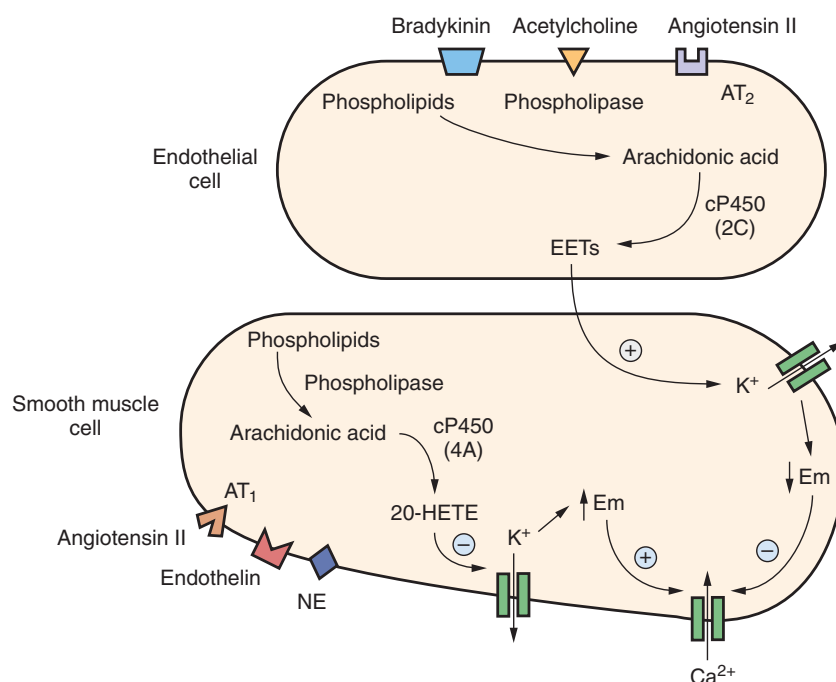


Fig. 13.14 Proposed interactions of CYP450 arachidonic acid metabolites derived from vascular endothelial cells and smooth muscle cells to regulate vascular tone. EET, Epoxyeicosatrienoic acid; HETE, hydroxyeicosatetraenoic acid; NE, norepinephrine.

independently of COX activity, whereas 5,6-EET and 8,9-EET cause COX-dependent vasodilation or vasoconstriction.⁵¹⁵ It is possible that these COX-dependent effects are mediated by COX conversion of 5,6-EET and 8,9-EET to PG- or TX-like compounds.⁵¹⁶ EETs are produced primarily in the endothelial cells and exert their vasoactive effects on the adjacent smooth muscle cells. In this regard, it has been suggested that EETs, and specifically 11,12-EET, may serve as an endothelium-derived hyperpolarizing factor (EDHF) in the renal microcirculation.^{502,517} EET-induced vasodilation is mediated by activation of K_{Ca} channels through cAMP-dependent stimulation of protein kinase C.

CYP450 metabolites may serve as second messengers or as modulators of the actions of hormonal and paracrine agents. Vasopressin increases renal production of CYP450 metabolites,⁵¹⁸ and increases in intracellular calcium and proliferation in cultured renal mesangial cells are augmented by EET administration.⁵¹⁸ CYP450 metabolites also may serve to modulate the renal hemodynamic responses of endothelin-1, with 20-HETE as a possible mediator of the vasoconstrictive effects and EETs counteracting the vasoconstriction.^{502,519} The formation of 20-HETE does not affect the ability of ET-1 to increase free intracellular calcium transient in renal vascular smooth muscle but appears to enhance the sustained elevations that represent calcium influx through voltage-sensitive channels.

CYP450 metabolites have also been implicated in mediation of renal vascular responses to Ang II. In the presence of AT1 receptor blockers, Ang II produces an endothelial-dependent vasodilation in rabbit afferent arterioles that is dependent on CYP450 epoxygenase metabolite production by AT2 receptor activation.⁵²⁰ With intact AT1 receptors, Ang II increases 20-HETE release from isolated preglomerular microvessels through an endothelium-independent mechanism.⁵²¹ Ang II's vasoconstrictive effects are in part the result of 20-HETE-mediated inhibition of K_{Ca} , which enhances sustained increases in intracellular calcium concentration by calcium influx through voltage-sensitive channels. Inhibition of 20-HETE production reduces the vasoconstrictor response to Ang II by more than 50% in rat renal interlobular arteries in which the endothelium has been removed.⁵²¹

AUTOREGULATION

CYP450 metabolites of AA have been shown to be mediators of RBF autoregulatory mechanisms. When PG production was blocked in canine arcuate arteries, AA administration enhanced myogenic responsiveness, and renal blood flow autoregulation was blocked by CYP450 inhibitors.^{300,500} Similarly, in the rat juxtamedullary preparation, selective blockade of 20-HETE formation significantly decreased afferent arteriolar vasoconstrictor responses to elevations in perfusion pressure, and inhibition of epoxygenase activity enhanced vasoconstriction.⁵²² This suggests that 20-HETE is involved in afferent arteriolar autoregulatory adjustment, whereas release of vasodilatory epoxygenase metabolites in response to increases in renal perfusion pressure acts to attenuate the vasoconstriction. In vivo studies have also implicated 20-HETE as a mediator of the autoregulatory response to increased perfusion pressure.⁵²³ Bradykinin-induced efferent arteriolar vasodilation has been shown to be mediated in part by the direct release of EETs from this vascular segment. In addition, bradykinin-induced release of 20-HETE from the glomerulus

can modulate the EET-mediated vasodilation.⁵²⁴ Deficient 20-HETE production worsens ischemic kidney injury by impairing medullary blood flow.⁵²⁵

TUBULOGLOMERULAR FEEDBACK

CYP450 metabolites may also be involved in the tubuloglomerular feedback response.³⁰⁰ As noted, 20-HETE is produced by both the afferent arteriole and macula densa, and studies have suggested the possibility that HETEs may serve as a vasoconstrictive mediator of TGF released by the macula densa or as a second messenger in the afferent arteriole in response to mediators released by the macula densa, such as adenosine or ATP.⁵²⁶ 20-HETE may also be a mediator of regulation of intrarenal distribution of blood flow.^{527,528} In addition, there is evidence for “connecting tubule–glomerular feedback,” in which increased sodium reabsorption in the connecting segment, which abuts the afferent arteriole, leads to increased AA release, leading to increased production of EETs and vasodilatory PGs. These then diffuse to the adjacent afferent arteriole and dilate it.⁵²⁹

TUBULES

Both 20-HETE and EETs inhibit tubular sodium reabsorption.^{300,498} Renal cortical interstitial infusion of the nonselective CYP450 inhibitor 17-ODYA increases papillary blood flow, renal interstitial hydrostatic pressure, and sodium excretion without affecting total RBF or GFR. High dietary salt intake in rats increases expression of the renal epoxygenase 2C23 and production and urinary excretion of EETs while decreasing 20-HETE production in the renal cortex.^{497,510} 14,15-EET has also been shown to inhibit renin secretion;⁵³⁰ however, clotrimazole, which is a relatively selective epoxygenase inhibitor, induced hypertension in rats fed a high-salt diet, suggesting a role in the regulation of blood pressure.⁵¹⁰

Proximal Tubule

The proximal tubule contains the highest concentration of CYP450 in the mammalian kidney and expresses minimal COX and LOX activity.⁴⁹⁷ The 4A CYP450 family of hydroxylases that produce 19- and 20-HETE is highly expressed in the mammalian proximal tubule.³¹⁴ CYP450 enzymes of both the 2C and 2J family, which catalyze the formation of EETs, are also expressed in the proximal tubule.⁴⁹⁷ Both EETs and 20-HETE have been shown to be produced in the proximal tubule and have been proposed to be modulators of sodium reabsorption in the proximal tubule.

Studies in isolated perfused proximal tubule have indicated that 20-HETE inhibits sodium transport, whereas 19-HETE stimulates sodium transport, suggesting that 19-HETE may serve as a competitive antagonist of 20-HETE.^{508,531} Administration of EETs inhibits amiloride-sensitive sodium transport in primary cultures of proximal tubule cells⁵³² and in LLC-PK1 cells, a nontransformed, immortalized cell line from pig kidney with proximal tubule characteristics.^{533,534}

It has been proposed that 20-HETE can be a mediator of the hormonal inhibition of proximal tubule reabsorption by parathyroid hormone (PTH), dopamine, Ang II, and EGF. Although the mechanisms of 20-HETE's inhibition have not yet been completely elucidated, there is evidence that it can inhibit Na^+K^+ -ATPase activity by phosphorylation of the Na^+K^+ -ATPase alpha subunit through a protein kinase C-dependent pathway.^{535,536}

EETs may also serve as second messengers in the proximal tubule for EGF.⁵³⁷ The mechanisms whereby CYP450 AA metabolites modulate proximal tubule reabsorption have not been completely elucidated; they may involve both luminal (NHE3) and basolateral (Na⁺-K⁺-ATPase) transporters.^{532,535} CYP450 AA metabolites may modulate the proximal tubule component of the pressure-natriuresis response.⁵³⁸ Of note, intrarenal dopamine, which originates from the proximal tubule, induces the production of EETs and, if EET production is inhibited genetically, dopamine-mediated diuresis and natriuresis are inhibited.⁵³⁹

THICK ASCENDING LIMB OF HENLE

In addition, 20-HETE serves as a second messenger to regulation transport in the TAL. It is produced in this nephron segment⁵⁰⁴ and can inhibit net Na-K-Cl cotransport by direct inhibition of the transporter and by blocking the 70-pS apical K⁺ channel.⁵⁴⁰ In addition, 20-HETE has been implicated as a mediator of the inhibitory effects of Ang II⁵⁴¹ and bradykinin⁵⁴² on TALH transport.

COLLECTING DUCT

In the collecting duct, EETs and/or their diol metabolites serve as inhibitors of the hydroosmotic effects of vasopressin, as well as inhibitors of sodium transport in this segment.^{509,543} Patch clamp studies have indicated that the eNaC sodium channel activity in the cortical collecting duct is inhibited by 11,12-EET.^{544,545} Studies using mice with selective deletion of CYP2C44, the major kidney epoxygenase, have confirmed that EETs modulate eNaC activity and that EET production is mediated in part by activation of collecting duct epidermal growth factor receptors and ERK1/2 activation.^{546–549} There is also an intriguing association of 20-HETE with circadian clock sodium regulation in the collecting duct.⁵⁵⁰

ROLE IN ACUTE AND CHRONIC KIDNEY DISEASE

EET-mediated increases in rat mesangial cell proliferation was the first direct evidence that CYP450 AA metabolites are cellular mitogens.⁵⁵¹ In cultured rabbit proximal tubule cells, CYP450 inhibitors blunted EGF-stimulated proliferation in proximal tubule cells.⁵³⁷ In LLC-PK₁, EETs were found to be potent mitogens, cytoprotective agents, and second messengers for EGF signaling. 14,15-EET-mediated signaling and mitogenesis are dependent on EGF receptor transactivation, which is mediated by the metalloproteinase-dependent release of heparin-binding (HB)-EGF.⁵⁵² In addition to the EETs, 20-HETE has been shown to increase thymidine incorporation in primary cultures of rat proximal tubule and LLC-PK₁ cells⁵⁵³ and vascular smooth muscle cells.⁵⁵⁴ EETs are also pro-angiogenic factors.⁵⁵⁵

There is increasing evidence that the activation of EETs or the administration of EET analogues can protect against acute kidney injury.^{556–558} Conversely, inhibition of 20-HETE is beneficial in AKI.⁵⁵⁹

EETs may also be protective in chronic models of renal injury, such as diabetic nephropathy, and in a 5/6 nephrectomy model.^{560,561} Increasing EET levels by the inhibition of soluble epoxide hydrolase decreases the inflammation and fibrosis in a model of unilateral ureteral obstruction.⁵⁶² Similarly, an EET analogue was found to inhibit the development of radiation-induced renal fibrosis.⁵⁶³

ROLE IN HYPERTENSION

There is increasing evidence that the renal production of CYP450 AA metabolites is altered in a variety of models of hypertension and that blockade of the formation of compounds can alter blood pressure in several of these models. CYP450 AA metabolites may have both pro- and antihypertensive properties. At the level of the renal tubule, both 20-HETE and EETs inhibit sodium transport. However, in the vasculature, 20-HETE promotes vasoconstriction and hypertension, whereas EETs are endothelial-derived vasodilators that have antihypertensive properties. Rats fed a high-salt diet increase expression of the CYP450 epoxygenase 2C23⁵⁶⁴ and develop hypertension if treated with a relatively selective epoxygenase inhibitor. Because EETs have antihypertensive properties, efforts are underway to develop selective inhibitors of soluble epoxide hydrolase (sEH), which converts active EETs to their inactive metabolites, DHETs, and thereby increase EET levels. Studies in rats have indicated that one such sEH inhibitor, 1-cyclohexyl-3-dodecylurea, lowers blood pressure and reduces glomerular and tubulointerstitial injury in an Ang II-mediated model of hypertension in rats.⁵⁶⁵ Furthermore, genetic deletion of CYP2C44, the major kidney epoxygenase, leads to the development of salt-sensitive hypertension.⁵⁴⁷

In deoxycorticosterone acetate (DOCA)-salt hypertension, the administration of a CYP450 inhibitor prevented the development of hypertension.^{519,566} Ang II stimulates the formation of 20-HETE in the renal circulation,⁵⁶⁷ and 20-HETE synthesis inhibition attenuates Ang II-mediated renal vasoconstriction^{521,568} and reduces Ang II-mediated hypertension.⁵⁶⁶

The CYP450 4A2 gene is regulated by salt and is overexpressed in spontaneously hypertensive rats (SHRs);⁵⁶⁹ the production of both 20-HETEs and diHETEs is increased and the production of EETs is reduced.^{314,570} CYP450 inhibitors or antisense oligonucleotides directed against CYP4A1 and 4A2 lowered blood pressure in SHRs.^{514,571} Conversely, studies in humans have indicated that a variant of the human CYP4A11, with reduced 20-HETE synthase activity, is associated with hypertension.⁵⁷²

In Dahl salt-sensitive rats (Dahl S), pressure natriuresis in response to salt loading is shifted so that the kidney requires a higher perfusion pressure to excrete the same amount of sodium as normotensive salt-resistant (Dahl R) rats;^{300,497,498} this is because, at least in part, of increased TALH reabsorption. The production of 20-HETE and expression of CYP4A protein are reduced in the outer medulla and TALH of Dahl S rats relative to Dahl R rats, which is consistent with the observed effect of 20-HETE to inhibit TALH transport. In addition, Dahl S rats do not increase EET production in response to salt loading.

Studies have indicated that Ang II acts on AT₂ receptors on renal vascular endothelial cells to release EETs that may then counteract AT₁-induced renal vasoconstriction and influence pressure natriuresis.^{515,573,574} AT₂ receptor knockout mice develop hypertension,⁵⁷⁵ which is associated with blunted pressure natriuresis, reduced RBF and GFR, and defects in kidney 20-HETE production.⁵⁷⁵ There is also evidence that the natriuretic effects of dopamine are mediated by EETs and 20-HETE.^{539,576}

There has been recent interest in the role of soluble epoxide hydrolase (sEH), which is the major enzyme

mediating the metabolism of EETs to the inactive dHETEs in the regulation of blood pressure. Ang II induces sEH in the vasculature, which may contribute to the hypertensive effects by increasing EET metabolism.⁵⁷⁷ Progressively more selective sEH inhibitors are being developed and have been shown to be effective in reducing blood pressure in a number of experimental models of hypertension.⁵⁷⁸

ACKNOWLEDGMENTS

The writing of this chapter was supported by grants from the Veterans Administration to Raymond C. Harris, National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) to Raymond C. Harris and Ming-Zhi Zhang (DK62794 and DK95785), and National Heart, Lung, and Blood Institute (NHLBI) R56HL127218 and R01HL134895 to Richard Breyer.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

9. Fitzpatrick FA, Soberman R. Regulated formation of eicosanoids. *J Clin Invest*. 2001;107(11):1347–1351.

35. Harris RC, McKanna JA, Akai Y, et al. Cyclooxygenase-2 is associated with the macula densa of rat kidney and increases with salt restriction. *J Clin Invest*. 1994;94(6):2504–2510.
36. Peti-Peterdi J, Harris RC. Macula densa sensing and signaling mechanisms of renin release. *J Am Soc Nephrol*. 2010;21(7):1093–1096.
79. Stegbauer J, Chen D, Herrera M, et al. Resistance to hypertension mediated by intercalated cells of the collecting duct. *JCI Insight*. 2017;2(7):e92720.
86. Zhang MZ, Yao B, Wang Y, et al. Inhibition of cyclooxygenase-2 in hematopoietic cells results in salt-sensitive hypertension. *J Clin Invest*. 2015;125(11):4281–4294.
126. Fitzgerald GA. Coxibs and cardiovascular disease. *N Engl J Med*. 2004;351(17):1709–1711.
237. Breyer MD, Breyer RM. G protein-coupled prostanoid receptors and the kidney. *Annu Rev Physiol*. 2001;63:579–605.
300. Roman RJ. P-450 metabolites of arachidonic acid in the control of cardiovascular function. *Physiol Rev*. 2002;82(1):131–185.
420. Wang X, Yao B, Wang Y, et al. Macrophage cyclooxygenase-2 protects against development of diabetic nephropathy. *Diabetes*. 2017;66(2):494–504.
464. Imig JD. Eicosanoid regulation of the renal vasculature. *Am J Physiol Renal Physiol*. 2000;279(6):F965–F981.

REFERENCES

- Harris RC, Breyer MD. Arachidonic acid metabolites and the kidney. In: Brenner BM, ed. *The Kidney*. 7th ed. 2004:727-776.
- Murakami M, Kudo I. Phospholipase A2. *J Biochem*. 2002;131(3):285-292.
- Boulven I, Palmier B, Robin P, et al. Platelet-derived growth factor stimulates phospholipase C-gamma 1, extracellular signal-regulated kinase, and arachidonic acid release in rat myometrial cells: contribution to cyclic 3',5'-adenosine monophosphate production and effect on cell proliferation. *Biol Reprod*. 2001;65(2):496-506.
- Rouzer CA, Marnett LJ. Endocannabinoid oxygenation by cyclooxygenases, lipoxygenases, and cytochromes P450: cross-talk between the eicosanoid and endocannabinoid signaling pathways. *Chem Rev*. 2011;111(10):5899-5921.
- Fujishima H, Sanchez Mejia RO, Bingham CO 3rd, et al. Cytosolic phospholipase A2 is essential for both the immediate and the delayed phases of eicosanoid generation in mouse bone marrow-derived mast cells. *Proc Natl Acad Sci USA*. 1999;96(9):4803-4807.
- Balsinde J, Winstead MV, Dennis EA. Phospholipase A(2) regulation of arachidonic acid mobilization. *FEBS Lett*. 2002;531(1):2-6.
- Murakami M, Yoshihara K, Shimbara S, et al. Cellular arachidonate-releasing function and inflammation-associated expression of group IIF secretory phospholipase A2. *J Biol Chem*. 2002;277(21):19145-19155.
- Smith WL, Langenbach R. Why there are two cyclooxygenase isozymes. *J Clin Invest*. 2001;107(12):1491-1495.
- Fitzpatrick FA, Soberman R. Regulated formation of eicosanoids. *J Clin Invest*. 2001;107(11):1347-1351.
- FitzGerald GA, Patrono C. The coxibs, selective inhibitors of cyclooxygenase-2. *N Engl J Med*. 2001;345(6):433-442.
- Bonazzi A, Mastuygin V, Mieyal PA, et al. Regulation of cyclooxygenase-2 by hypoxia and peroxisome proliferators in the corneal epithelium. *J Biol Chem*. 2000;275(4):2837-2844.
- Hayama M, Inoue R, Akiba S, et al. ERK and p38 MAP kinase are involved in arachidonic acid release induced by H(2)O(2) and PDGF in mesangial cells. *Am J Physiol Renal Physiol*. 2002;282(3):F485-F491.
- Basavappa S, Pedersen SF, Jorgensen NK, et al. Swelling-induced arachidonic acid release via the 85-kDa cPLA2 in human neuroblastoma cells. *J Neurophysiol*. 1998;79(3):1441-1449.
- ...3 fatty acids are those in which the double-bond is three carbons from the terminal (omega) carbon, i.e. that furthest from the carboxy-group atom. AA is thus an n-6 fatty acid.
- Hansen RA, Ogilvie GK, Davenport DJ, et al. Duration of effects of dietary fish oil supplementation on serum eicosapentaenoic acid and docosahexaenoic acid concentrations in dogs. *Am J Vet Res*. 1998;59(7):864-868.
- Grande JP, Donadio JV Jr. Dietary fish oil supplementation in IgA nephropathy: a therapy in search of a mechanism? *Nutrition*. 1998;14(2):240-242.
- Kujubu DA, Fletcher BS, Varnum BC, et al. TIS10, a phorbol ester tumor promoter-inducible mRNA from Swiss 3T3 cells, encodes a novel prostaglandin synthase/cyclooxygenase homologue. *J Biol Chem*. 1991;266(20):12866-12872.
- O'Banion M, Winn V, Young D. cDNA cloning and functional activity of a glucocorticoid-regulated inflammatory cyclooxygenase. *Proc Natl Acad Sci U S A*. 1992;89:4888-4892.
- Jang BC, Munoz-Najar U, Paik JH, et al. Leptomycin B, an inhibitor of the nuclear export receptor CRM1, inhibits COX-2 expression. *J Biol Chem*. 2002.
- Dixon DA, Tolley ND, King PH, et al. Altered expression of the mRNA stability factor HuR promotes cyclooxygenase-2 expression in colon cancer cells. *J Clin Invest*. 2001;108(11):1657-1665.
- Inoue H, Taba Y, Miwa Y, et al. Transcriptional and posttranscriptional regulation of cyclooxygenase-2 expression by fluid shear stress in vascular endothelial cells. *Arterioscler Thromb Vasc Biol*. 2002;22(9):1415-1420.
- Hla T, Bishop-Bailey D, Liu CH, et al. Cyclooxygenase-1 and -2 isoenzymes. *Int J Biochem Cell Biol*. 1999;31(5):551-557.
- Mestre JR, Mackrell PJ, Rivadeneira DE, et al. Redundancy in the signaling pathways and promoter elements regulating cyclooxygenase-2 gene expression in endotoxin-treated macrophage/monocytic cells. *J Biol Chem*. 2001;276(6):3977-3982.
- Tanabe T, Tohnai N. Cyclooxygenase isozymes and their gene structures and expression. *Prostaglandins Other Lipid Mediat*. 2002;68-69:95-114.
- Inoue H, Tanabe T. Transcriptional role of the nuclear factor kappa B site in the induction by lipopolysaccharide and suppression by dexamethasone of cyclooxygenase-2 in U937 cells. *Biochem Biophys Res Commun*. 1998;244(1):143-148.
- Dixon DA, Kaplan CD, McIntyre TM, et al. Post-transcriptional control of cyclooxygenase-2 gene expression. The role of the 3'-untranslated region. *J Biol Chem*. 2000;275(16):11750-11757.
- Veza R, Mezzasoma AM, Venditti G, et al. Prostaglandin endoperoxides and thromboxane A2 activate the same receptor isoforms in human platelets. *Thromb Haemost*. 2002;87(1):114-121.
- Garavito MR, Malkowski MG, DeWitt DL. The structures of prostaglandin endoperoxide H synthases-1 and -2. *Prostaglandins Other Lipid Mediat*. 2002;68-69:129-152.
- Kalgutkar AS, Crews BC, Rowlinson SW, et al. Aspirin-like molecules that covalently inactivate cyclooxygenase-2. *Science*. 1998;280(5367):1268-1270.
- Yu Y, Fan J, Chen X-S, et al. Genetic model of selective COX2 inhibition reveals novel heterodimer signaling. *Nat Med*. 2006;12(6):699-704.
- Crofford LJ. Specific cyclooxygenase-2 inhibitors: what have we learned since they came into widespread clinical use? *Curr Opin Rheumatol*. 2002;14(3):225-230.
- Li S, Ballou LR, Morham SG, et al. Cyclooxygenase-2 mediates the febrile response of mice to interleukin-1beta. *Brain Res*. 2001;910(1-2):163-173.
- Turini ME, DuBois RN. Cyclooxygenase-2: a therapeutic target. *Annu Rev Med*. 2002;53:35-57.
- Pasinetti GM. From epidemiology to therapeutic trials with anti-inflammatory drugs in Alzheimer's disease: the role of NSAIDs and cyclooxygenase in beta-amyloidosis and clinical dementia. *J Alzheimers Dis*. 2002;4(5):435-445.
- Harris RC, McKanna JA, Akai Y, et al. Cyclooxygenase-2 is associated with the macula densa of rat kidney and increases with salt restriction. *J Clin Invest*. 1994;94(6):2504-2510.
- Peti-Peterdi J, Harris RC. Macula densa sensing and signaling mechanisms of renin release. *J Am Soc Nephrol*. 2010;21(7):1093-1096.
- Guan Y, Chang M, Cho W, et al. Cloning, expression, and regulation of rabbit cyclooxygenase-2 in renal medullary interstitial cells. *Am J Physiol*. 1997;273(1 Pt 2):F18-F26.
- Yang T, Schnermann JB, Briggs JP. Regulation of cyclooxygenase-2 expression in renal medulla by tonicity in vivo and in vitro. *Am J Physiol*. 1999;277(1 Pt 2):F1-F9.
- Nantel F, Meadows E, Denis D, et al. Immunolocalization of cyclooxygenase-2 in the macula densa of human elderly. *FEBS Lett*. 1999;457:475-477.
- Adegboyega PA, Ololade O. Immunohistochemical expression of cyclooxygenase-2 in normal kidneys. *Appl Immunohistochem Mol Morphol*. 2004;12(1):71-74.
- Khan KN, Stanfield KM, Harris RK, et al. Expression of cyclooxygenase-2 in the macula densa of human kidney in hypertension, congestive heart failure, and diabetic nephropathy. *Ren Fail*. 2001;23(3-4):321-330.
- Komhoff M, Jeck ND, Seyberth HW, et al. Cyclooxygenase-2 expression is associated with the renal macula densa of patients with Bartter-like syndrome. *Kidney Int*. 2000;58(6):2420-2424.
- Harris RC, Breyer MD. Physiological regulation of cyclooxygenase-2 in the kidney. *Am J Physiol Renal Physiol*. 2001;281(1):F1-F11.
- Schnermann J. Juxtaglomerular cell complex in the regulation of renal salt excretion. *Am J Physiol*. 1998;274:R263-R279.
- Schnermann J, Traynor T, Pohl H, et al. Vasoconstrictor responses in thromboxane receptor knockout mice: tubuloglomerular feedback and ureteral obstruction. *Acta Physiol Scand*. 2000;168(1):201-207.
- Qi Z, Hao CM, Langenbach RI, et al. Opposite effects of cyclooxygenase-1 and -2 activity on the pressor response to angiotensin II. *J Clin Invest*. 2002;110(1):61-69.
- Ichihara A, Imig JD, Inscho EW, et al. Cyclooxygenase-2 participates in tubular flow-dependent afferent arteriolar tone: interaction with neuronal NOS. *Am J Physiol*. 1998;275(4 Pt 2):F605-F612.
- Perazella MA, Tray K. Selective cyclooxygenase-2 inhibitors: a pattern of nephrotoxicity similar to traditional nonsteroidal anti-inflammatory drugs. *Am J Med*. 2001;111(1):64-67.
- Peti-Peterdi J, Komlosi P, Fuson AL, et al. Luminal NaCl delivery regulates basolateral PGE2 release from macula densa cells. *J Clin Invest*. 2003;112(1):76-82.

50. Cheng HF, Wang JL, Zhang MZ, et al. Role of p38 in the regulation of renal cortical cyclooxygenase-2 expression by extracellular chloride. *J Clin Invest.* 2000;106(5):681–688.
51. Yang T, Park JM, Arend L, et al. Low chloride stimulation of prostaglandin E2 release and cyclooxygenase-2 expression in a mouse macula densa cell line. *J Biol Chem.* 2000;275(48):37922–37929.
52. Zhang MZ, Yao B, McKanna JA, et al. Cross talk between the intrarenal dopaminergic and cyclooxygenase-2 systems. *Am J Physiol Renal Physiol.* 2005;288(4):F840–F845.
53. Zhang MZ, Yao B, Fang X, et al. Intrarenal dopaminergic system regulates renin expression. *Hypertension.* 2009;53(3):564–570.
54. Hanner F, Chambrey R, Bourgeois S, et al. Increased renal renin content in mice lacking the Na⁺/H⁺ exchanger NHE2. *Am J Physiol Renal Physiol.* 2008;294(4):F937–F944.
55. Schnermann J, Briggs JP. Tubular control of renin synthesis and secretion. *Pflügers Arch.* 2013;465(1):39–51.
56. Cheng HF, Wang JL, Zhang MZ, et al. Angiotensin II attenuates renal cortical cyclooxygenase-2 expression. *J Clin Invest.* 1999;103(7):953–961.
57. Harding P, Sigmon DH, Alfie ME, et al. Cyclooxygenase-2 mediates increased renal renin content induced by low-sodium diet. *Hypertension.* 1997;29(1 Pt 2):297–302.
58. Stüchtenoth DO, Marhauer V, Tsikas D, et al. Effects of specific COX-2-inhibition on renin release and renal and systemic prostanoid synthesis in healthy volunteers. *Kidney Int.* 2005;68(5):2197–2207.
59. Traynor TR, Smart A, Briggs JP, et al. Inhibition of macula densa-stimulated renin secretion by pharmacological blockade of cyclooxygenase-2. *Am J Physiol.* 1999;277(5 Pt 2):F706–F710.
60. Cheng HF, Wang JL, Zhang MZ, et al. Genetic deletion of COX-2 prevents increased renin expression in response to ACE inhibition. *Am J Physiol Renal Physiol.* 2001;280(3):F449–F456.
61. Yang T, Endo Y, Huang YG, et al. Renin expression in COX-2-knockout mice on normal or low-salt diets. *Am J Physiol Renal Physiol.* 2000;279(5):F819–F825.
62. Cheng HF, Wang SW, Zhang MZ, et al. Prostaglandins that increase renin production in response to ACE inhibition are not derived from cyclooxygenase-1. *Am J Physiol Regul Integr Comp Physiol.* 2002;283(3):R638–R646.
63. Athirakul K, Kim HS, Audoly LP, et al. Deficiency of COX-1 causes natriuresis and enhanced sensitivity to ACE inhibition. *Kidney Int.* 2001;60(6):2324–2329.
64. Facemire CS, Nguyen M, Jania L, et al. A major role for the EP4 receptor in regulation of renin. *Am J Physiol Renal Physiol.* 2011;301(5):F1035–F1041.
65. Matzdorf C, Kurtz A, Hoehnerl K. COX-2 activity determines the level of renin expression but is dispensable for acute upregulation of renin expression in rat kidneys. *Am J Physiol Renal Physiol.* 2007;292(6):F1782–F1790.
66. Kim HS, Kim MS, Hancock AL, et al. Identification of novel Wilms' tumor suppressor gene target genes implicated in kidney development. *J Biol Chem.* 2007;282(22):16278–16287.
67. Yang Y, Gomez JA, Herrera M, et al. Salt restriction leads to activation of adult renal mesenchymal stromal cell-like cells via prostaglandin E2 and E-prostanoid receptor 4. *Hypertension.* 2015;65(5):1047–1054.
68. Nguyen G, Burckle CA. The (pro)renin receptor: biology and functional significance. *Bull Acad Natl Med.* 2004;188(4):621–628, discussion 8-9.
69. Wang JL, Cheng HF, Harris RC. Cyclooxygenase-2 inhibition decreases renin content and lowers blood pressure in a model of renovascular hypertension. *Hypertension.* 1999;34(1):96–101.
70. Fujino T, Nakagawa N, Yuhki K, et al. Decreased susceptibility to renovascular hypertension in mice lacking the prostaglandin I2 receptor IP. *J Clin Invest.* 2004;114(6):805–812.
71. El-Achkar TM, Plotkin Z, Marcic B, et al. Sepsis induces an increase in thick ascending limb Cox-2 that is TLR4 dependent. *Am J Physiol Renal Physiol.* 2007;293(4):F1187–F1196.
72. Ichihara A, Imig JD, Navar LG. Cyclooxygenase-2 modulates afferent arteriolar responses to increases in pressure. *Hypertension.* 1999;34(4 Pt 2):843–847.
73. Araujo M, Welch WJ. Cyclooxygenase 2 inhibition suppresses tubuloglomerular feedback: roles of thromboxane receptors and nitric oxide. *Am J Physiol Renal Physiol.* 2009;296(4):F790–F794.
74. He W, Miao FJ, Lin DC, et al. Citric acid cycle intermediates as ligands for orphan G-protein-coupled receptors. *Nature.* 2004;429(6988):188–193.
75. Toma I, Kang JJ, Sipos A, et al. Succinate receptor GPR91 provides a direct link between high glucose levels and renin release in murine and rabbit kidney. *J Clin Invest.* 2008;118(7):2526–2534.
76. Vargas SL, Toma I, Kang JJ, et al. Activation of the succinate receptor GPR91 in macula densa cells causes renin release. *J Am Soc Nephrol.* 2009;20(5):1002–1011.
77. Hao CM, Yull F, Blackwell T, et al. Dehydration activates an NF-kappaB-driven, COX2-dependent survival mechanism in renal medullary interstitial cells. *J Clin Invest.* 2000;106(8):973–982.
78. Gonzalez AA, Luffman C, Bourgeois CR, et al. Angiotensin II-independent upregulation of cyclooxygenase-2 by activation of the (Pro)renin receptor in rat renal inner medullary cells. *Hypertension.* 2013;61(2):443–449.
79. Stegbauer J, Chen D, Herrera M, et al. Resistance to hypertension mediated by intercalated cells of the collecting duct. *JCI Insight.* 2017;2(7):e92720.
80. Ichitani Y, Holmberg K, Maunsbach AB, et al. Cyclooxygenase-1 and cyclooxygenase-2 expression in rat kidney and adrenal gland after stimulation with systemic lipopolysaccharide: in situ hybridization and immunocytochemical studies. *Cell Tissue Res.* 2001;303(2):235–252.
81. He W, Xie Q, Wang Y, et al. Generation of a tenascin-C-CreER2 knockin mouse line for conditional DNA recombination in renal medullary interstitial cells. *PLoS ONE.* 2013;8(11):e79839.
82. Yang T, Zhang A, Pasumarthy A, et al. Nitric oxide stimulates COX-2 expression in cultured collecting duct cells through MAP kinases and superoxide but not cGMP. *Am J Physiol Renal Physiol.* 2006;291(4):F891–F895.
83. Kuper C, Bartels H, Fraek ML, et al. Ectodomain shedding of pro-TGF-alpha is required for COX-2 induction and cell survival in renal medullary cells exposed to osmotic stress. *Am J Physiol Cell Physiol.* 2007;293(6):C1971–C1982.
84. Yang Z, Asico LD, Yu P, et al. D5 dopamine receptor regulation of phospholipase D. *Am J Physiol Heart Circ Physiol.* 2005;288(1):H55–H61.
85. Jaimes EA, Zhou MS, Pearce DD, et al. Upregulation of cortical COX-2 in salt-sensitive hypertension: role of angiotensin II and reactive oxygen species. *Am J Physiol Renal Physiol.* 2008;294(2):F385–F392.
86. Zhang MZ, Yao B, Wang Y, et al. Inhibition of cyclooxygenase-2 in hematopoietic cells results in salt-sensitive hypertension. *J Clin Invest.* 2015;125(11):4281–4294.
87. Rocca B, Secchiero P, Ciabattini G, et al. Cyclooxygenase-2 expression is induced during human megakaryopoiesis and characterizes newly formed platelets. *Proc Natl Acad Sci U S A.* 2002;99(11):7634–7639.
88. Zhang MZ, Sanchez Lopez P, McKanna JA, et al. Regulation of cyclooxygenase expression by vasopressin in rat renal medulla. *Endocrinology.* 2004;145(3):1402–1409.
89. Taniura S, Kamitani H, Watanabe T, et al. Transcriptional regulation of cyclooxygenase-1 by histone deacetylase inhibitors in normal human astrocyte cells. *J Biol Chem.* 2002;277(19):16823–16830.
90. Brater DC. Effects of nonsteroidal anti-inflammatory drugs on renal function: focus on cyclooxygenase-2-selective inhibition. *Am J Med.* 1999;107(6A):65S–70S, discussion S-1S.
91. Catella-Lawson F, McAdam B, Morrison BW, et al. Effects of specific inhibition of cyclooxygenase-2 on sodium balance, hemodynamics, and vasoactive eicosanoids. *J Pharmacol Exp Ther.* 1999;289(2):735–741.
92. Whelton A, Fort JG, Puma JA, et al. Cyclooxygenase-2-specific inhibitors and cardiorenal function: a randomized, controlled trial of celecoxib and rofecoxib in older hypertensive osteoarthritis patients. *Am J Ther.* 2001;8(2):85–95.
93. Swan SK, Rudy DW, Lasseter KC, et al. Effect of cyclooxygenase-2 inhibition on renal function in elderly persons receiving a low-salt diet. A randomized, controlled trial. *Ann Intern Med.* 2000;133(1):1–9.
94. Rossat J, Maillard M, Nussberger J, et al. Renal effects of selective cyclooxygenase-2 inhibition in normotensive salt-depleted subjects. *Clin Pharmacol Ther.* 1999;66(1):76–84.
95. Stokes JB. Effect of prostaglandin E2 on chloride transport across the rabbit thick ascending limb of Henle. *J Clin Invest.* 1979;64:495–502.
96. Whelton A, Schulman G, Wallemark C, et al. Effects of celecoxib and naproxen on renal function in the elderly. *Arch Intern Med.* 2000;160(10):1465–1470.
97. Muscara MN, Vergnolle N, Lovren F, et al. Selective cyclo-oxygenase-2 inhibition with celecoxib elevates blood pressure and promotes leukocyte adherence. *Br J Pharmacol.* 2000;129(7):1423–1430.

98. Yao B, Harris RC, Zhang MZ. Interactions between 11 β -hydroxysteroid dehydrogenase and COX-2 in kidney. *Am J Physiol Regul Integr Comp Physiol*. 2005;288(6):R1767–R1773.
99. Aljadhey H, Tu W, Hansen RA, et al. Risk of hyperkalemia associated with selective COX-2 inhibitors. *Pharmacoevidenciol Drug Saf*. 2010;19(11):1194–1198.
100. Atta MG, Whelton A. Acute renal papillary necrosis induced by ibuprofen. *Am J Ther*. 1997;4(1):55–60.
101. DeBroe M, Elseviers M. Analgesic nephropathy. *N Engl J Med*. 1998;338(7):446–452.
102. Segasothy M, Samad S, Zulfigar A, et al. Chronic renal disease and papillary necrosis associated with the long-term use of nonsteroidal anti-inflammatory drugs as the sole or predominant analgesic. *Am J Kid Dis*. 1994;24(1):17–24.
103. Black HE. Renal toxicity of non-steroidal anti-inflammatory drugs. *Toxicol Pathol*. 1986;14(1):83–90.
104. Hao CM, Redha R, Morrow J, et al. Peroxisome proliferator-activated receptor delta activation promotes cell survival following hypertonic stress. *J Biol Chem*. 2002;277(24):21341–21345.
105. Price SR, Klein JD. Cyclooxygenase-2 in the kidney: good, BAD, or both? *Kidney Int*. 2011;80(9):905–907.
106. Akhund L, Quinet RJ, Ishaq S. Celecoxib-related renal papillary necrosis. *Arch Intern Med*. 2003;163(1):114–115.
107. Ahmad SR, Kortepeter C, Brinker A, et al. Renal failure associated with the use of celecoxib and rofecoxib. *Drug Saf*. 2002;25(7):537–544.
108. Kleinknecht D. Interstitial nephritis, the nephrotic syndrome, and chronic renal failure secondary to nonsteroidal anti-inflammatory drugs. *Semin Nephrol*. 1995;15(3):228–235.
109. Henao J, Hisamuddin I, Nzerue CM, et al. Celecoxib-induced acute interstitial nephritis. *Am J Kidney Dis*. 2002;39(6):1313–1317.
110. Alper AB Jr, Meleg-Smith S, Krane NK. Nephrotic syndrome and interstitial nephritis associated with celecoxib. *Am J Kidney Dis*. 2002;40(5):1086–1090.
111. Tietjen DP. Recurrence and specificity of nephrotic syndrome due to tolmetin. *Am J Med*. 1989;87(3):354–355.
112. Radford MG Jr, Holley KE, Grande JP, et al. Reversible membranous nephropathy associated with the use of nonsteroidal anti-inflammatory drugs. *JAMA*. 1996;276(6):466–469.
113. Peruzzi L, Gianoglio B, Porcellini MG, et al. Neonatal end-stage renal failure associated with maternal ingestion of cyclo-oxygenase-type-2 selective inhibitor nimesulide as tocolytic [letter; comment]. *Lancet*. 1999;354(9190):1615.
114. Smith FG, Wade AW, Lewis ML, et al. Cyclooxygenase (COX) inhibitors and the newborn kidney. *Pharmaceuticals*. 2012;5(11):1160–1176.
115. Komhoff M, Wang JL, Cheng HF, et al. Cyclooxygenase-2-selective inhibitors impair glomerulogenesis and renal cortical development. *Kidney Int*. 2000;57(2):414–422.
116. Dinchuk JE, Car BD, Focht RJ, et al. Renal abnormalities and an altered inflammatory response in mice lacking cyclooxygenase II. *Nature*. 1995;378(6555):406–409.
117. Morham SG, Langenbach R, Loftin CD, et al. Prostaglandin synthase 2 gene disruption causes severe renal pathology in the mouse. *Cell*. 1995;83(3):473–482.
118. Langenbach R, Morham SG, Tian HF, et al. Prostaglandin synthase 1 gene disruption in mice reduces arachidonic acid-induced inflammation and indomethacin-induced gastric ulceration. *Cell*. 1995;83(3):483–492.
119. Zhang MZ, Wang JL, Cheng HF, et al. Cyclooxygenase-2 in rat nephron development. *Am J Physiol*. 1997;273(6 Pt 2):F994–F1002.
120. Avner ED, Sweeney WE Jr, Piesco NP, et al. Growth factor requirements of organogenesis in serum-free metanephric organ culture. *In Vitro Cell Dev Biol*. 1985;21(5):297–304.
121. Pouretezadi SJ, Cheng CN, Chambers JM, et al. Prostaglandin signaling regulates nephron segment patterning of renal progenitors during zebrafish kidney development. *Elife*. 2016;5.
122. Frölich S, Slattery P, Thomas D, et al. Angiotensin II-AT1-receptor signaling is necessary for cyclooxygenase-2-dependent postnatal nephron generation. *Kidney Int*. 2017;91(4):818–829.
123. Bagai S, Rubio E, Cheng JF, et al. Fibroblast growth Factor-10 is a mitogen for urothelial cells. *J Biol Chem*. 2002;277(26):23828–23837.
124. Gimbrone MA Jr, Topper JN, Nagel T, et al. Endothelial dysfunction, hemodynamic forces, and atherogenesis. *Ann N Y Acad Sci*. 2000;902:230–239, discussion 9–40.
125. McAdam BF, Catella-Lawson F, Mardini IA, et al. Systemic biosynthesis of prostacyclin by cyclooxygenase (COX)-2: the human pharmacology of a selective inhibitor of COX-2. *Proc Natl Acad Sci USA*. 1999;96(1):272–277.
126. Fitzgerald GA. Coxibs and cardiovascular disease. *N Engl J Med*. 2004;351(17):1709–1711.
127. Bea F, Blessing E, Bennett BJ, et al. Chronic inhibition of cyclooxygenase-2 does not alter plaque composition in a mouse model of advanced unstable atherosclerosis. *Cardiovasc Res*. 2003;60(1):198–204.
128. Pratico D, Tillmann C, Zhang ZB, et al. Acceleration of atherogenesis by COX-1-dependent prostanoid formation in low density lipoprotein receptor knockout mice. *Proc Natl Acad Sci USA*. 2001;98(6):3358–3363.
129. Burleigh ME, Babaev VR, Oates JA, et al. Cyclooxygenase-2 promotes early atherosclerotic lesion formation in LDL receptor-deficient mice. *Circulation*. 2002;105(15):1816–1823.
130. Belton OA, Duffy A, Toomey S, et al. Cyclooxygenase isoforms and platelet vessel wall interactions in the apolipoprotein E knockout mouse model of atherosclerosis. *Circulation*. 2003;108(24):3017–3023.
131. Burleigh ME, Babaev VR, Yancey PG, et al. Cyclooxygenase-2 promotes early atherosclerotic lesion formation in ApoE-deficient and C57BL/6 mice. *J Mol Cell Cardiol*. 2005;39(3):443–452.
132. Egan KM, Wang M, Fries S, et al. Cyclooxygenases, thromboxane, and atherosclerosis: plaque destabilization by cyclooxygenase-2 inhibition combined with thromboxane receptor antagonism. *Circulation*. 2005;111(3):334–342.
133. Hansson GK. Inflammation, atherosclerosis, and coronary artery disease. *N Engl J Med*. 2005;352(16):1685–1695.
134. Sato T, Sawada S, Tsuda Y, et al. The mechanism of thrombin-induced prostacyclin synthesis in human endothelial cells with reference to the gene transcription of prostacyclin-related enzymes and Ca²⁺ kinetics. *J Pharmacol Toxicol Methods*. 1999;41(4):173–182.
135. Okahara K, Sun B, Kambayashi J. Upregulation of prostacyclin synthesis-related gene expression by shear stress in vascular endothelial cells. *Arterioscler Thromb Vasc Biol*. 1998;18(12):1922–1926.
136. Guan Z, Buckman SY, Miller BW, et al. Interleukin-1 β -induced cyclooxygenase-2 expression requires activation of both c-Jun NH₂-terminal kinase and p38 MAPK signal pathways in rat renal mesangial cells. *J Biol Chem*. 1998;273(44):28670–28676.
137. Soler M, Camacho M, Sola R, et al. Mesangial cells release untransformed prostaglandin H₂ as a major prostanoid. *Kidney Int*. 2001;59(4):1283–1289.
138. Guan Y, Zhang Y, Schneider A, et al. Urogenital distribution of a mouse membrane-associated prostaglandin E(2) synthase. *Am J Physiol Renal Physiol*. 2001;281(6):F1173–F1177.
139. Hao CM, Komhoff M, Guan Y, et al. Selective targeting of cyclooxygenase-2 reveals its role in renal medullary interstitial cell survival. *Am J Physiol*. 1999;277(3 Pt 2):F352–F359.
140. Chevalier D, Lo-Guidice JM, Sergeant E, et al. Identification of genetic variants in the human thromboxane synthase gene (CYP5A1). *Mutat Res*. 2001;432(3–4):61–67.
141. Nusing R, Fehr PM, Gudat F, et al. The localization of thromboxane synthase in normal and pathological human kidney tissue using a monoclonal antibody Tu 300. *Virchows Arch*. 1994;424(1):69–74.
142. Wilcox CS, Welch WJ. Thromboxane synthase and TP receptor mRNA in rat kidney and brain: effects of salt intake and ANG II. *Am J Physiol Renal Physiol*. 2003.
143. Quest DW, Wilson TW. Effects of ridogrel, a thromboxane synthase inhibitor and receptor antagonist, on blood pressure in the spontaneously hypertensive rat. *Jpn J Pharmacol*. 1998;78(4):479–486.
144. Yokoyama C, Yabuki T, Shimonishi M, et al. Prostacyclin-deficient mice develop ischemic renal disorders, including nephrosclerosis and renal infarction. *Circulation*. 2002;106(18):2397–2403.
145. Murata T, Ushikubi F, Matsuo K, et al. Altered pain perception and inflammatory response in mice lacking prostacyclin receptor. *Nature*. 1997;388(6643):678–682.
146. Urade Y, Eguchi N. Lipocalin-type and hematopoietic prostaglandin D synthases as a novel example of functional convergence. *Prostaglandins Other Lipid Mediat*. 2002;68-69:375–382.
147. Urade Y, Hayaishi O. Prostaglandin D synthase: structure and function. *Vitam Horm*. 2000;58:89–120.
148. Eguchi N, Minami T, Shirafuji N, et al. Lack of tactile pain (allodynia) in lipocalin-type prostaglandin D synthase-deficient mice. *Proc Natl Acad Sci USA*. 1999;96(2):726–730.
149. Vitzthum H, Abt I, Einhellig S, et al. Gene expression of prostanoid forming enzymes along the rat nephron. *Kidney Int*. 2002;62(5):1570–1581.

150. Ogawa M, Hirawa N, Tsuchida T, et al. Urinary excretions of lipocalin-type prostaglandin D2 synthase predict the development of proteinuria and renal injury in OLETF rats. *Nephrol Dial Transplant*. 2006;21(4):924–934.
151. Ragolia L, Palaia T, Hall CE, et al. Accelerated glucose intolerance, nephropathy, and atherosclerosis in prostaglandin D2 synthase knock-out mice. *J Biol Chem*. 2005;280(33):29946–29955.
152. Watanabe K. Prostaglandin F synthase. *Prostaglandins Other Lipid Mediat*. 2002;68-69:401–407.
153. Roberts LJ 2nd, Seibert K, Liston TE, et al. PGD2 is transformed by human coronary arteries to 9 alpha, 11 beta- PGF₂, which contracts human coronary artery rings. *Adv Prostaglandin Thromboxane Leukot Res*. 1987;17A:427–429.
154. Sharif NA, Xu SX, Williams GW, et al. Pharmacology of [3H] prostaglandin E1/[3H]prostaglandin E2 and [3H]prostaglandin F2alpha binding to EP3 and FP prostaglandin receptor binding sites in bovine corpus luteum: characterization and correlation with functional data. *J Pharmacol Exp Ther*. 1998;286(2):1094–1102.
155. Wallner EI, Wada J, Tramonti G, et al. Relevance of aldo-keto reductase family members to the pathobiology of diabetic nephropathy and renal development. *Ren Fail*. 2001;23(3–4):311–320.
156. Siragy HM, Inagami T, Ichiki T, et al. Sustained hypersensitivity to angiotensin II and its mechanism in mice lacking the subtype-2 (AT2) angiotensin receptor. *Proc Natl Acad Sci USA*. 1999;96(11):6506–6510.
157. Siragy HM, Senbonmatsu T, Ichiki T, et al. Increased renal vasodilator prostanoins prevent hypertension in mice lacking the angiotensin subtype-2 receptor. *J Clin Invest*. 1999;104(2):181–188.
158. Siragy HM, Carey RM. The subtype 2 angiotensin receptor regulates renal prostaglandin F2 alpha formation in conscious rats. *Am J Physiol*. 1997;273(3 Pt 2):R1103–R1107.
159. Tanikawa N, Ohmiya Y, Ohkubo H, et al. Identification and characterization of a novel type of membrane-associated prostaglandin E synthase. *Biochem Biophys Res Commun*. 2002;291(4):884–889.
160. Jakobsson PJ, Thoren S, Morgenstern R, et al. Identification of human prostaglandin E synthase: a microsomal, glutathione-dependent, inducible enzyme, constituting a potential novel drug target. *Proc Natl Acad Sci USA*. 1999;96(13):7220–7225.
161. Trebino CE, Stock JL, Gibbons CP, et al. Impaired inflammatory and pain responses in mice lacking an inducible prostaglandin E synthase. *Proc Natl Acad Sci USA*. 2003;100(15):9044–9049.
162. Engblom D, Saha S, Engstrom L, et al. Microsomal prostaglandin E synthase-1 is the central switch during immune-induced pyresis. *Nat Neurosci*. 2003;6(11):1137–1138.
163. Ouellet M, Falgout JP, Hien Ear P, et al. Purification and characterization of recombinant microsomal prostaglandin E synthase-1. *Protein Expr Purif*. 2002;26(3):489–495.
164. Uematsu S, Matsumoto M, Takeda K, et al. Lipopolysaccharide-dependent prostaglandin E(2) production is regulated by the glutathione-dependent prostaglandin E(2) synthase gene induced by the Toll-like receptor 4/MyD88/NF-IL6 pathway. *J Immunol*. 2002;168(11):5811–5816.
165. Dinchuk JE, Car BD, Focht RJ, et al. Renal abnormalities and an altered inflammatory response in mice lacking cyclooxygenase II. *Nature*. 1995;378(6555):406–409.
166. Nguyen M, Camenisch T, Snouwaert JN, et al. The prostaglandin receptor EP4 triggers remodelling of the cardiovascular system at birth. *Nature*. 1997;390(6655):78–81.
167. Tanioka T, Nakatani Y, Semmyo N, et al. Molecular identification of cytosolic prostaglandin E2 synthase that is functionally coupled with cyclooxygenase-1 in immediate prostaglandin E2 biosynthesis. *J Biol Chem*. 2000;275(42):32775–32782.
168. Zhang Y, Schneider A, Rao R, et al. Genomic structure and genitourinary expression of mouse cytosolic prostaglandin E(2) synthase gene. *Biochim Biophys Acta*. 2003;1634(1–2):15–23.
169. Murakami M, Nakatani Y, Tanioka T, et al. Prostaglandin E synthase. *Prostaglandins Other Lipid Mediat*. 2002;68-69:383–399.
170. Hirata M, Hayashi Y, Ushikubi F, et al. Cloning and expression of cDNA for a human thromboxane A2 receptor. *Nature*. 1991;349:617–620.
171. Raychowdhury MK, Yukawa M, Collins IJ, et al. Alternative splicing produces a divergent cytoplasmic tail in the human endothelial thromboxane A2 receptor [published erratum appears in *J Biol Chem* 1995 Mar 24;270(12):7011]. *J Biol Chem*. 1994;269(30):19256–19261.
172. Pierce KL, Regan JW. Prostanoid receptor heterogeneity through alternative mRNA splicing. *Life Sci*. 1998;62(17–18):1479–1483.
173. Wilson RJ, Rhodes SA, Wood RL, et al. Functional pharmacology of human prostanoid EP2 and EP4 receptors. *Eur J Pharmacol*. 2004;501(1–3):49–58.
174. Morinelli TA, Oatis JE Jr, Okwu AK, et al. Characterization of an 125I-labeled thromboxane A2/prostaglandin H2 receptor agonist. *J Pharmacol Exp Ther*. 1989;251(2):557–562.
175. Narumiya S, Sugimoto Y, Ushikubi F. Prostanoid receptors: structures, properties, and functions. *Physiol Rev*. 1999;79(4):1193–1226.
176. Abramovitz M, Adam M, Boie Y, et al. The utilization of recombinant prostanoid receptors to determine the affinities and selectivities of prostaglandins and related analogs [in process citation]. *Biochim Biophys Acta*. 2000;1483(2):285–293.
177. Audoly LP, Rocca B, Fabre JE, et al. Cardiovascular responses to the isoprostanes iPF(2alpha)-III and iPE(2)-III are mediated via the thromboxane A(2) receptor in vivo. *Circulation*. 2000;101(24):2833–2840.
178. Welch WJ. Effects of isoprostane on tubuloglomerular feedback: roles of TP receptors, NOS, and salt intake. *Am J Physiol Renal Physiol*. 2005;288(4):F757–F762.
179. Morrow JD. Quantification of isoprostanes as indices of oxidant stress and the risk of atherosclerosis in humans. *Arterioscler Thromb Vasc Biol*. 2005;25(2):279–286.
180. Abe T, Takeuchi K, Takahashi N, et al. Rat kidney thromboxane A2 receptor: molecular cloning signal transduction and intrarenal expression localization. *J Clin Invest*. 1995;96:657–664.
181. Regard JB, Sato IT, Coughlin SR. Anatomical profiling of G protein-coupled receptor expression. *Cell*. 2008;135(3):561–571.
182. Hirata T, Kakizuka A, Ushikubi F, et al. Arg60 to Leu mutation of the human thromboxane A2 receptor in a dominantly inherited bleeding disorder. *J Clin Invest*. 1994;94:1662–1667.
183. Mannon RB, Coffman TM, Mannon PJ. Distribution of binding sites for thromboxane A2 in the mouse kidney. *Am J Physiol*. 1996;271(6 Pt 2):F1131–F1138.
184. Thomas DW, Mannon RB, Mannon PJ, et al. Coagulation defects and altered hemodynamic responses in mice lacking receptors for thromboxane A2. *J Clin Invest*. 1998;102(11):1994–2001.
185. Spurney RF, Onorato JJ, Albers FJ, et al. Thromboxane binding and signal transduction in rat glomerular mesangial cells. *Am J Physiol*. 1993;264:F292–F299.
186. Nasjletti A, Arthur C. Corcoran memorial lecture. The role of eicosanoids in angiotensin-dependent hypertension. *Hypertension*. 1998;31(1 Pt 2):194–200.
187. Kawada N, Dennehy K, Solis G, et al. TP receptors regulate renal hemodynamics during angiotensin II slow pressor response. *Am J Physiol Renal Physiol*. 2004;287(4):F753–F759.
188. Boffa JJ, Just A, Coffman TM, et al. Thromboxane receptor mediates renal vasoconstriction and contributes to acute renal failure in endotoxemic mice. *J Am Soc Nephrol*. 2004;15(9):2358–2365.
189. Welch WJ, Peng B, Takeuchi K, et al. Salt loading enhances rat renal TXA2/PGH2 receptor expression and TGF response to U-46,619. *Am J Physiol*. 1997;273(6 Pt 2):F976–F983.
190. Welch WJ, Wilcox CS. Potentiation of tubuloglomerular feedback in the rat by thromboxane mimetic. Role of macula densa. *J Clin Invest*. 1992;89(6):1857–1865.
191. Coffman TM, Spurney RF, Mannon RB, et al. Thromboxane A2 modulates the fibrinolytic system in glomerular mesangial cells. *Am J Physiol*. 1998;275(2 Pt 2):F262–F269.
192. Francois H, Coffman TM. Prostanoids and blood pressure: which way is up? *J Clin Invest*. 2004;114(6):757–759.
193. Francois H, Makhanova N, Ruiz P, et al. A role for the thromboxane receptor in L-NAME hypertension. *Am J Physiol Renal Physiol*. 2008;295(4):F1096–F1102.
194. Kiriya M, Ushikubi F, Kobayashi T, et al. Ligand binding specificities of the eight types and subtypes of the mouse prostanoid receptors expressed in chinese hamster ovary cells. *Br J Pharmacol*. 1997;122:217–224.
195. Namba T, Oida H, Sugimoto Y, et al. cDNA cloning of a mouse prostacyclin receptor: multiple signaling pathways and expression in thymic medulla. *J Biol Chem*. 1994;269(13):9986–9992.
196. Boie Y, Rushmore TH, Darmon-Goodwin A, et al. Cloning and expression of a cDNA for the human prostanoid IP receptor. *J Biol Chem*. 1994;269(16):12173–12178.
197. Oida H, Namba T, Sugimoto Y, et al. In situ hybridization studies on prostacyclin receptor mRNA expression in various mouse organs. *Br J Pharmacol*. 1995;116:2828–2837.

198. Nasrallah R, Hebert RL. Prostacyclin signaling in the kidney: implications for health and disease. *Am J Physiol Renal Physiol*. 2005;289(2):F235–F246.
199. Edwards A, Silldorff EP, Pallone TL. The renal medullary microcirculation. *Front Biosci*. 2000;5:E36–E52.
200. Bugge JF, Stokke ES, Vikse A, et al. Stimulation of renin release by PGE₂ and PGI₂ infusion in the dog: enhancing effect of ureteral occlusion or administration of ethacrynic acid. *Acta Physiol Scand*. 1990;138(2):193–201.
201. Ito S, Carretero OA, Abe K, et al. Effect of prostanoids on renin release from rabbit afferent arterioles with and without macula densa. *Kidney Int*. 1989;35(5):1138–1144.
202. Chaudhari A, Gupta S, Kirschenbaum M. Biochemical evidence for PGI₂ and PGE₂ receptors in the rabbit renal preglomerular microvasculature. *Biochim Biophys Acta*. 1990;1053(2–3):156–161.
203. Francois H, Athirakul K, Howell D, et al. Prostacyclin protects against elevated blood pressure and cardiac fibrosis. *Cell Metab*. 2005;2(3):201–207.
- 203a. Batchu SN, Majumder S, Bowskill BB, et al. Prostaglandin I₂ receptor agonism preserves β -cell function and attenuates albuminuria through nephrin dependent mechanisms. *Diabetes*. 2016;65:1398–1409.
204. Hébert R, Regnier L, Peterson L. Rabbit cortical collecting ducts express a novel prostacyclin receptor. *Am J Physiol*. 1995;268:F145–F154.
205. Komhoff M, Lesener B, Nakao K, et al. Localization of the prostacyclin receptor in human kidney. *Kidney Int*. 1998;54(6):1899–1908.
206. Tone Y, Inoue H, Hara S, et al. The regional distribution and cellular localization of mRNA encoding rat prostacyclin synthase. *Eur J Cell Biol*. 1997;72(3):268–277.
207. Hirata M, Kakizuka A, Aizawa M, et al. Molecular characterization of a mouse prostaglandin D receptor and functional expression of the cloned gene. *Proc Natl Acad Sci USA*. 1994;91(23):11192–11196.
208. Coleman RA, Grix SP, Head SA, et al. A novel inhibitory prostanoid receptor in piglet saphenous vein. *Prostaglandins*. 1994;47:151–168.
209. Oida H, Hirata M, Sugimoto Y, et al. Expression of messenger RNA for the prostaglandin D receptor in the leptomeninges of the mouse brain. *FEBS Lett*. 1997;417(1):53–56.
210. Boie Y, Sawyer N, Slipetz DM, et al. Molecular cloning and characterization of the human prostanoid DP receptor. *J Biol Chem*. 1995;270(32):18910–18916.
211. Urade Y, Hayaishi O. Prostaglandin D₂ and sleep regulation. *Biochim Biophys Acta*. 1999;1436(3):606–615.
212. Sri Kantha S, Matsumura H, Kubo E, et al. Effects of prostaglandin D₂, lipoxins and leukotrienes on sleep and brain temperature of rats. *Prostaglandins Leukot Essent Fatty Acids*. 1994;51(2):87–93.
213. Matsuoka T, Hirata M, Tanaka H, et al. Prostaglandin D₂ as a mediator of allergic asthma. *Science*. 2000;287(5460):2013–2017.
214. Sturino CF, O'Neill G, Lachance N, et al. Discovery of a potent and selective prostaglandin D₂ receptor antagonist, [(3R)-4-(4-chlorobenzyloxy)-7-fluoro-5-(methylsulfonyl)-1,2,3,4-tetrahydrocyclopent a[b] indol-3-yl]-acetic acid (MK-0524). *J Med Chem*. 2007;50(4):794–806.
215. Rao PS, Cavanagh D, Dietz JR, et al. Dose-dependent effects of prostaglandin D₂ on hemodynamics, renal function, and blood gas analyses. *Am J Obstet Gynecol*. 1987;156(4):843–851.
216. Hirai H, Tanaka K, Takano S, et al. Cutting edge: agonistic effect of indomethacin on a prostaglandin D₂ receptor, CRTH2. *J Immunol*. 2002;168(3):981–985.
217. Ly TW, Bacon KB. Small-molecule CRTH2 antagonists for the treatment of allergic inflammation: an overview. *Expert Opin Investig Drugs*. 2005;14(7):769–773.
218. Ito H, Yan X, Nagata N, et al. PGD₂-CRTH2 pathway promotes tubulointerstitial fibrosis. *J Am Soc Nephrol*. 2012;23(11):1797–1809.
219. Abramovitz M, Boie Y, Nguyen T, et al. Cloning and expression of a cDNA for the human prostanoid FP receptor. *J Biol Chem*. 1994;269(4):2632–2636.
220. Woodward DF, Fairbairn CE, Lawrence RA. Identification of the FP-receptor as a discrete entity by radioligand binding in biosystems that exhibit different functional rank orders of potency in response to prostanoids. *Adv Exp Med Biol*. 1997;400A:223–227.
221. Pierce KL, Bailey TJ, Hoyer PB, et al. Cloning of a carboxyl-terminal isoform of the prostanoid FP receptor. *J Biol Chem*. 1997;272(2):883–887.
222. Pierce KL, Fujino H, Srinivasan D, et al. Activation of FP prostanoid receptor isoforms leads to Rho-mediated changes in cell morphology and in the cell cytoskeleton. *J Biol Chem*. 1999;274(50):35944–35949.
223. Fujino H, Srinivasan D, Regan JW. Cellular conditioning and activation of beta-Catenin signaling by the FPB prostanoid receptor. *J Biol Chem*. 2002;277(50):48786–48795.
224. Hasumoto K, Sugimoto Y, Gotoh M, et al. Characterization of the mouse prostaglandin F receptor gene: a transgenic mouse study of a regulatory region that controls its expression in the stomach and kidney but not in the ovary. *Genes Cells*. 1997;2(9):571–580.
225. Chen J, Champa-Rodriguez ML, Woodward DF. Identification of a prostanoid FP receptor population producing endothelium-dependent vasorelaxation in the rabbit jugular vein. *Br J Pharmacol*. 1995;116(7):3035–3041.
226. Muller K, Krieg P, Marks F, et al. Expression of PGF₂(α) receptor mRNA in normal, hyperplastic and neoplastic skin. *Carcinogenesis*. 2000;21(5):1063–1066.
227. Linden C, Alm A. Prostaglandin analogues in the treatment of glaucoma. *Drugs Aging*. 1999;14(5):387–398.
228. Hebert RL, Carmosino M, Saito O, et al. Characterization of a rabbit PGF₂(α) (FP) receptor exhibiting Gi-restricted signaling and that inhibits water absorption in renal collecting duct. *J Biol Chem*. 2005.
229. Hebert RL, Jacobson HR, Fredin D, et al. Evidence that separate PGE₂ receptors modulate water and sodium transport in rabbit cortical collecting duct. *Am J Physiol*. 1993;265(5 Pt 2):F643–F650.
230. Funk C, Furchi L, FitzGerald G, et al. Cloning and expression of a cDNA for the human prostaglandin E receptor EP1 subtype. *J Biol Chem*. 1993;268:26767–26772.
231. Breyer MD, Jacobson HR, Davis LS, et al. In situ hybridization and localization of mRNA for the rabbit prostaglandin EP3 receptor. *Kidney Int*. 1993;44(6):1372–1378.
232. Bek M, Nusing R, Kowark P, et al. Characterization of prostanoid receptors in podocytes. *J Am Soc Nephrol*. 1999;10(10):2084–2093.
233. Breshnahan BA, Kelefiotis D, Stratidakis I, et al. PGF₂(α)-induced signaling events in glomerular mesangial cells. *Proc Soc Exp Biol Med*. 1996;212(2):165–173.
234. Yu Y, Lucitt MB, Stubbe J, et al. Prostaglandin F₂(α) elevates blood pressure and promotes atherosclerosis. *Proc Natl Acad Sci USA*. 2009;106(19):7985–7990.
235. Hata AN, Breyer RM. Pharmacology and signaling of prostaglandin receptors: multiple roles in inflammation and immune modulation. *Pharmacol Ther*. 2004;103(2):147–166.
236. Toh H, Ichikawa A, Narumiya S. Molecular evolution of receptors for eicosanoids. *FEBS Lett*. 1995;361:17–21.
237. Breyer MD, Breyer RM. G protein-coupled prostanoid receptors and the kidney. *Annu Rev Physiol*. 2001;63:579–605.
238. Guan Y, Zhang Y, Breyer RM, et al. Prostaglandin E₂ inhibits renal collecting duct Na⁺ absorption by activating the EP1 receptor. *J Clin Invest*. 1998;102(1):194–201.
239. Coleman RA, Kennedy I, Humphrey PPA, et al. In: Emmet JC, ed. *Comprehensive Medicinal Chemistry*. Oxford: Pergamon Press; 1990:643–714.
240. Ishibashi R, Tanaka I, Kotani M, et al. Roles of prostaglandin E receptors in mesangial cells under high-glucose conditions. *Kidney Int*. 1999;56(2):589–600.
241. Inscho E, Carmine P, Navar L. Prostaglandin influences on afferent arteriolar responses to vasoconstrictor agonists. *Am J Physiol*. 1990;259:F157–F163.
242. Purdy KE, Arendshorst WJ. EP(1) and EP(4) receptors mediate prostaglandin E(2) actions in the microcirculation of rat kidney. *Am J Physiol Renal Physiol*. 2000;279(4):F755–F764.
243. Zhang Y, Guan Y, Scheider A, et al. Characterization of murine vasopressor and vasodepressor prostaglandin E₂ receptors. *Hypertension*. 2000;35:1129–1134.
244. Stock JL, Shinjo K, Burkhardt J, et al. The prostaglandin E₂ EP1 receptor mediates pain perception and regulates blood pressure. *J Clin Invest*. 2001;107(3):325–331.
245. Tsuboi K, Sugimoto Y, Ichikawa A. Prostanoid receptor subtypes. *Prostaglandins Other Lipid Mediat*. 2002;68-69:535–556.
246. Regan JW, Bailey TJ, Pepperl DJ, et al. Cloning of a novel human prostaglandin receptor with characteristics of the pharmacologically defined EP2 subtype. *Mol Pharmacol*. 1994;46:213–220.
247. Nishigaki N, Negishi M, Honda A, et al. Identification of prostaglandin E receptor 'EP2' cloned from mastocytoma cells as EP4 subtype. *FEBS Lett*. 1995;364:339–341.
248. Guan Y, Stillman BA, Zhang Y, et al. Cloning and expression of the rabbit prostaglandin EP2 receptor. *BMC Pharmacol*. 2002;2(1):14.

249. Kabashima K, Saji T, Murata T, et al. The prostaglandin receptor EP4 suppresses colitis, mucosal damage and CD4 cell activation in the gut. *J Clin Invest.* 2002;109(7):883–893.
250. af Forselles KJ, Root J, Clarke T, et al. In vitro and in vivo characterization of PF-04418948, a novel, potent and selective prostaglandin EP(2) receptor antagonist. *Br J Pharmacol.* 2011;164(7):1847–1856.
251. Jiang J, Ganesh T, Du Y, et al. Small molecule antagonist reveals seizure-induced mediation of neuronal injury by prostaglandin E2 receptor subtype EP2. *Proc Natl Acad Sci USA.* 2012;109(8):3149–3154.
252. Katsuyama M, Ikegami R, Karahashi H, et al. Characterization of the LPS-stimulated expression of EP2 and EP4 prostaglandin E receptors in mouse macrophage-like cell line, J774.1 [in process citation]. *Biochem Biophys Res Commun.* 1998;251(3):727–731.
253. Kennedy C, Schneider A, Young-Sieglar A, et al. Regulation of renin and aldosterone levels in mice lacking the prostaglandin EP2 receptor. *J Am Soc Nephrol.* 1999;10(348A).
254. Chen J, Zhao M, He W, et al. Increased dietary NaCl induces renal medullary PGE2 production and natriuresis via the EP2 receptor. *Am J Physiol Renal Physiol.* 2008;295(3):F818–F825.
255. Coleman RA, Smith WL, Narumiya SVIII. International union of pharmacology classification of prostanoid receptors: properties, distribution, and structure of the receptors and their subtypes. *Pharmacol Rev.* 1994;46(2):205–229.
256. Boie Y, Stocco R, Sawyer N, et al. Molecular cloning and characterization of the four rat prostaglandin E2 prostanoid receptor subtypes. *Eur J Pharmacol.* 1997;340(2–3):227–241.
257. Breyer RM, Emeson RB, Tamg JL, et al. Alternative splicing generates multiple isoforms of a rabbit prostaglandin E2 receptor. *J Biol Chem.* 1994;269(8):6163–6169.
258. Kotani M, Tanaka I, Ogawa Y, et al. Molecular cloning and expression of multiple isoforms of human prostaglandin E receptor EP3 subtype generated by alternative messenger RNA splicing: multiple second messenger systems and tissue-specific distributions. *Mol Pharmacol.* 1995;48(5):869–879.
259. Irie A, Sugimoto Y, Namba T, et al. Third isoform of the Prostaglandin-E-receptor EP3 subtype with different C-terminal tail coupling to both stimulation and inhibition of adenylate cyclase. *Eur J Biochem.* 1993;217:313–318.
260. Aoki J, Katoh H, Yasui H, et al. Signal transduction pathway regulating prostaglandin EP3 receptor-induced neurite retraction: requirement for two different tyrosine kinases. *Biochem J.* 1999;340(Pt 2):365–369.
261. Kimple ME, Keller MP, Rabaglia MR, et al. Prostaglandin E2 receptor, EP3, is induced in diabetic islets and negatively regulates glucose- and hormone-stimulated insulin secretion. *Diabetes.* 2013;62(6):1904–1912.
262. Hasegawa H, Negishi M, Ichikawa A. Two isoforms of the prostaglandin E receptor EP3 subtype different in agonist-independent constitutive activity. *J Biol Chem.* 1996;271(4):1857–1860.
263. Breyer MD, Davis L, Jacobson HR, et al. Differential localization of prostaglandin E receptor subtypes in human kidney. *Am J Physiol.* 1996;270(5 Pt 2):F912–F918.
264. Taniguchi S, Watanabe T, Nakao A, et al. Detection and quantitation of EP3 prostaglandin E2 receptor mRNA along mouse nephron segments by RT-PCR. *Am J Physiol.* 1994;266:C1453–C1458.
265. Good DW, George T. Regulation of HCO₃⁻ absorption by prostaglandin E2 and G-proteins in rat medullary thick ascending limb. *Am J Physiol.* 1996;270:F711–F717.
266. Good D. PGE2 reverses AVP inhibition of HCO₃⁻ absorption in rat MTAL by activation of protein kinase C. *Am J Physiol.* 1996;270:F978–F985.
267. Sakairi Y, Jacobson HR, Noland TD, et al. Luminal prostaglandin E receptors regulate salt and water transport in rabbit cortical collecting duct. *Am J Physiol.* 1995;269(2 Pt 2):F257–F265.
268. Ushikubi F, Segi E, Sugimoto Y, et al. Impaired febrile response in mice lacking the prostaglandin E receptor subtype EP3. *Nature.* 1998;395(6699):281–284.
269. Audoly LP, Ruan X, Wagner VA, et al. Role of EP(2) and EP(3) PGE(2) receptors in control of murine renal hemodynamics. *Am J Physiol Heart Circ Physiol.* 2001;280(1):H327–H333.
270. van Rodijnen WF, Korstjens IJ, Legersteet N, et al. Direct vasoconstrictor effect of prostaglandin E2 on renal interlobular arteries: role of the EP3 receptor. *Am J Physiol Renal Physiol.* 2007;292(3):F1094–F1101.
271. Castleberry TA, Lu B, Smock SL, et al. Molecular cloning and functional characterization of the canine prostaglandin E(2) receptor EP4 subtype. *Prostaglandins.* 2001;65(4):167–187.
272. Yokoyama U, Iwatsubo K, Umemura M, et al. The prostanoid EP4 receptor and its signaling pathway. *Pharmacol Rev.* 2013;65(3):1010–1052.
273. Bastien L, Sawyer N, Grygorczyk R, et al. Cloning, functional expression, and characterization of the human prostaglandin E₂ receptor EP2 subtype. *J Biol Chem.* 1994;269:11873–11877.
274. Li JH, Chou CL, Li B, et al. A selective EP4 PGE2 receptor agonist alleviates disease in a new mouse model of X-linked nephrogenic diabetes insipidus. *J Clin Invest.* 2009;119(10):3115–3126.
275. Breyer RM, Davis LS, Nian C, et al. Cloning and expression of the rabbit prostaglandin EP4 receptor. *Am J Physiol.* 1996;270(3 Pt 2):F485–F493.
- 275a. Schneider A, Guan YF, Zhang YF, et al. Generation of a conditional allele of the mouse prostaglandin EP4 receptor. *Genesis.* 2004;40:7–14.
276. Segi E, Sugimoto Y, Yamasaki A, et al. Patent ductus arteriosus and neonatal death in prostaglandin receptor EP4-deficient mice. *Biochem Biophys Res Commun.* 1998;246(1):7–12.
277. Audoly LP, Tilley SL, Goulet J, et al. Identification of specific EP receptors responsible for the hemodynamic effects of PGE2. *Am J Physiol.* 1999;277(3 Pt 2):H924–H930.
278. Csukas S, Hanke C, Rewolinski D, et al. Prostaglandin E2-induced aldosterone release is mediated by an EP2 receptor. *Hypertension.* 1998;31:575–581.
279. Sugimoto Y, Namba T, Shigemoto R, et al. Distinct cellular localization of mRNAs for three subtypes of prostaglandin E receptor in kidney. *Am J Physiol.* 1994;266:F823–F828.
280. Jensen BL, Stubbe J, Hansen PB, et al. Localization of prostaglandin E(2) EP2 and EP4 receptors in the rat kidney. *Am J Physiol Renal Physiol.* 2001;280(6):F1001–F1009.
281. Nusing RM, Treude A, Weissenberger C, et al. Dominant role of prostaglandin E2 EP4 receptor in furosemide-induced salt-losing tubulopathy: a model for hyperprostaglandin E syndrome/antenatal Bartter syndrome. *J Am Soc Nephrol.* 2005;16(8):2354–2362.
282. Faour WH, Gomi K, Kennedy CR. PGE(2) induces COX-2 expression in podocytes via the EP(4) receptor through a PKA-independent mechanism. *Cell Signal.* 2008;20(11):2156–2164.
283. Kopp UC, Cicha MZ, Nakamura K, et al. Activation of EP4 receptors contributes to prostaglandin E2-mediated stimulation of renal sensory nerves. *Am J Physiol Renal Physiol.* 2004;287(6):F1269–F1282.
284. Sildorf E, Yang S, Pallone T. Prostaglandin E2 abrogates endothelin-induced vasoconstriction in renal outer medullary descending vasa recta of the rat. *J Clin Invest.* 1995;95:2734–2740.
285. Breyer M, Breyer R, Fowler B, et al. EP1 receptor antagonists block PGE2 dependent inhibition of Na⁺ absorption in the cortical collecting duct. *J Am Soc Nephrol.* 1996;7(9):1645.
286. Schlondorff D. Renal complications of nonsteroidal anti-inflammatory drugs. *Kidney Int.* 1993;44:643–653.
287. Jensen B, Schmid C, Kurtz A. Prostaglandins stimulate renin secretion and renin mRNA in mouse renal juxtaglomerular cells. *Am J Physiol.* 1996;271:F659–F669.
288. Hockel G, Cowley A. Prostaglandin E2-induced hypertension in conscious dogs. *Am J Physiol.* 1979;237:H449–H454.
289. Francisco L, Osborn J, Dibona G. Prostaglandins in renin release during sodium deprivation. *Am J Physiol.* 1982;243:F537–F542.
290. Imig JD, Breyer MD, Breyer RM. Contribution of prostaglandin EP(2) receptors to renal microvascular reactivity in mice. *Am J Physiol Renal Physiol.* 2002;283(3):F415–F422.
291. Schnermann J. Cyclooxygenase-2 and macula densa control of renin secretion. *Nephrol Dial Transplant.* 2001;16(9):1735–1738.
292. Imanishi M, Tsuji T, Nakamura S, et al. Prostaglandin i(2)/e(2) ratios in unilateral renovascular hypertension of different severities. *Hypertension.* 2001;38(1):23–29.
293. Jensen BL, Mann B, Skott O, et al. Differential regulation of renal prostaglandin receptor mRNAs by dietary salt intake in the rat. *Kidney Int.* 1999;56(2):528–537.
294. Tilley SL, Audoly LP, Hicks EH, et al. Reproductive failure and reduced blood pressure in mice lacking the EP2 prostaglandin E2 receptor. *J Clin Invest.* 1999;103(11):1539–1545.
295. Guyton A. Blood pressure control-special role of the kidneys and body fluids. *Science.* 1991;252:1813–1816.
296. Carmines P, Bell P, Roman R, et al. Prostaglandins in the sodium excretory response to altered renal arterial pressure in dogs. *Am J Physiol.* 1985;248:F8–F14.
297. Roman R, Lianos E. Influence of prostaglandins on papillary blood flow and pressure-natriuretic response. *Hypertension.* 1990;15:29–35.

298. Pallone TL, Sildorff EP. Pericyte regulation of renal medullary blood flow. *Exp Nephrol*. 2001;9(3):165–170.
299. Breyer MD, Breyer RM. Prostaglandin E receptors and the kidney. *Am J Physiol Renal Physiol*. 2000;279(1):F12–F23.
300. Roman RJ. P-450 metabolites of arachidonic acid in the control of cardiovascular function. *Physiol Rev*. 2002;82(1):131–185.
301. Syal A, Schiavi S, Chakravarty S, et al. Fibroblast growth Factor-23 increases mouse PGE2 production in vivo and in vitro. *Am J Physiol Renal Physiol*. 2005.
302. Baum M, Loleh S, Saini N, et al. Correction of proximal tubule phosphate transport defect in Hyp mice in vivo and in vitro with indomethacin. *Proc Natl Acad Sci USA*. 2003;100(19):11098–11103.
303. Eriksson LO, Larsson B, Andersson KE. Biochemical characterization and autoradiographic localization of [3H]PGE2 binding sites in rat kidney. *Acta Physiol Scand*. 1990;139(3):405–415.
304. Hebert RL, Jacobson HR, Breyer MD. Prostaglandin E2 inhibits sodium transport in rabbit cortical collecting duct by increasing intracellular calcium. *J Clin Invest*. 1991;87(6):1992–1998.
305. Breyer MD, Jacobson HR, Hebert RL. Cellular mechanisms of prostaglandin E2 and vasopressin interactions in the collecting duct. *Kidney Int*. 1990;38(4):618–624.
- 305a. Grantham JJ, Orloff J. Effect of prostaglandin E1 on the permeability response of the isolated collecting tubule to vasopressin, adenosine 3',5'-monophosphate, and theophylline. *J Clin Invest*. 1968;47(5):1154–1161.
306. Nadler SP, Hebert SC, Brenner BM. PGE2, forskolin, and cholera toxin interactions in rabbit cortical collecting tubule. *Am J Physiol*. 1986;250:F127–F135.
307. Hebert RL, Jacobson HR, Breyer MD. PGE2 inhibits AVP-induced water flow in cortical collecting ducts by protein kinase C activation. *Am J Physiol*. 1990;259:F318–F325.
308. Gao M, Cao R, Du S, et al. Disruption of prostaglandin E2 receptor EP4 impairs urinary concentration via decreasing aquaporin 2 in renal collecting ducts. *Proc Natl Acad Sci USA*. 2015;112(27):8397–8402.
309. Tai HH, Ensor CM, Tong M, et al. Prostaglandin catabolizing enzymes. *Prostaglandins Other Lipid Mediat*. 2002;68-69:483–493.
310. Sakuma S, Fujimoto Y, Hikita E, et al. Effects of metal ions on 15-hydroxy prostaglandin dehydrogenase activity in rabbit kidney cortex. *Prostaglandins*. 1990;40(5):507–514.
311. Yao B, Xu J, Harris RC, et al. Renal localization and regulation of 15-hydroxyprostaglandin dehydrogenase. *Am J Physiol Renal Physiol*. 2008;294(2):F433–F439.
312. Coggins KG, Latour A, Nguyen MS, et al. Metabolism of PGE2 by prostaglandin dehydrogenase is essential for remodeling the ductus arteriosus. *Nat Med*. 2002;8(2):91–92.
313. Oliw E. Oxygenation of polyunsaturated fatty acids by cytochrome P450 monooxygenases. *Prog Lipid Res*. 1994;33(3):329–354.
314. Schwartzman ML, da Silva JL, Lin F, et al. Cytochrome P450 4A expression and arachidonic acid omega-hydroxylation in the kidney of the spontaneously hypertensive rat. *Nephron*. 1996;73(4):652–663.
315. Stec DE, Flasch A, Roman RJ, et al. Distribution of cytochrome P-450 4A and 4F isoforms along the nephron in mice. *Am J Physiol Renal Physiol*. 2003;284(1):F95–F102.
316. Yu K, Bayona W, Kallen CB, et al. Differential activation of peroxisome proliferator activated receptors by eicosanoids. *J Biol Chem*. 1995;270(41):23975–23983.
317. Forman B, Tontonoz P, Chen J, et al. 15-deoxy- Δ 12,14-Prostaglandin J2 is a ligand for the adipocyte determination factor PPAR- γ . *Cell*. 1995;83:803–812.
318. Kliewer S, Lenhard J, Wilson T, et al. A prostaglandin J₂ metabolite binds peroxisome proliferator-activated receptor gamma and promotes adipocyte differentiation. *Cell*. 1995;83:813–819.
319. Witzensbichler B, Asahara T, Murohara T, et al. Vascular endothelial growth factor-C (VEGF-C/VEGF-2) promotes angiogenesis in the setting of tissue ischemia. *Am J Pathol*. 1998;153(2):381–394.
320. Hata AN, Zent R, Breyer MD, et al. Expression and molecular pharmacology of the mouse CRTH2 receptor. *J Pharmacol Exp Ther*. 2003;306(2):463–470.
321. Stamatakis K, Sanchez-Gomez FJ, Perez-Sala D. Identification of novel protein targets for modification by 15-Deoxy-[Delta]12,14-Prostaglandin J2 in mesangial cells reveals multiple interactions with the cytoskeleton. *J Am Soc Nephrol*. 2006;17(1):89–98.
322. Straus DS, Glass CK. Cyclopentenone prostaglandins: new insights on biological activities and cellular targets. *Med Res Rev*. 2001;21(3):185–210.
323. Negishi M, Katoh H. Cyclopentenone prostaglandin receptors. *Prostaglandins Other Lipid Mediat*. 2002;68-69:611–617.
324. Rossl A, Kapahl P, Natoli G, et al. Anti-inflammatory cyclopentenone prostaglandins are direct inhibitors of I κ B kinase. *Nature*. 2000;403:103–108.
325. Shibata T, Kondo M, Osawa T, et al. 15-Deoxy-Delta 12,14-prostaglandin J2. A prostaglandin D2 metabolite generated during inflammatory processes. *J Biol Chem*. 2002;277(12):10459–10466.
326. Fam SS, Murphey LJ, Terry ES, et al. Formation of highly reactive A-ring and J-ring Isoprostane-like compounds (A4/J4-neuroprostanes) in vivo from docosahexaenoic acid. *J Biol Chem*. 2002;277(39):36076–36084.
327. Morrow JD, Hill KE, Burk RF, et al. A series of prostaglandin F2-like compounds are produced in vivo in humans by a non-cyclooxygenase, free radical-catalyzed mechanism. *Proc Natl Acad Sci USA*. 1990;87(23):9383–9387.
328. Roberts LJ 2nd, Morrow JD. Products of the isoprostane pathway: unique bioactive compounds and markers of lipid peroxidation. *Cell Mol Life Sci*. 2002;59(5):808–820.
329. Takahashi K, Kato T, Schreiner GF, et al. Essential fatty acid deficiency normalizes function and histology in rat nephrotoxic nephritis. *Kidney Int*. 1992;41(5):1245–1253.
330. Schuster VL. Prostaglandin transport. *Prostaglandins Other Lipid Mediat*. 2002;68-69:633–647.
331. Chan BS, Satriano JA, Pucci M, et al. Mechanism of prostaglandin E2 transport across the plasma membrane of HeLa cells and Xenopus oocytes expressing the prostaglandin transporter “PGT”. *J Biol Chem*. 1998;273(12):6689–6697.
332. Lu R, Kanai N, Bao Y, et al. Cloning, in vitro expression, and tissue distribution of a human prostaglandin transporter cDNA(hPGT). *J Clin Invest*. 1996;98(5):1142–1149.
333. Kanai N, Lu R, Satriano JA, et al. Identification and characterization of a prostaglandin transporter. *Science*. 1995;268(5212):866–869.
334. Bao Y, Pucci ML, Chan BS, et al. Prostaglandin transporter PGT is expressed in cell types that synthesize and release prostanoids. *Am J Physiol Renal Physiol*. 2002;282(6):F1103–F1110.
335. Chi Y, Khersonsky SM, Chang Y-T, et al. Identification of a new class of prostaglandin transporter inhibitors and characterization of their biological effects on prostaglandin E2 transport. *J Pharmacol Exp Ther*. 2006;316(3):1346–1350.
336. Nomura T, Chang HY, Lu R, et al. Prostaglandin signaling in the renal collecting duct: release, reuptake, and oxidation in the same cell. *J Biol Chem*. 2005;280(31):28424–28429.
337. Chi Y, Pucci ML, Schuster VL. Dietary salt induces transcription of the prostaglandin transporter gene in renal collecting ducts. *Am J Physiol Renal Physiol*. 2008;295(3):F765–F771.
338. Kimura H, Takeda M, Narikawa S, et al. Human organic anion transporters and human organic cation transporters mediate renal transport of prostaglandins. *J Pharmacol Exp Ther*. 2002;301(1):293–298.
339. Sauvage C, Holzinger H, Gekle M. Prostaglandin E2 inhibits its own renal transport by downregulation of organic anion transporters rOAT1 and rOAT3. *J Am Soc Nephrol*. 2006;17(1):46–53.
340. Touhey S, O'Connor R, Plunkett S, et al. Structure-activity relationship of indomethacin analogues for MRP-1, COX-1 and COX-2 inhibition. identification of novel chemotherapeutic drug resistance modulators. *Eur J Cancer*. 2002;38(12):1661–1670.
341. Homem de Bittencourt PI Jr, Curi R. Antiproliferative prostaglandins and the MRP/GS-X pump role in cancer immunosuppression and insight into new strategies in cancer gene therapy. *Biochem Pharmacol*. 2001;62(7):811–819.
342. Jedlitschky G, Keppler D. Transport of leukotriene C4 and structurally related conjugates. *Vitam Horm*. 2002;64:153–184.
343. Nies AT, König J, Cui Y, et al. Structural requirements for the apical sorting of human multidrug resistance protein 2 (ABCC2). *Eur J Biochem*. 2002;269(7):1866–1876.
344. Van Aubel RA, Peters JG, Masereeuw R, et al. Multidrug resistance protein mrp2 mediates ATP-dependent transport of classic renal organic anion p-aminohippurate. *Am J Physiol Renal Physiol*. 2000;279(4):F713–F717.
345. Klahr S, Morrissey JJ. The role of growth factors, cytokines, and vasoactive compounds in obstructive nephropathy. *Semin Nephrol*. 1998;18(6):622–632.
346. Channugam P, Feng L, Liou S, et al. Radicol, a protein tyrosine kinase inhibitor, suppresses the expression of mitogen-inducible cyclooxygenase in macrophages stimulated with lipopolysaccharide

- and in experimental glomerulonephritis. *J Biol Chem.* 1995;270(10):5418–5426.
347. Hirose S, Yamamoto T, Feng L, et al. Expression and localization of cyclooxygenase isoforms and cytosolic phospholipase A2 in anti-Thy-1 glomerulonephritis. *J Am Soc Nephrol.* 1998;9(3):408–416.
 348. Yang T, Sun D, Huang YG, et al. Differential regulation of COX-2 expression in the kidney by lipopolysaccharide: role of CD14. *Am J Physiol.* 1999;277(1 Pt 2):F10–F16.
 349. Tomasoni S, Noris M, Zappella S, et al. Upregulation of renal and systemic cyclooxygenase-2 in patients with active lupus nephritis. *J Am Soc Nephrol.* 1998;9(7):1202–1212.
 350. Zoja C, Benigni A, Noris M, et al. Mycophenolate mofetil combined with a cyclooxygenase-2 inhibitor ameliorates murine lupus nephritis. *Kidney Int.* 2001;60(2):653–663.
 351. Nagamatsu T, Imai H, Yokoi M, et al. Protective effect of prostaglandin EP4-receptor agonist on anti-glomerular basement membrane antibody-associated nephritis. *J Pharmacol Sci.* 2006;102(2):182–188.
 352. Hartner A, Pahl A, Brune K, et al. Upregulation of cyclooxygenase-1 and the PGE2 receptor EP2 in rat and human mesangioproliferative glomerulonephritis. *Inflamm Res.* 2000;49(7):345–354.
 353. Kitahara M, Eitner F, Ostendorf T, et al. Selective cyclooxygenase-2 inhibition impairs glomerular capillary healing in experimental glomerulonephritis. *J Am Soc Nephrol.* 2002;13(5):1261–1270.
 354. Schneider A, Harendza S, Zahner G, et al. Cyclooxygenase metabolites mediate glomerular monocyte chemoattractant protein-1 formation and monocyte recruitment in experimental glomerulonephritis [see comments]. *Kidney Int.* 1999;55(2):430–441.
 355. Wang J-L, Cheng H-F, Zhang M-Z, et al. Selective increase of cyclooxygenase-2 expression in a model of renal ablation. *Am J Physiol.* 1998;275:F613–F622.
 356. Weichert W, Paliege A, Provoost AP, et al. Upregulation of juxtaglomerular NOS1 and COX-2 precedes glomerulosclerosis in fawn-hooded hypertensive rats. *Am J Physiol Renal Physiol.* 2001;280(4):F706–F714.
 357. Cheng H, Fan X, Guan Y, et al. Distinct roles for basal and induced COX-2 in podocyte injury. *J Am Soc Nephrol.* 2009;20(9):1953–1962.
 358. Jo YI, Cheng H, Wang S, et al. Puromycin induces reversible proteinuric injury in transgenic mice expressing cyclooxygenase-2 in podocytes. *Nephron Exp Nephrol.* 2007;107(3):e87–e94.
 359. Cheng H, Wang S, Jo YI, et al. Overexpression of cyclooxygenase-2 predisposes to podocyte injury. *J Am Soc Nephrol.* 2007;18(2):551–559.
 360. Komers R, Lindsley JN, Oyama TT, et al. Immunohistochemical and functional correlations of renal cyclooxygenase-2 in experimental diabetes. *J Clin Invest.* 2001;107(7):889–898.
 361. Bing Y, Xu J, Qi Z, et al. The role of renal cortical cyclooxygenase-2 (COX-2) expression in hyperfiltration in rats with high protein intake. *Am J Physiol Renal Physiol.* 2006.
 362. Vogt L, de Zeeuw D, Woittiez AJ, et al. Selective cyclooxygenase-2 (COX-2) inhibition reduces proteinuria in renal patients. *Nephrol Dial Transplant.* 2009;24(4):1182–1189.
 363. Wilcox CS, Welch WJ, Murad F, et al. Nitric oxide synthase in macula densa regulates glomerular capillary pressure. *Proc Natl Acad Sci USA.* 1992;89(24):11993–11997.
 364. Welch WJ, Wilcox CS, Thomson SC. Nitric oxide and tubuloglomerular feedback. *Semin Nephrol.* 1999;19(3):251–262.
 365. Thorup C, Erik A, Persson G. Macula densa derived nitric oxide in regulation of glomerular capillary pressure. *Kidney Int.* 1996;49(2):430–436.
 366. Wang JL, Cheng HF, Shappell S, et al. A selective cyclooxygenase-2 inhibitor decreases proteinuria and retards progressive renal injury in rats. *Kidney Int.* 2000;57(6):2334–2342.
 367. Goncalves AR, Fujihara CK, Mattar AL, et al. Renal expression of COX-2, ANG II, and AT1 receptor in remnant kidney: strong renoprotection by therapy with losartan and a nonsteroidal anti-inflammatory. *Am J Physiol Renal Physiol.* 2004;286(5):F945–F954.
 368. Fujihara CK, Malheiros DM, Donato JL, et al. Nitroflurbiprofen, a new nonsteroidal anti-inflammatory, ameliorates structural injury in the remnant kidney. *Am J Physiol.* 1998;274(3 Pt 2):F573–F579.
 369. Wang LH, Kulmacz RJ. Thromboxane synthase: structure and function of protein and gene. *Prostaglandins Other Lipid Mediat.* 2002;68-69:409–422.
 370. Cheng HF, Wang CJ, Moeckel GW, et al. Cyclooxygenase-2 inhibitor blocks expression of mediators of renal injury in a model of diabetes and hypertension. *Kidney Int.* 2002;62(3):929–939.
 371. Dey A, Maric C, Kaesemeyer WH, et al. Rofecoxib decreases renal injury in obese Zucker rats. *Clin Sci.* 2004;107(6):561–570.
 372. Schmitz PG, Krupa SM, Lane PH, et al. Acquired essential fatty acid depletion in the remnant kidney: amelioration with U-63557A. *Kidney Int.* 1994;46(4):1184–1191.
 373. Takano T, Cybulsky AV. Complement C5b-9-mediated arachidonic acid metabolism in glomerular epithelial cells: role of cyclooxygenase-1 and -2. *Am J Pathol.* 2000;156(6):2091–2101.
 374. Villa E, Martinez J, Ruilope L, et al. Cicaprost, a prostacyclin analog, protects renal function in uninephrectomized dogs in the absence of changes in blood pressure. *Am J Hypertension.* 1992;6:253–257.
 375. Sankaran D, Bankovic-Calic N, Ogborn MR, et al. 2 inhibition markedly slows disease progression and attenuates altered prostanoid production in Han:SPRD-cy rats with inherited kidney disease. *Am J Physiol Renal Physiol.* 2007;293(3):F821–F830.
 376. Elberg G, Elberg D, Lewis TV, et al. EP2 receptor mediates PGE2-induced cystogenesis of human renal epithelial cells. *Am J Physiol Renal Physiol.* 2007;293(5):F1622–F1632.
 377. Studer R, Negrete H, Craven P, et al. Protein kinase C signals thromboxane induced increases in fibronectin synthesis and TGF-beta bioactivity in mesangial cells. *Kidney Int.* 1995;48:422–430.
 378. Zahner G, Disser M, Thaiss F, et al. The effect of prostaglandin E2 on mRNA expression and secretion of collagens I, III, and IV and fibronectin in cultured rat mesangial cells. *J Am Soc Nephrol.* 1994;4:1778–1785.
 379. Singhal P, Sagar S, Garg P, et al. Vasoactive agents modulate matrix metalloproteinase-2 activity by mesangial cells. *Am J Med Sci.* 1995;310:235–241.
 380. Varga J, Diaz-Perez A, Rosenbloom J, et al. PGE2 causes a coordinate decrease in the steady state levels of fibronectin and types I and III procollagen mRNAs in normal human dermal fibroblasts. *Biochem Biophys Res Commun.* 1987;147:1282–1288.
 381. Willborn J, Crofford LJ, Burdick MD, et al. Cultured lung fibroblasts isolated from patients with idiopathic pulmonary fibrosis have a diminished capacity to synthesize prostaglandin E2 and to express cyclooxygenase-2. *J Clin Invest.* 1995;95(4):1861–1868.
 382. Wise WC, Cook JA, Tempel GE, et al. The rat in sepsis and endotoxic shock. *Prog Clin Biol Res.* 1989;299:243–252.
 383. Ruschitzka F, Shaw S, Noll G, et al. Endothelial vasoconstrictor prostanoids, vascular reactivity, and acute renal failure. *Kidney Int Suppl.* 1998;67:S199–S201.
 384. Chaudhari A, Kirschenbaum MA. Altered glomerular eicosanoid biosynthesis in uranyl nitrate-induced acute renal failure. *Biochim Biophys Acta.* 1984;792(2):135–140.
 385. Hardie WD, Ebert J, Frazer M, et al. The effect of thromboxane A2 receptor antagonism on amphotericin B-induced renal vasoconstriction in the rat. *Prostaglandins.* 1993;45(1):47–56.
 386. Higa EM, Schor N, Boim MA, et al. Role of the prostaglandin and kallikrein-kinin systems in aminoglycoside-induced acute renal failure. *Braz J Med Biol Res.* 1985;18(3):355–365.
 387. Papanicolaou N, Hatziantoniou C, Bariety J. Selective inhibition of thromboxane synthesis partially protected while inhibition of angiotensin II formation did not protect rats against acute renal failure induced with glycerol. *Prostaglandins Leukot Med.* 1986;21(1):29–35.
 388. Vargas AV, Krishnamurthi V, Masih R, et al. Prostaglandin E1 attenuation of ischemic renal reperfusion injury in the rat. *J Am Coll Surg.* 1995;180(6):713–717.
 389. Patel NS, Cuzzocrea S, Collino M, et al. The role of cyclooxygenase-2 in the rodent kidney following ischaemia/reperfusion injury in vivo. *Eur J Pharmacol.* 2007;562(1–2):148–154.
 390. Villanueva S, Cespedes C, Gonzalez AA, et al. Effect of ischemic acute renal damage on the expression of COX-2 and oxidative stress-related elements in rat kidney. *Am J Physiol Renal Physiol.* 2007;292(5):F1364–F1371.
 391. Feitoza CQ, Goncalves GM, Semedo P, et al. Inhibition of COX 1 and 2 prior to renal ischemia/reperfusion injury decreases the development of fibrosis. *Mol Med.* 2008;14(11–12):724–730.
 392. Hsu YH, Chen CH, Hou CC, et al. Prostacyclin protects renal tubular cells from gentamicin-induced apoptosis via a PPARalpha-dependent pathway. *Kidney Int.* 2008;73(5):578–587.
 393. Miyajima A, Ito K, Asano T, et al. Does cyclooxygenase-2 inhibitor prevent renal tissue damage in unilateral ureteral obstruction? *J Urol.* 2001;166(3):1124–1129.
 394. Ozturk H, Ozdemir E, Otcu S, et al. Renal effects on a solitary kidney of specific inhibition of cyclooxygenase-2 after 24 h of complete ureteric obstruction in rats. *Urol Res.* 2002;30(4):223–226.

395. Nakagawa N, Yuhki K, Kawabe J, et al. The intrinsic prostaglandin E2-EP4 system of the renal tubular epithelium limits the development of tubulointerstitial fibrosis in mice. *Kidney Int.* 2012;82(2):158–171.
396. Norregaard R, Jensen BL, Topcu SO, et al. COX-2 activity transiently contributes to increased water and NaCl excretion in the polyuric phase after release of ureteral obstruction. *Am J Physiol Renal Physiol.* 2007;292(5):F1322–F1333.
397. Coffman TM, Yarger WE, Klotman PE. Functional role of thromboxane production by acutely rejecting renal allografts in rats. *J Clin Invest.* 1985;75(4):1242–1248.
398. Tonshoff B, Busch C, Schweer H, et al. In vivo prostanoid formation during acute renal allograft rejection. *Nephrol Dial Transplant.* 1993;8(7):631–636.
399. Coffman TM, Yohay D, Carr DR, et al. Effect of dietary fish oil supplementation on eicosanoid production by rat renal allografts. *Transplantation.* 1988;45(2):470–474.
400. Coffman TM, Carr DR, Yarger WE, et al. Evidence that renal prostaglandin and thromboxane production is stimulated in chronic cyclosporine nephrotoxicity. *Transplantation.* 1987;43(2):282–285.
401. Hocherl K, Dreher F, Vitzthum H, et al. Cyclosporine A suppresses cyclooxygenase-2 expression in the rat kidney. *J Am Soc Nephrol.* 2002;13(10):2427–2436.
402. Laffi G, La Villa G, Pinzani M, et al. Arachidonic acid derivatives and renal function in liver cirrhosis. *Semin Nephrol.* 1997;17(6):530–548.
403. Lopez-Parra M, Claria J, Planaguma A, et al. Cyclooxygenase-1 derived prostaglandins are involved in the maintenance of renal function in rats with cirrhosis and ascites. *Br J Pharmacol.* 2002;135(4):891–900.
404. Bosch-Marce M, Claria J, Titos E, et al. Selective inhibition of cyclooxygenase 2 spares renal function and prostaglandin synthesis in cirrhotic rats with ascites. *Gastroenterology.* 1999;116(5):1167–1175.
405. Medina JF, Prieto J, Guarner F, et al. Effect of spironolactone on renal prostaglandin excretion in patients with liver cirrhosis and ascites. *J Hepatol.* 1986;3(2):206–211.
406. Epstein M, Lifschitz M. Renal eicosanoids as determinants of renal function in liver disease. *Hepatology.* 1987;7(6):1359–1367.
407. Moore K, Ward PS, Taylor GW, et al. Systemic and renal production of thromboxane A2 and prostacyclin in decompensated liver disease and hepatorenal syndrome. *Gastroenterology.* 1991;100(4):1069–1077.
408. Cheng H, Fan X, Moeckel GW, et al. Podocyte COX-2 exacerbates diabetic nephropathy by increasing podocyte (pro)renin receptor expression. *J Am Soc Nephrol.* 2011;22(7):1240–1251.
409. Komers R, Mar D, Denisenko O, et al. Epigenetic changes in renal genes dysregulated in mouse and rat models of type 1 diabetes. *Lab Invest.* 2013;93(5):543–552.
410. DeRubertis FR, Craven PA. Eicosanoids in the pathogenesis of the functional and structural alterations of the kidney in diabetes. *Am J Kidney Dis.* 1993;22(5):727–735.
411. Asakura J, Hasegawa H, Takayanagi K, et al. Renoprotective effect of pioglitazone by the prevention of glomerular hyperfiltration through the possible restoration of altered macula densa signaling in rats with type 2 diabetic nephropathy. *Nephron Exp Nephrol.* 2012;122(3–4):83–94.
412. Cherney DZ, Miller JA, Scholey JW, et al. Renal hyperfiltration is a determinant of endothelial function responses to cyclooxygenase 2 inhibition in type 1 diabetes. *Diabetes Care.* 2010;33(6):1344–1346.
413. Sasson AN, Cherney DZ. Renal hyperfiltration related to diabetes mellitus and obesity in human disease. *World J Diabetes.* 2012;3(1):1–6.
414. Yao B, Xu J, Qi Z, et al. Role of renal cortical cyclooxygenase-2 expression in hyperfiltration in rats with high-protein intake. *Am J Physiol Renal Physiol.* 2006;291(2):F368–F374.
415. Li J, Chen YJ, Quilley J. Effect of tempol on renal cyclooxygenase expression and activity in experimental diabetes in the rat. *J Pharmacol Exp Ther.* 2005;314(2):818–824.
416. Quilley J, Santos M, Pedraza P. Renal protective effect of chronic inhibition of COX-2 with SC-58236 in streptozotocin-diabetic rats. *Am J Physiol Heart Circ Physiol.* 2011;300(6):H2316–H2322.
417. Zhang MZ, Yao B, Yang S, et al. Intrarenal dopamine inhibits progression of diabetic nephropathy. *Diabetes.* 2012;61(10):2575–2584.
418. Makino H, Tanaka I, Mukoyama M, et al. Prevention of diabetic nephropathy in rats by prostaglandin E receptor EP1-selective antagonist. *J Am Soc Nephrol.* 2002;13(7):1757–1765.
419. Mohamed R, Jayakumar C, Ramesh G. Chronic administration of EP4-selective agonist exacerbates albuminuria and fibrosis of the kidney in streptozotocin-induced diabetic mice through IL-6. *Lab Invest.* 2013;93(8):933–945.
420. Wang X, Yao B, Wang Y, et al. Macrophage cyclooxygenase-2 protects against development of diabetic nephropathy. *Diabetes.* 2017;66(2):494–504.
421. Baylis C. Cyclooxygenase products do not contribute to the gestational renal vasodilation in the nitric oxide synthase inhibited pregnant rat. *Hypertens Pregnancy.* 2002;21(2):109–114.
422. Khalil RA, Granger JP. Vascular mechanisms of increased arterial pressure in preeclampsia: lessons from animal models. *Am J Physiol Regul Integr Comp Physiol.* 2002;283(1):R29–R45.
423. Okawara M, Seki H, Matsuoka K, et al. Examination of the expression of cyclooxygenase-2 in placenta villi from sufferers of pregnancy induced hypertension. *Biol Pharm Bull.* 2009;32(12):2053–2056.
424. Keith JC Jr, Thatcher CD, Schaub RG. Beneficial effects of U-63,557A, a thromboxane synthetase inhibitor, in an ovine model of pregnancy-induced hypertension. *Am J Obstet Gynecol.* 1987;157(1):199–203.
425. Klockenbusch W, Rath W. Prevention of pre-eclampsia by low-dose acetylsalicylic acid—a critical appraisal. *Z Geburtshilfe Neonatol.* 2002;206(4):125–130.
426. Heyborne KD. Preeclampsia prevention: lessons from the low-dose aspirin therapy trials. *Am J Obstet Gynecol.* 2000;183(3):523–528.
427. Rao R, Zhang MZ, Zhao M, et al. Lithium treatment inhibits renal GSK-3 activity and promotes cyclooxygenase 2-dependent polyuria. *Am J Physiol Renal Physiol.* 2005;288(4):F642–F649.
428. Kim B, Lee JH, Yang MS, et al. Retinoic acid enhances prostaglandin E2 production through increased expression of cyclooxygenase-2 and microsomal prostaglandin E synthase-1 in rat brain microglia. *J Neurosci Res.* 2008;86(6):1353–1360.
429. Virdis A, Colucci R, Fornai M, et al. Cyclooxygenase-2 inhibition improves vascular endothelial dysfunction in a rat model of endotoxic shock: role of inducible nitric-oxide synthase and oxidative stress. *J Pharmacol Exp Ther.* 2005;312(3):945–953.
430. Xu MZ, Lee WS, Han JM, et al. Antioxidant and anti-inflammatory activities of N-acetyldopamine dimers from periostracum cicadae. *Bioorg Med Chem.* 2006;14(23):7826–7834.
431. Marnett LJ, Kalutkar AS. Cyclooxygenase 2 inhibitors: discovery, selectivity and the future. *Trends Pharmacol Sci.* 1999;20(11):465–469.
432. Rockwell P, Martinez J, Papa L, et al. Redox regulates COX-2 upregulation and cell death in the neuronal response to cadmium. *Cell Signal.* 2004;16(3):343–353.
433. Kiritoshi S, Nishikawa T, Sonoda K, et al. Reactive oxygen species from mitochondria induce cyclooxygenase-2 gene expression in human mesangial cells: potential role in diabetic nephropathy. *Diabetes.* 2003;52(10):2570–2577.
434. Lee SH, Williams MV, Dubois RN, et al. Cyclooxygenase-2-mediated DNA damage. *J Biol Chem.* 2005;280(31):28337–28346.
435. Mouithys-Mickalad A, Deby-Dupont G, Dogne JM, et al. Effects of COX-2 inhibitors on ROS produced by Chlamydia pneumoniae-infected human promonocytic cells (THP-1). *Biochem Biophys Res Commun.* 2004;325(4):1122–1130.
436. Wilcox CS. Oxidative stress and nitric oxide deficiency in the kidney: a critical link to hypertension? *Am J Physiol Regul Integr Comp Physiol.* 2005;289(4):R913–R935.
437. Jaimes EA, Tian RX, Pearce D, et al. Up-regulation of glomerular COX-2 by angiotensin II: role of reactive oxygen species. *Kidney Int.* 2005;68(5):2143–2153.
438. Serezani CH, Chung J, Ballinger MN, et al. Prostaglandin E2 suppresses bacterial killing in alveolar macrophages by inhibiting NADPH oxidase. *Am J Respir Cell Mol Biol.* 2007;37(5):562–570.
439. Jia Z, Guo X, Zhang H, et al. Microsomal prostaglandin synthase-1-derived prostaglandin E2 protects against angiotensin II-induced hypertension via inhibition of oxidative stress. *Hypertension.* 2008;52(5):952–959.
440. Kim HJ, Kim KW, Yu BP, et al. The effect of age on cyclooxygenase-2 gene expression: NF-kappaB activation and IkappaBalpha degradation. *Free Radic Biol Med.* 2000;28(5):683–692.
441. Hermanz R, Briones AM, Salaices M, et al. New roles for old pathways? A circuitous relationship between reactive oxygen species and cyclo-oxygenase in hypertension. *Clin Sci.* 2014;126(2):111–121.
442. Reinhold SW, Vitzthum H, Filbeck T, et al. Gene expression of 5-, 12-, and 15-lipoxygenases and leukotriene receptors along the rat nephron. *Am J Physiol Renal Physiol.* 2006;290(4):F864–F872.
443. Claxson MR, McGinty A, Godson C, et al. Leukotrienes and lipoxins: lipoxygenase-derived modulators of leukocyte recruitment

- and vascular tone in glomerulonephritis. *Nephrol Dial Transplant*. 1998;13(12):3043–3051.
444. Dixon RA, Diehl RE, Opas E, et al. Requirement of a 5-lipoxygenase-activating protein for leukotriene synthesis. *Nature*. 1990;343(6255):282–284.
 445. Albrightson CR, Short B, Dytko G, et al. Selective inhibition of 5-lipoxygenase attenuates glomerulonephritis in the rat. *Kidney Int*. 1994;45(5):1301–1310.
 446. Lynch KR, O'Neill GP, Liu Q, et al. Characterization of the human cysteinyl leukotriene CysLT1 receptor. *Nature*. 1999;399(6738):789–793.
 447. Sarau HM, Ames RS, Chambers J, et al. Identification, molecular cloning, expression, and characterization of a cysteinyl leukotriene receptor. *Mol Pharmacol*. 1999;56(3):657–663.
 448. Hui Y, Funk CD. Cysteinyl leukotriene receptors. *Biochem Pharmacol*. 2002;64(11):1549–1557.
 449. Bigby TD. The yin and the yang of 5-lipoxygenase pathway activation. *Mol Pharmacol*. 2002;62(2):200–202.
 450. Hallstrand TS, Henderson WR Jr. Leukotriene modifiers. *Med Clin North Am*. 2002;86(5):1009–1033, vi.
 451. Yokomizo T, Kato K, Terawaki K, et al. A second leukotriene B₄ receptor, BLT2. A new therapeutic target in inflammation and immunological disorders. *J Exp Med*. 2000;192(3):421–432.
 452. Noiri E, Taguchi J, Nakao A, et al. MTHFR gene polymorphism as an exacerbation factor of diabetic nephropathy in type 2 diabetes. Analysis in Japanese male hemodialysis patients. *Diabetes Care*. 2000;23(2):260.
 453. Suzuki S, Kuroda T, Kazama JI, et al. The leukotriene B₄ receptor antagonist ONO-4057 inhibits nephrotoxic serum nephritis in WKY rats. *J Am Soc Nephrol*. 1999;10(2):264–270.
 454. Chiang N, Gronert K, Clish CB, et al. Leukotriene B₄ receptor transgenic mice reveal novel protective roles for lipoxins and aspirin-triggered lipoxins in reperfusion. *J Clin Invest*. 1999;104(3):309–316.
 455. Devchand P, Keller H, Peters J, et al. The PPAR α -leukotriene B₄ pathway to inflammation control. *Nature*. 1996;384:39–43.
 456. Badr KF. Glomerulonephritis: roles for lipoxygenase pathways in pathophysiology and therapy. *Curr Opin Nephrol Hypertens*. 1997;6(2):111–118.
 457. Papayianni A, Serhan CN, Brady HR. Lipoxin A₄ and B₄ inhibit leukotriene-stimulated interactions of human neutrophils and endothelial cells. *J Immunol*. 1996;156(6):2264–2272.
 458. Nassar GM, Badr KF. Role of leukotrienes and lipoxygenases in glomerular injury. *Miner Electrolyte Metab*. 1995;21(4–5):262–270.
 459. Katoh T, Takahashi K, DeBoer DK, et al. Renal hemodynamic actions of lipoxins in rats: a comparative physiological study. *Am J Physiol*. 1992;263(3 Pt 2):F436–F442.
 460. Brady HR, Lamas S, Papayianni A, et al. Lipoxygenase product formation and cell adhesion during neutrophil-glomerular endothelial cell interaction. *Am J Physiol*. 1995;268(1 Pt 2):F1–F12.
 461. Chiang N, Fierro IM, Gronert K, et al. Activation of lipoxin A₄ receptors by aspirin-triggered lipoxins and select peptides evokes ligand-specific responses in inflammation. *J Exp Med*. 2000;191(7):1197–1208.
 462. Pawloski JR, Chapnick BM. Leukotrienes C₄ and D₄ are potent endothelium-dependent relaxing agents in canine splanchnic venous capacitance vessels. *Circ Res*. 1993;73(2):395–404.
 463. Claria J, Lee MH, Serhan CN. Aspirin-triggered lipoxins (15-epi-LX) are generated by the human lung adenocarcinoma cell line (A549)-neutrophil interactions and are potent inhibitors of cell proliferation. *Mol Med*. 1996;2(5):583–596.
 464. Imig JD. Eicosanoid regulation of the renal vasculature. *Am J Physiol Renal Physiol*. 2000;279(6):F965–F981.
 465. Nie D, Lamberti M, Zacharek A, et al. Thromboxane A₂ regulation of endothelial cell migration, angiogenesis, and tumor metastasis. *Biochem Biophys Res Commun*. 2000;267(1):245–251.
 466. Ma J, Natarajan R, LaPage J, et al. 12/15-lipoxygenase inhibitors in diabetic nephropathy in the rat. *Prostaglandins Leukot Essent Fatty Acids*. 2005;72(1):13–20.
 467. Kim YS, Xu ZG, Reddy MA, et al. Novel interactions between TGF- β 1 actions and the 12/15-lipoxygenase pathway in mesangial cells. *J Am Soc Nephrol*. 2005;16(2):352–362.
 468. Imig JD, Deichmann PC. Afferent arteriolar responses to ANG II involve activation of PLA₂ and modulation by lipoxygenase and P-450 pathways. *Am J Physiol*. 1997;273(2 Pt 2):F274–F282.
 469. Wu JN, Edwards D, Berecek KH. Changes in renal angiotensin II receptors in spontaneously hypertensive rats by early treatment with the angiotensin-converting enzyme inhibitor captopril. *Hypertension*. 1994;23(6 Pt 2):819–822.
 470. Stern N, Nozawa K, Kisch E, et al. Tonic inhibition of renin secretion by the 12 lipoxygenase pathway: augmentation by high salt intake. *Endocrinology*. 1996;137(5):1878–1884.
 471. Antonipillai I, Nadler J, Vu EJ, et al. A 12-lipoxygenase product, 12-hydroxyeicosatetraenoic acid, is increased in diabetics with incipient and early renal disease. *J Clin Endocrinol Metab*. 1996;81(5):1940–1945.
 472. Brady HR, Papayianni A, Serhan CN. Transcellular pathways and cell adhesion as potential contributors to leukotriene and lipoxin biosynthesis in acute glomerulonephritis. *Adv Exp Med Biol*. 1997;631–640.
 473. Kang SW, Adler SG, Nast CC, et al. 12-lipoxygenase is increased in glucose-stimulated mesangial cells and in experimental diabetic nephropathy. *Kidney Int*. 2001;59(4):1354–1362.
 474. Rahman MA, Nakazawa M, Emancipator SN, et al. Increased leukotriene B₄ synthesis in immune injured rat glomeruli. *J Clin Invest*. 1988;81(6):1945–1952.
 475. Badr KF. Five-lipoxygenase products in glomerular immune injury. *J Am Soc Nephrol*. 1992;3(4):907–915.
 476. Papayianni A, Serhan CN, Phillips ML, et al. Transcellular biosynthesis of lipoxin A₄ during adhesion of platelets and neutrophils in experimental immune complex glomerulonephritis. *Kidney Int*. 1995;47(5):1295–1302.
 477. Katoh T, Lianos EA, Fukunaga M, et al. Leukotriene D₄ is a mediator of proteinuria and glomerular hemodynamic abnormalities in passive Heymann nephritis. *J Clin Invest*. 1993;91(4):1507–1515.
 478. Butterly DW, Spurney RF, Ruiz P, et al. A role for leukotrienes in cyclosporine nephrotoxicity. *Kidney Int*. 2000;57(6):2586–2593.
 479. Goulet JL, Griffiths RC, Ruiz P, et al. Deficiency of 5-lipoxygenase accelerates renal allograft rejection in mice. *J Immunol*. 2001;167(11):6631–6636.
 480. Fauler J, Wiemeyer A, Marx KH, et al. LTB₄ in nephrotoxic serum nephritis in rats. *Kidney Int*. 1989;36(1):46–50.
 481. Lianos EA. Synthesis of hydroxyeicosatetraenoic acids and leukotrienes in rat nephrotoxic serum glomerulonephritis. Role of anti-glomerular basement membrane antibody dose, complement, and neutrophils. *J Clin Invest*. 1988;82(2):427–435.
 482. Ferrante JV, Huang ZH, Nandopkar M, et al. Altered responses of human macrophages to lipopolysaccharide by hydroperoxy eicosatetraenoic acid, hydroxy eicosatetraenoic acid, and arachidonic acid. Inhibition of tumor necrosis factor production. *J Clin Invest*. 1997;99(6):1445–1452.
 483. Munger KA, Montero A, Fukunaga M, et al. Transfection of rat kidney with human 15-lipoxygenase suppresses inflammation and preserves function in experimental glomerulonephritis. *Proc Natl Acad Sci USA*. 1999;96(23):13375–13380.
 484. Guasch A, Zayas CF, Badr KF. MK-591 acutely restores glomerular size selectivity and reduces proteinuria in human glomerulonephritis. *Kidney Int*. 1999;56(1):261–267.
 485. Makita K, Takahashi K, Karara A, et al. Experimental and/or genetically controlled alterations of the renal microsomal cytochrome P450 epoxide synthase induce hypertension in rats fed a high salt diet. *J Clin Invest*. 1994;94(6):2414–2420.
 486. Yared A, Albrightson-Winslow C, Griswold D, et al. Functional significance of leukotriene B₄ in normal and glomerulonephritic kidneys. *J Am Soc Nephrol*. 1991;2(1):45–56.
 487. Patel NS, Cuzzocrea S, Chatterjee PK, et al. Reduction of renal ischemia-reperfusion injury in 5-lipoxygenase knockout mice and by the 5-lipoxygenase inhibitor zileuton. *Mol Pharmacol*. 2004;66(2):220–227.
 488. Xu ZG, Miao LN, Cui YC, et al. Angiotensin II type 1 receptor expression is increased via 12-lipoxygenase in high glucose-stimulated glomerular cells and type 2 diabetic glomeruli. *Nephrol Dial Transplant*. 2009;24(6):1744–1752.
 489. Xu HZ, Wang WN, Zhang YY, et al. Effect of angiotensin II type 1 receptor blocker on 12-lipoxygenase activity and slit diaphragm protein expression in type 2 diabetic rat glomeruli. *J Nephrol*. 2016;29(6):775–782.
 490. Faulkner J, Pye C, Al-Shabraway M, et al. Inhibition of 12/15-Lipoxygenase reduces renal inflammation and injury in streptozotocin-induced diabetic mice. *J Diabetes Metab*. 2015;6(6).
 491. Yuan H, Reddy MA, Deshpande S, et al. Epigenetic histone modifications involved in profibrotic gene regulation by 12/15-Lipoxygenase

- and its oxidized lipid products in diabetic nephropathy. *Antioxid Redox Signal*. 2016;24(7):361–375.
492. Alkhamees OA, Alroujaye AS, Abuhashish HM, et al. Possible involvement of the lipoxygenase and leukotriene signaling pathways in cisplatin-mediated renal toxicity. *Cancer Chemother Pharmacol*. 2017;80(1):55–64.
 493. Reinhold SW, Scherl T, Stolcker B, et al. Lipoxygenase products in the urine correlate with renal function and body temperature but not with acute transplant rejection. *Lipids*. 2013;48(2):167–175.
 494. Aukema HM, Lu J, Borthwick F, et al. Dietary fish oil reduces glomerular injury and elevated renal hydroxyeicosatetraenoic acid levels in the JCR:LA-cp rat, a model of the metabolic syndrome. *Br J Nutr*. 2013;110(1):11–19.
 495. Vilaysane A, Chun J, Seamone ME, et al. The NLRP3 inflammasome promotes renal inflammation and contributes to CKD. *J Am Soc Nephrol*. 2010;21(10):1732–1744.
 496. Li G, Chen Z, Bhat OM, et al. NLRP3 inflammasome as a novel target for docosahexaenoic acid metabolites to abrogate glomerular injury. *J Lipid Res*. 2017;58(6):1080–1090.
 497. Capdevila JH, Harris RC, Falck JR. Microsomal cytochrome P450 and eicosanoid metabolism. *Cell Mol Life Sci*. 2002;59(5):780–789.
 498. Capdevila JH, Falck JR, Harris RC. Cytochrome P450 and arachidonic acid bioactivation: molecular and functional properties of the arachidonate monooxygenase. *J Lipid Res*. 2000;41(2):163–181.
 499. Capdevila JH, Falck JR. Biochemical and molecular characteristics of the cytochrome P450 arachidonic acid monooxygenase. *Prostaglandins Other Lipid Mediat*. 2000;62(3):271–292.
 500. Imig JD, Pham BT, LeBlanc EA, et al. Cytochrome P450 and cyclooxygenase metabolites contribute to the endothelin-1 afferent arteriolar vasoconstrictor and calcium responses. *Hypertension*. 2000;35(1 Pt 2):307–312.
 501. McGiff JC, Quilley J. 20-HETE and the kidney: resolution of old problems and new beginnings. *Am J Physiol*. 1999;277(3 Pt 2):R607–R623.
 502. Imig JD. Epoxygenase metabolites. Epithelial and vascular actions. *Mol Biotechnol*. 2000;16(3):233–251.
 503. Ma J, Qu W, Scarborough PE, et al. Molecular cloning, enzymatic characterization, developmental expression, and cellular localization of a mouse cytochrome P450 highly expressed in kidney. *J Biol Chem*. 1999;274(25):17777–17788.
 504. Ito O, Alonso-Galicia M, Hopp KA, et al. Localization of cytochrome p-450 4A isoforms along the rat nephron. *Am J Physiol*. 1998;274:F395–F404.
 505. Yokose T, Doy M, Taniguchi T, et al. Immunohistochemical study of cytochrome P450 2C and 3A in human non-neoplastic and neoplastic tissues. *Virchows Arch*. 1999;434(5):401–411.
 506. Imig JD, Zou AP, Stec DE, et al. Formation and actions of 20-hydroxyeicosatetraenoic acid in rat renal arterioles. *Am J Physiol*. 1996;270(1 Pt 2):R217–R227.
 507. Ito O, Roman RJ. Role of 20-HETE in elevating chloride transport in the thick ascending limb of Dahl SS/Jr rats. *Hypertension*. 1999;33(1 Pt 2):419–423.
 508. Quigley R, Baum M, Reddy KM, et al. Effects of 20-HETE and 19(S)-HETE on rabbit proximal straight tubule volume transport. *Am J Physiol Renal Physiol*. 2000;278(6):F949–F953.
 509. Hirt DL, Capdevila J, Falck JR, et al. Cytochrome P450 metabolites of arachidonic acid are potent inhibitors of vasopressin action on rabbit cortical collecting duct. *J Clin Invest*. 1989;84(6):1805–1812.
 510. Makita K, Falck J, Capdevila J. Cytochrome P450, the arachidonic acid cascade, and hypertension: new vistas for an old enzyme system. *FASEB J*. 1996;10:1456–1463.
 511. Chen JK, Chen J, Imig JD, et al. Identification of novel endogenous cytochrome p450 arachidonate metabolites with high affinity for cannabinoid receptors. *J Biol Chem*. 2008;283(36):24514–24524.
 512. Gangadhariah MH, Luther JM, Garcia V, et al. Hypertension is a major contributor to 20-hydroxyeicosatetraenoic acid-mediated kidney injury in diabetic nephropathy. *J Am Soc Nephrol*. 2015;26(3):597–610.
 513. Alonso-Galicia M, Sun CW, Falck JR, et al. Contribution of 20-HETE to the vasodilator actions of nitric oxide in renal arteries. *Am J Physiol*. 1998;275(3 Pt 2):F370–F378.
 514. Sun CW, Alonso-Galicia M, Taheri MR, et al. Nitric oxide-20-hydroxyeicosatetraenoic acid interaction in the regulation of K⁺ channel activity and vascular tone in renal arterioles. *Circ Res*. 1998;83(11):1069–1079.
 515. Imig JD, Navar LG, Roman RJ, et al. Actions of epoxygenase metabolites on the preglomerular vasculature. *J Am Soc Nephrol*. 1996;7(11):2364–2370.
 516. Katoh T, Takahashi K, Capdevila J, et al. Glomerular stereospecific synthesis and hemodynamic actions of 8,9-epoxyeicosatrienoic acid in rat kidney. *Am J Physiol*. 1991;261:F578–F586.
 517. Campbell WB, Gauthier KM. What is new in endothelium-derived hyperpolarizing factors? *Curr Opin Nephrol Hypertens*. 2002;11(2):177–183.
 518. Vazquez B, Rios A, Escalante B. Arachidonic acid metabolism modulates vasopressin-induced renal vasoconstriction. *Life Sci*. 1995;56(18):1455–1466.
 519. Oyekan AO, McAward K, Conetta J, et al. Endothelin-1 and CYP450 arachidonate metabolites interact to promote tissue injury in DOCA-salt hypertension. *Am J Physiol*. 1999;276(3 Pt 2):R766–R775.
 520. Arima S, Endo Y, Yaoita H, et al. Possible role of P-450 metabolite of arachidonic acid in vasodilator mechanism of angiotensin II type 2 receptor in the isolated microperfused rabbit afferent arteriole. *J Clin Invest*. 1997;100(11):2816–2823.
 521. Alonso-Galicia M, Maier KG, Greene AS, et al. Role of 20-hydroxyeicosatetraenoic acid in the renal and vasoconstrictor actions of angiotensin II. *Am J Physiol Regul Integr Comp Physiol*. 2002;283(1):R60–R68.
 522. Imig JD, Inscho EW, Deichmann PC, et al. Afferent arteriolar vasodilation to the sulfonimide analog of 11, 12- epoxyeicosatrienoic acid involves protein kinase A. *Hypertension*. 1999;33(1 Pt 2):408–413.
 523. Zou A, Drummond H, Roman R. Role of 20-HETE in elevating loop diuretic reabsorption in Dahl SS/Jr rats. *Hypertension*. 1996;27:631–635.
 524. Wang H, Garvin JL, Falck JR, et al. Glomerular cytochrome P-450 and cyclooxygenase metabolites regulate efferent arteriole resistance. *Hypertension*. 2005;46(5):1175–1179.
 525. Muroya Y, Fan F, Regner KR, et al. Deficiency in the formation of 20-hydroxyeicosatetraenoic acid enhances renal ischemia-reperfusion injury. *J Am Soc Nephrol*. 2015;26(10):2460–2469.
 526. Schnermann J. Adenosine mediates tubuloglomerular feedback. *Am J Physiol Regul Integr Comp Physiol*. 2002;283(1):R276–R277, discussion R8–9.
 527. Zou AP, Imig JD, Kaldunski M, et al. Inhibition of renal vascular 20-HETE production impairs autoregulation of renal blood flow. *Am J Physiol*. 1994;266(2 Pt 2):F275–F282.
 528. Hercule HC, Oyekan AO. Cytochrome P450 omega/omega-1 hydroxylase-derived eicosanoids contribute to endothelin(A) and endothelin(B) receptor-mediated vasoconstriction to endothelin-1 in the rat preglomerular arteriole. *J Pharmacol Exp Ther*. 2000;292(3):1153–1160.
 529. Ren Y, Garvin JL, Liu R, et al. Cross-talk between arterioles and tubules in the kidney. *Pediatr Nephrol*. 2009;24(1):31–35.
 530. Henrich WL, Falck JR, Campbell WB. Inhibition of renin release by 14,15-epoxyeicosatrienoic acid in renal. *Am J Physiol*. 1990.
 531. Alonso-Galicia M, Falck JR, Reddy KM, et al. 20-HETE agonists and antagonists in the renal circulation. *Am J Physiol*. 1999;277(5 Pt 2):F790–F796.
 532. Romero MF, Madhup ZT, Hopfer U, et al. An epoxygenase metabolite of arachidonic acid 5,6 epoxy-eicosatrienoic acid mediates angiotensin-induced natriuresis in proximal tubular epithelium. *Adv Prostaglandin Thromboxane Leukot Res*. 1991.
 533. Escalante BA, Staudinger R, Schwartzman M, et al. Amiloride-sensitive ion transport inhibition by epoxyeicosatrienoic acids in renal epithelial cells. *Adv Prostaglandin Thromboxane Leukot Res*. 1995;23:207–209.
 534. Staudinger R, Escalante B, Schwartzman ML, et al. Effects of epoxyeicosatrienoic acids on 86Rb uptake in renal epithelial cells. *J Cell Physiol*. 1994;160(1):69–74.
 535. Nowicki S, Chen SL, Aizman O, et al. 20-Hydroxyeicosatetraenoic acid (20 HETE) activates protein kinase C. Role in regulation of rat renal Na⁺,K⁺-ATPase. *J Clin Invest*. 1997;99(6):1224–1230.
 536. Carroll M, Balazy M, Margiotta P, et al. Cytochrome p-450 dependent HETEs: profile of biological activity and stimulation by vasoactive peptides. *Am J Physiol*. 1996;271:R863–R869.
 537. Burns K, Capdevila J, Wei S, et al. Role of cytochrome P-450 epoxygenase metabolites in EGF signaling in renal proximal tubules. *Am J Physiol*. 1995;269:C831–C840.
 538. Zhang YB, Magyar CE, Holstein-Rathlou NH, et al. The cytochrome P-450 inhibitor cobalt chloride prevents inhibition of renal

- Na,K-ATPase and redistribution of apical NHE-3 during acute hypertension. *J Am Soc Nephrol*. 1998;9(4):531–537.
539. Zhang MZ, Wang Y, Yao B, et al. Role of epoxyeicosatrienoic acids (EETs) in mediation of dopamine's effects in the kidney. *Am J Physiol Renal Physiol*. 2013;305(12):F1680–F1686.
 540. Gu R, Wei Y, Jiang H, et al. Role of 20-HETE in mediating the effect of dietary K intake on the apical K channels in the mTAL. *Am J Physiol Renal Physiol*. 2001;280(2):F223–F230.
 541. Good DW, George T, Wang DH. Angiotensin II inhibits HCO₃ absorption via a cytochrome P-450-dependent pathway in MTAL. *Am J Physiol*. 1999;276(5 Pt 2):F726–F736.
 542. Grider JS, Falcone JC, Kilpatrick EL, et al. P450 arachidonate metabolites mediate bradykinin-dependent inhibition of NaCl transport in the rat thick ascending limb. *Can J Physiol Pharmacol*. 1997;75(2):91–96.
 543. Sakairi Y, Jacobson HR, Noland TD, et al. 5,6-EET inhibits ion transport in collecting duct by stimulating endogenous prostaglandin synthesis. *Am J Physiol*. 1995;268(5 Pt 2):F931–F939.
 544. Wei Y, Lin DH, Kemp R, et al. Arachidonic acid inhibits epithelial Na channel via cytochrome P450 (CYP) epoxygenase-dependent metabolic pathways. *J Gen Physiol*. 2004;124(6):719–727.
 545. Nakagawa K, Holla VR, Wei Y, et al. Salt-sensitive hypertension is associated with dysfunctional Cyp4a10 gene and kidney epithelial sodium channel. *J Clin Invest*. 2006;116(6):1696–1702.
 546. Capdevila JH, Pidkova N, Mei S, et al. The Cyp2c44 epoxygenase regulates epithelial sodium channel activity and the blood pressure responses to increased dietary salt. *J Biol Chem*. 2013.
 547. Capdevila JH, Pidkova N, Mei S, et al. The Cyp2c44 epoxygenase regulates epithelial sodium channel activity and the blood pressure responses to increased dietary salt. *J Biol Chem*. 2014;289(7):4377–4386.
 548. Pavlov TS, Ilatovskaya DV, Levchenko V, et al. Effects of cytochrome P-450 metabolites of arachidonic acid on the epithelial sodium channel (ENaC). *Am J Physiol Renal Physiol*. 2011;301(3):F672–F681.
 549. Pidkova N, Rao R, Mei S, et al. Epoxyeicosatrienoic acids (EETs) regulate epithelial sodium channel activity by extracellular signal-regulated kinase 1/2 (ERK1/2)-mediated phosphorylation. *J Biol Chem*. 2013;288(7):5223–5231.
 550. Nikolaeva S, Pradervand S, Centeno G, et al. The circadian clock modulates renal sodium handling. *J Am Soc Nephrol*. 2012;23(6):1019–1026.
 551. Harris RC, Homma T, Jacobson HR, et al. Epoxyeicosatrienoic acids activate Na⁺/H⁺ exchange and are mitogenic in cultured rat glomerular mesangial cells. *J Cell Physiol*. 1990;144:429–437.
 552. Chen JK, Capdevila J, Harris RC. Heparin-binding EGF-like growth factor mediates the biological effects of P450 arachidonate epoxygenase metabolites in epithelial cells. *Proc Natl Acad Sci USA*. 2002;99(9):6029–6034.
 553. Lin F, Rios A, Falck JR, et al. 20-Hydroxyeicosatetraenoic acid is formed in response to EGF and is a mitogen in rat proximal tubule. *Am J Physiol*. 1995;269(6 Pt 2):F806–F816.
 554. Uddin MR, Muthalif MM, Karzoun NA, et al. Cytochrome P-450 metabolites mediate norepinephrine-induced mitogenic signaling. *Hypertension*. 1998;31(1 Pt 2):242–247.
 555. Yang J, Cui Z, He D, et al. Insulin increases D5 dopamine receptor expression and function in renal proximal tubule cells from Wistar-Kyoto rats. *Am J Hypertens*. 2009;22(7):770–776.
 556. Hye Khan MA, Neckar J, Manthali V, et al. Orally active epoxyeicosatrienoic acid analog attenuates kidney injury in hypertensive Dahl salt-sensitive rat. *Hypertension*. 2013;62(5):905–913.
 557. Khan MA, Liu J, Kumar G, et al. Novel orally active epoxyeicosatrienoic acid (EET) analogs attenuate cisplatin nephrotoxicity. *FASEB J*. 2013;27(8):2946–2956.
 558. Panigrahy D, Kalish BT, Huang S, et al. Epoxyeicosanoids promote organ and tissue regeneration. *Proc Natl Acad Sci USA*. 2013;110(33):13528–13533.
 559. Hoff U, Lukitsch I, Chaykovska L, et al. Inhibition of 20-HETE synthesis and action protects the kidney from ischemia/reperfusion injury. *Kidney Int*. 2011;79(1):57–65.
 560. Chen G, Xu R, Wang Y, et al. Genetic disruption of soluble epoxide hydrolase is protective against streptozotocin-induced diabetic nephropathy. *Am J Physiol Endocrinol Metab*. 2012;303(5):E563–E575.
 561. Zhao G, Tu L, Li X, et al. Delivery of AAV2-CYP2J2 protects remnant kidney in the 5/6-nephrectomized rat via inhibition of apoptosis and fibrosis. *Hum Gene Ther*. 2012;23(7):688–699.
 562. Kim J, Imig JD, Yang J, et al. Inhibition of soluble epoxide hydrolase prevents renal interstitial fibrosis and inflammation. *Am J Physiol Renal Physiol*. 2014;307(8):F971–F980.
 563. Hye Khan MA, Fish B, Wahl G, et al. Epoxyeicosatrienoic acid analogue mitigates kidney injury in a rat model of radiation nephropathy. *Clin Sci*. 2016;130(8):587–599.
 564. Holla VR, Makita K, Zaphiropoulos PG, et al. The kidney cytochrome P-450 2C23 arachidonic acid epoxygenase is upregulated during dietary salt loading. *J Clin Invest*. 1999;104(6):751–760.
 565. Imig JD. Epoxide hydrolase and epoxygenase metabolites as therapeutic targets for renal diseases. *Am J Physiol Renal Physiol*. 2005;289(3):F496–F503.
 566. Muthalif MM, Benter IF, Khandekar Z, et al. Contribution of ras GTPase/MAP kinase and cytochrome P450 metabolites to deoxycorticosterone-salt-induced hypertension. *Hypertension*. 2000;35(1 Pt 2):457–463.
 567. Croft KD, McGiff JC, Sanchez-Mendoza A, et al. Angiotensin II releases 20-HETE from rat renal microvessels. *Am J Physiol Renal Physiol*. 2000;279(3):F544–F551.
 568. Joly E, Seqqat R, Flamion B, et al. Increased renal vascular reactivity to ANG II after unilateral nephrectomy in the rat involves 20-HETE. *Am J Physiol Regul Integr Comp Physiol*. 2006;291(4):R977–R986.
 569. Iwai N, Inagami T. Identification of a candidate gene responsible for the high blood pressure of spontaneously hypertensive rats. *J Hypertens*. 1992;10(10):1155–1157.
 570. Stec DE, Trollet MR, Krieger JE, et al. Renal cytochrome P4504A activity and salt sensitivity in spontaneously hypertensive rats. *Hypertension*. 1996;27(6):1329–1336.
 571. Wang MH, Zhang F, Marji J, et al. CYP4A1 antisense oligonucleotide reduces mesenteric vascular reactivity and blood pressure in SHR. *Am J Physiol Regul Integr Comp Physiol*. 2001;280(1):R255–R261.
 572. Gainer JV, Bellamine A, Dawson EP, et al. Functional variant of CYP4A11 20-hydroxyeicosatetraenoic acid synthase is associated with essential hypertension. *Circulation*. 2005;111(1):63–69.
 573. Imig JD, Zou AP, Ortiz de Montellano PR, et al. Cytochrome P-450 inhibitors alter afferent arteriolar responses to elevations in pressure. *Am J Physiol*. 1994;266(5 Pt 2):H1879–H1885.
 574. Muller C, Endlich K, Helwig JJ. AT2 antagonist-sensitive potentiation of angiotensin II-induced constriction by NO blockade and its dependence on endothelium and P450 eicosanoids in rat renal vasculature. *Br J Pharmacol*. 1998;124(5):946–952.
 575. Gross V, Schunck WH, Honeck H, et al. Inhibition of pressure natriuresis in mice lacking the AT2 receptor. *Kidney Int*. 2000;57(1):191–202.
 576. Fernandez MM, Gonzalez D, Williams JM, et al. Inhibitors of 20-hydroxyeicosatetraenoic acid (20-HETE) formation attenuate the natriuretic effect of dopamine. *Eur J Pharmacol*. 2012;686(1–3):97–103.
 577. Ai D, Fu Y, Guo D, et al. Angiotensin II up-regulates soluble epoxide hydrolase in vascular endothelium in vitro and in vivo. *Proc Natl Acad Sci USA*. 2007;104(21):9018–9023.
 578. Chiamvimonvat N, Ho CM, Tsai HJ, et al. The soluble epoxide hydrolase as a pharmaceutical target for hypertension. *J Cardiovasc Pharmacol*. 2007;50(3):225–237.

BOARD REVIEW QUESTIONS

1. Prostaglandins are produced from arachidonic acid by:
 - a. Lipoxygenases
 - b. Cytochrome P450
 - c. Cyclooxygenases
 - d. Soluble epoxide hydrolases
 - e. All of the above

Answer: c
2. Prostaglandins in the kidney can modulate which of the following functions (more than one answer possible)?
 - a. Regulation of blood flow
 - b. Regulation of renin secretion
 - c. Regulation of vitamin D synthesis
 - d. Regulation of collecting duct function
 - e. Regulation of erythropoietin production

Answer: a, b, e
3. Choose the Prostanoid Receptor(s) that are vasodilatory (more than one answer possible).
 - a. EP1
 - b. IP
 - c. EP4
 - d. TP
 - e. EP2

Answer: b, c, e
4. The lipoxygenase enzymes metabolize arachidonic acid to make all of the below except:
 - a. HETEs
 - b. Lipoxins
 - c. EETs
 - d. Leukotrienes

Answer: c